

Teacher Professional Learning

Drawing Upon International Practice for a Future Vision for Ukraine



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DRAWING UPON INTERNATIONAL PRACTICE
FOR A FUTURE VISION FOR UKRAINE

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Preface

Nowhere is the power of professional resilience more visible than in Ukraine. Despite the devastation caused by Russia's war of aggression against Ukraine, Ukraine's educators have refused to let learning stop. Through the *New Ukrainian School* reform, the country is reshaping its schools to help young people think for themselves and collaborate with others, to act ethically amid uncertainty, and to navigate tensions – whether between equity and freedom, autonomy and community, innovation and continuity, or efficiency and democratic process.

But this vision hinges on teachers who can put educational ideas and concepts into action. Around the world, everyone agrees: teachers are the beating heart of education. They don't just shape what students *know* – they shape how they *think*, *feel*, and *act*. Yet too often, we talk about great teachers as if they simply “appear”, rather than as the result of smart systems that invest in their growth every step of the way.

If education is to meet the demands of the 21st century, we can't leave teacher quality to chance. We need to take a lifelong view of what it means to grow as a professional – from the moment someone first dreams of teaching to the day they mentor others. Because when teachers flourish, schools flourish. And when schools flourish, societies rise.

A strong teaching profession starts with attraction and preparation. For too long, we've imagined teaching as a narrow career path – a few hoops to jump through before stepping into the classroom. But today's world demands more. Young people expect multiple careers, and our classrooms need diverse skills – from digital fluency to emotional intelligence to climate literacy. That means creating multiple pathways into teaching, inviting experts from many fields to share their knowledge with the next generation.

Preparation for teaching must blend deep theory with the realities of the classroom. It must help new teachers find their purpose – and give them the research skills, curiosity, and collaborative mindset to keep growing. Mentorship, coaching, and a smooth transition into real classrooms are not luxuries; they're the bridge between good intentions and great teaching.

But building a strong profession doesn't stop at entry. Teachers need room to grow – and recognition when they do. A clear, inspiring career structure can reward excellence while preserving professional autonomy. Yes, pay matters, but so does purpose. Teachers want to learn from each other, to co-create knowledge, and to shape their craft together. Collaboration isn't a side activity – it's the engine of innovation and motivation. Investing in professional learning isn't just good policy; it's a retention strategy.

A thriving profession also depends on connections – within schools, across schools, and through digital networks that allow knowledge to flow freely. When teachers learn from one another, when research and classroom practice meet, systems evolve. That's why creating structures that mobilise educational research and translate it into classroom impact is essential. Teachers' research literacy shouldn't end at graduation – it should grow through every stage of their career.

Every education system stands before the same challenge – and the same opportunity. If we want students who can build the future, we must build the teachers who can inspire them to do it. That's not just education policy. That's nation-building. And in no country will this matter more than in Ukraine.



Andreas Schleicher

Foreword

This report presents a holistic conceptual model for teacher professional learning from teachers' initial teacher education through to their ongoing, collective growth. It was prepared following the request of the Ukrainian Ministry for Education and Science to learn from leading international practice on teacher professional learning. The work was undertaken as part of the OECD Ukraine Country Programme. Alongside the Ukrainian Ministry for Education and Science and EdCamp Ukraine NGO, the OECD helped organise an international peer-learning event on 13-14 May 2025 in Kyiv, Ukraine, bringing together key Ukrainian stakeholders to examine teacher professional learning. The OECD prepared a background document to support this event. A draft of these chapters served as the foundation of this background document.

The report aims to support Ukraine's ongoing efforts to re-imagine their approach to pre- and in-service teacher professional learning in the context of the New Ukrainian School reform effort. Chapters 1-5 were prepared by the OECD Secretariat and present four high-level ambitions and accompanying levers for change for policymakers. These ambitions and levers are grounded in research and OECD reviews of leading international practice. They may be informative for all systems seeking to strengthen the infrastructure of teacher professional learning. The report concludes with Chapter 6, drafted by external experts on the Ukrainian context, which focusses on the current challenges and opportunities in professional learning in Ukraine and some of the main lessons learned from the international peer-learning seminar.

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The OECD Secretariat is also grateful for the participation of Chile, England (United Kingdom), Finland, Ireland and Singapore, as well as the European School Heads Association and the European Training Foundation, in the international peer-learning seminar. The OECD Secretariat is also grateful for the valuable input of Chile, England (United Kingdom), Estonia, Finland, Ireland, Japan and Singapore during the preparation of the work for this seminar and this publication.

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Abbreviations and acronyms

MEXT	Ministry of Education, Culture, Sports, Science and Technology (Japan)
MoES	Ministry of Education and Science (Ukraine)
NUS	New Ukrainian School
OECD	Organisation for Economic Co-operation and Development

Executive summary

What goes into making great teachers and a thriving teaching profession? There is agreement around the world that teachers are essential in shaping the knowledge, skills and values and attitudes of young people. However, there can sometimes be a tendency to think of great teachers simply ‘appearing’, without fully acknowledging the important structural features that shape their learning and growth. As education systems look to strengthen to meet the demands of the 21st century, a holistic view of teacher professional learning across a teacher’s career is essential to supporting the flourishing of our teachers, schools and, ultimately, students.

A strong teaching profession begins with attracting the best possible candidates and providing them with quality preparation. Whilst historically we have tended to primarily think about a series of specific ‘hoops’ for new teachers to jump through, policymakers across systems now also need to ensure that there are multiple routes for new teachers to enter the classroom. This is because it is now far more common for young people to imagine having multiple careers, but it is also a reflection of the diversity of skills that we expect our classrooms to nurture. Building strong digital literacy, social-emotional skills and climate literacy may mean finding ways of supporting individuals with existing expertise in these fields to become teachers.

This new reality of the teaching workforce only increases the importance of ensuring that preparation for the classroom is both theoretically sound and informed by the practical realities of the classroom. It is also about supporting the holistic development of teachers from day one; nurturing teachers’ sense of purpose and their ability to research, collaborate and reflect. These can provide teachers with a ‘compass’ to navigate the complexities and demands of teaching as their careers progress. Meanwhile, attending to the transition to the classroom, such as through ongoing mentoring and coaching from expert teachers, is essential for all teachers, regardless of their initial pathway to the classroom.

Building a strong teaching profession is also about ensuring that from day one there are both opportunities to grow, and recognition of this. A clear and attractive career structure underpins this; one that incentivises and rewards excellence, but that also offers a degree of agency and choice. One part of this is getting the financial incentives right, but it is also important to not lose sight of the fact that key to the attractiveness and strength of the teaching profession is how it works. Teachers desperately want opportunities to meaningfully collaborate together in their schools to build and exchange knowledge. In this respect, rich collaborative professional learning is essential for both capacity-building in the profession – particularly during periods such as today of rapid and unpredictable change – and for the sense of motivation that the profession feels. Quality professional learning can be an investment in retention.

This invites consideration of how professional learning connects as part of a wider ecosystem. A thriving profession is shaped by not only what happens within schools and the learning interactions that occur there, but also that between schools. Networks – both in-person and in the digital space – can be powerful means of creating new knowledge and ensuring more equitable access to it. In particular, as research on teaching continues to advance and grow, ensuring that system-level approaches are in place to mobilise this body of scientific knowledge and support its effective use is essential too. Indeed, this is a reminder of how interconnected professional learning across a teacher’s career is; teachers’ research literacy begins

in the skills and attitudes they develop in initial education and runs through their career as they are supported to keep up-to-date with the latest developments.

Despite the context of Russia's full-scale invasion of Ukraine, Ukraine has shown a remarkable tenacity and will-power to transform its education system through the New Ukrainian School (NUS) reform. Ukraine has continued even in the most challenging of circumstances to continue the reform efforts started in 2016. This has concentrated on fostering a richer, more holistic education experience where students develop depth of knowledge and transformative skills such as critical thinking and social-emotional skills. This demands that classrooms shift to including more competency-based pedagogy, which is why efforts to support the professional learning and growth of its extraordinary teachers are so important for the long-term success and sustainability of the NUS.

Whilst Ukraine's unwavering efforts to strive for a better education system even in the context of Russia's war of aggression is unique, many of the questions that they are confronting echo across systems. After all, the quality of an education system hinges upon its teachers. The challenges of nurturing a workforce that can foster students' holistic development and navigate the unpredictability of technological progress are not unique to Ukraine. The insights from research and country examples included in this report can serve as points for reflection for all systems, and perhaps also spur action as they are in Ukraine.

1

Background to reforming teacher professional learning in Ukraine

This opening chapter briefly introduces the goal of this report in the context of the New Ukrainian School reform. It outlines the structure of the report and how it approaches teacher professional learning as a holistic whole – from the initial selection of teachers to the workforce and their preparation, to how they are supported to learn and grow throughout their career as part of a flourishing professional ecosystem.

A system's approach to pre-service and in-service professional learning has a profound influence on the quality of teaching in its classroom. Moreover, pre- and in-service professional learning are increasingly seen as an inter-connected continuum that, along with the wider career structure, also contribute to creating a rich, rewarding and impactful career as a teacher.

In the 21st century, professional learning is not only about supporting the development of teachers' knowledge and skills as it may have historically been commonly viewed. Rather, it is increasingly also about nurturing the type of professional identity that can allow teachers – and accordingly their students – to thrive in a complex, ever-changing future. The OECD Teaching Compass captures the breadth of competencies that teachers need going forward, ranging from particular types of knowledge and skills, but also values and attitudes (OECD, 2025^[1]). At the same time, the Compass also highlights the centrality of teacher well-being and teacher agency in their work. Indeed, at its core are the three anchors of teachers' being, belonging and becoming:

- Being: Establishing a solid and evolving sense of self
- Belonging: Building relationships in the classroom and school community
- Becoming: Embracing continuous growth and transformation

It helps point towards emerging challenges but also new opportunities in terms of how systems conceptualise teacher professional learning as a means to supporting the flourishing of students, schools and communities alike.

Background for reforming teacher professional learning in Ukraine

Despite the extremely testing context of Russia's full-scale invasion, Ukraine continues to advance reforms in its education system. The New Ukrainian School (NUS) reform, launched in 2016, aims to make schooling engaging and stimulating by focusing on each student's personal development and their acquisition of key competencies. The Ukrainian Ministry of Education and Science's Strategic Plan to 2027 aims to create a system where all children have a safe, modern learning environment with ample opportunities for self-realisation, and access to high-quality education (Ukrainian Ministry of Education and Science, 2024^[2]). Focusing initially on primary schools, the reform efforts have steadily expanded in scope and recently concentrated on the transformation of secondary education with the support of the OECD (OECD, 2025^[3]).

Teachers have a key role to play in the long-term success and sustainability of the NUS reform. After all, teaching quality is the most important school level variable in determining the success of an education system (OECD, 2019^[4]; OECD, 2005^[5]). Yet teaching is also very complex (OECD, 2025^[6]), and accordingly its mastery demands both quality preparation and ongoing learning – particularly in an era where the demands on classrooms and schools is evolving in step with wider societal transformation.

Ukraine's teachers have been working through an extraordinarily demanding period due to Russia's war of aggression against Ukraine. Their efforts have been nothing short of inspiring. According to data from the OECD's Programme for International Student Assessment (PISA), between 2018 and 2022, performance in Ukraine's 18 regions in mathematics fell less than the OECD average, despite the unprecedented disruption to learning (OECD, 2023^[7]). Similarly, the OECD's most recent Survey on Social-Emotional Skills suggests that 15-year-olds in Ukraine feel better at school than in most sites, reflecting the care and attention they receive from their teachers amidst the war (OECD, 2024^[8]). Although they report less motivation than the average, Ukrainian 15-year-olds report having felt more confident, less anxious and less angry while at school, compared to almost all other survey locations (OECD, 2024^[8]).

These are reflective of Ukraine's dedication to building an education system that allows all to flourish. Even in the most challenging of circumstances, Ukraine's ambition remains unwavering. As part of the NUS

reform, Ukraine is now exploring the ways to better recognise and reward teachers throughout their careers, and to re-evaluate the types of supports they need at different stages of their teaching journey.

A vision for teacher professional learning based on international insights and best practice

At the Ukrainian Ministry of Education and Science's request, the OECD is supporting the country as it engages in examining its models of teacher professional learning and the surrounding teacher career infrastructure.

Parts of this report were prepared as a background document for a hybrid international peer-learning event in Kyiv, Ukraine on 13-14 May 2025. This was organised by the OECD and EdCamp Ukraine at the request of the Ministry. The event aimed to serve as a spark for inspiring change in how policymaking on teachers' growth and development is approached. A key goal was to examine teachers' career structures and professional learning as a holistic whole, spanning the first steps of a teacher's career to their ongoing professional learning to the wider ecosystem of collective learning. Chile, England (UK), Finland, Ireland, and Singapore, as well as the European Schools Heads Association and European Training Foundation, kindly participated in this peer-learning event and shared their experiences.

Four high level ambitions serve as the spine of the report. Chapter 2 to 5 focus on these four high-level ambitions, taking each ambition one-by-one:

- Each ambition is broken down into three more specific '**levers**' for change (see Table 1.1).
- Within each of these levers, some of their **key design considerations** are explored. These design considerations can be viewed as particular decision points for reflection when enacting change.
- With a view to the process of change, each chapter also briefly outlines some **implementation considerations** for their respective ambition.

Throughout these chapters, a series of **illustrations from different national and sub-national systems** around the world are included. In the Annexes, more detailed deep-dives into a selection of systems are offered.

Reflection questions are also provided to support school and system leaders – as well as teachers themselves – to reflect on the aforementioned design and implementation considerations. As this paper is orientated around offering a 'vision' of professional learning, reflection questions are framed with a view to the future and what could be possible, but they could also be transformed to focus on the present and gauge the current state-of-play.

Chapter 6 then turns to consider more concretely the Ukrainian context and some of the current challenges it faces in terms of teacher professional learning, as well as existing opportunities. Chapter 6 was authored by Liliia Hrynevych, Volodymyr Bozhynskyi, Oleksandr Elkin, Halyna Kovalchuk, Yevhen Nikolaiev, Grygoriy Riy and EdCamp Ukraine NGO. Following these authors' assessment of the Ukrainian context, the authors present in Chapter 6 – informed also by the aforementioned peer-learning event – some of the potential directions for Ukraine to explore with a view to further strengthening its approach to teacher professional learning.

Table 1.1. Key elements of a vision for teacher professional learning in Ukraine

Ambition	Levers	Key design considerations	Key implementation considerations
1. Build the initial foundations of an effective and meaningful career	Process for selecting candidates for the profession <i>(Attracting and identifying high-quality candidates)</i>	Define the entry requirements to initial teacher education (ITE) Establish the entry requirements to entering the classroom Define pathways into teaching Incentivise and attract candidates	Facilitate buy-in from different stakeholders Co-ordinate across multiple stakeholders Attend to the quality of school-based experiences Dedicate resources for impactful induction
	Nature of preparation for the classroom and profession <i>(Preparing teachers for immediate effectiveness and lifelong learning)</i>	Ensure quality in the provision of ITE Develop a coherent body of theory and knowledge Move beyond theory to build an understanding of practice Attend to the wider knowledge, skills, values and attitudes for their career ahead	Attend to the capacity and needs of teacher educators
	Attending to the transition into the classroom and school <i>(Supporting beginning teachers through the most challenging early stages)</i>	Establish induction programmes with coaching and mentoring Align induction as a coherent continuum with professional learning Facilitate feedback loops between schools, ITE providers and stakeholders	
2. Develop an infrastructure that incentivises and recognises excellence	A clear and attractive vertical career structure <i>(Rewarding excellence and expertise)</i>	Define stages to be clear, coherent and comprehensible Align with competency frameworks Clarify connections between stages Leverage appropriate financial incentives	Translate existing structures to new structures Communicate changes and build ownership Ensure coherence with ITE and professional learning provision Build coherence with leaders' career structures
	Opportunities to specialise through horizontal diversification <i>(Recognising the variety of roles of teachers)</i>	Define specialisations Accredit and recognise specialisations Enable and incentivise specialisations	Understand the role of cultures
	Trusted evaluation mechanisms <i>(Ensuring fair promotions and identifying development needs)</i>	Protect the role of the 'external' in evaluation for certification and promotion Draw upon well-defined, multiple measures Calibrate and audit measures	

Ambition	Levers	Key design considerations	Key implementation considerations
		Use formative evaluation for development Facilitate connections between the summative and formative to identify learning needs	
3. Facilitate quality, meaningful ongoing learning that enables the profession to grow and face future challenges	Provision of relevant learning opportunities <i>(Responding to needs at the individual, school and system levels)</i>	Establish who are the providers Align provision with needs and competency frameworks	Foster coherence with the foundational content of ITE Develop facilitators' skills Understand the accompanying needs of school leaders Consider the role of market forces Attend to equality of access to learning opportunities across localities Troubleshoot challenges through pilot programmes
	Delivery of quality learning opportunities <i>(Ensuring impactful learning experiences for teachers)</i>	Support structured communities of learning Utilise coaching and mentoring Enable peer observation and feedback Implement quality assurance	
	Incentives to engage in professional learning <i>(Fostering a culture of effective self-regulated engagement)</i>	Balance mandatory and voluntary choices Facilitate the means of participation Recognise and track participation	
4. Create an ecosystem for collective learning and sharing to ensure opportunities for excellence for all	A culture of collaborative inquiry within and across schools <i>(Encouraging inquisitive and reflective practitioners)</i>	Promote tangible and specific focusses Delineate clear structures and roles Dedicate resources to support collaboration Create structures for school-to-school collaboration Tap into the role of additional actors	Support leaders Identify 'early adopters' Appreciate the time needed for new cultures to embed Attend to potential reform fatigue Celebrate successes
	Developing engagement with research <i>(Nurturing a culture of evidence-informed work)</i>	Supply high-quality, relevant and accessible research Build research engagement skills Create system-supported approaches for research engagement	
	Systematising and disseminating knowledge <i>(Ensuring the identification of and learning from best practices)</i>	Capture new knowledge in systematic, coherent ways Curate new emerging knowledge Ensure effective tools for dissemination at scale	

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2 Ambition One: Build the foundations of an effective and meaningful career

This chapter presents the first of the four ambitions that orientate this report. It focusses on the initial foundations of teachers' career, encompassing the process for selecting candidates for the profession, the nature of their preparation in initial teacher education, and how transitions from initial teacher education to the classroom unfold. In discussing these, the chapter presents key design considerations from research literature, insights on implementation, and examples from systems around the world.

A thriving career as a teacher is heavily shaped by how these teachers first enter into their roles and the classroom. Decisions around choosing who will become a teacher and how they are prepared and supported for this role are important foundations for teachers' subsequent growth and development.

The work of teachers is very complex and demands ongoing work to refine their craft, and that means that initial teacher education (ITE) can never achieve everything that teachers will need in their careers. Nevertheless, it can help prepare teachers with a strong base of knowledge and skills, as well as values and attitudes, for being effective with their students from the beginning and for ongoing learning and flourishing across a career.

Moreover, teacher selection and preparation are significant not only in terms of what it means for the growth trajectory of individual teachers but also its collective strength. The type of knowledge, skills, values and attitudes that teachers develop early in their careers shape the whole profession. When teachers are attuned to research and problem-solving together, this can shape how schools and system will face future challenges. In turn, the selection and preparation of teachers also shapes how the teaching profession is perceived and its status, and thus its future pipeline too.

This ambition can be understood as structured loosely around how candidates are first selected for the profession, how they are prepared for the classroom, and how they are transitioned into becoming fully independent teachers. The last point is an important one as the transition is really a continuum as the initial teacher education that teachers experience should be coherent with their ongoing professional learning. Indeed, all four of the ambitions and twelve levers below are interconnected; for instance, the research literacy and supportive school and system environment of Ambition 4 (see Chapter 5) are very much essential supports in ITE, as will become clear across the subsequent sections.

Key highlights

This ambition focusses on the goals of attracting and identifying high-quality candidates; preparing these candidates for immediate effectiveness in the classroom as well as their lifelong learning; and supporting beginning teachers through the most challenging early stages. Some of the considerations it draws particular attention to include how:

- There remains considerable variation across countries and debate around the most appropriate level of selectivity, the criteria used, and when identification of candidates takes place.
- Generalised differences between entry requirements for different levels of education are rarely justified.
- Gauging teacher quality is, in general, challenging; it often demands recourse to multiple measures at different time points to build a more rounded picture of candidates' suitability for the classroom.
- The criteria for accrediting or evaluation ITE providers are a key mechanism of quality assurance, such as in relation to research outputs, research-use and facilitation of quality grounded practice.
- Preparation should balance both theory and knowledge, alongside quality experiences in school to develop teachers' practice to support their readiness to be impactful.
- Preparation should attend to the wider mindsets and competencies of teachers.
- Induction programmes using coaching and mentoring remain primarily under-utilised despite promising potential to support beginning teachers.
- Meaningful feedback loops between stakeholders can support alignment between the needs of schools, ITE institutions and the wider system.

Lever 1: Process for selecting candidates for the profession

While there is agreement around how important the phase of selection is for shaping the profession, there remains considerable variation and debate across countries around the most appropriate level of selectivity, the criteria used, and when identification of candidates takes place.

At the heart of this is the nature of entry barriers and where these are positioned. Both of these are ultimately interconnected; policymakers need to consider when to put the selection barriers that teacher candidates must clear, as well as how high or low to put these barriers.

Of particular note here is that for many systems this sits within the context of high levels of autonomy for higher education institutions. These institutions are often key actors in delivering ITE, and systems need to have trust and confidence in their ability to deliver content and prepare quality teachers. Systems often need to balance respecting the autonomy of such institutions, such as in their design of bespoke content and delivery methods, with efforts to encourage high, consistent standards.

It is also worth noting that the process for selecting candidates sits within wider challenges that systems may face. The process of selecting and certifying teachers shapes the overall supply of teachers and impacts who and how many apply to join the profession. For those systems facing acute shortages of teachers – as many systems around the world are today – this can be particularly significant.

Define the entry requirements to ITE

In general, there are two distinct time points where entry requirements need to be considered. The first relates to the entry requirements to enter into ITE programmes.

- Joining an ITE programme may incur the same requirements as that of joining other equivalent higher education programmes. Hence, to join a concurrent model ITE programme (see below), an upper secondary qualification at a certain level may be necessary.
- In approaches where a consecutive approach is followed in ITE, a bachelor's degree in a particular field may be necessary. Notably, this may be in the same subject as the ITE programme (e.g. a mathematics bachelor's for pursuing an ITE programme on mathematics teaching).
- Alternatively, some systems have sought to address shortages in certain subjects by showing more leniency; it may be that prospective candidates can pursue an ITE programme in mathematics teaching having obtained a minimum number of credits in this subject in their bachelor's degree (a similar approach here is the major/minor split in bachelor's degree).

Minimum qualifications may be coupled with additional requirements. These may seek to screen weaker candidates, particularly if ITE programmes are highly competitive.

- To this end, systems may make use of competitive examinations or interviews prior to admitting individuals to an ITE programme.
- For instance, in Japan and Korea competitive examinations determine entry to ITE programmes (they are also used to determine certification afterwards, as set out below) (OECD, 2019^[1]).
- As mentioned, this is heavily shaped by the overall demand for ITE programmes. If teaching is a highly attractive career with a surplus of applicants a degree of screening is necessary.
- This partly explains why there is no clear pattern to the approach of systems performing highly in the OECD's Programme for International Student Assessment (PISA); as mentioned above some systems like Japan and Korea make use of competitive examinations – either or both at the start and end of ITE programmes – while others make no use of competitive examinations at either point (e.g. England (United Kingdom), Estonia, Singapore) (OECD, 2019^[1]).

- More broadly, the association of certain criteria with future performance in teaching is modest, and its effectiveness has been questioned by some researchers (Kane, Rockoff and Staiger, 2008^[2]).

In navigating this decision, it is important to note that generalised differences between different levels of education are rarely justified.

- Differences in degree requirements across levels of education remain widespread in OECD countries (OECD, 2019^[1]).
- This particularly effects the early years or primary level that typically see lower entry requirements.
- Not only does this build an innate inflexibility into a system's pool of teachers and how they can be allocated, but it risks sending signals that run counter to the considerable body of evidence on the critical role of high-quality early education to students' development and future success (OECD, 2019^[1]).
- Indeed, such differences are also challenged by the very demanding nature of the work at the early years and primary level.

Establish the entry requirements to certification and entering the classroom

The second key time point to consider is entering into teaching itself and what are the associated entry requirements here.

One essential question is what steps, if any, are required for certification as a teacher following ITE in order to actually start in the classroom.

- In some cases, entry requirements after ITE may not exist. Thus, successful candidates from ITE programmes are fully qualified and can start teaching directly (e.g. Norway).
- An alternative may see the need to be certified by a specific national, sub-national or local authority. This could be a mere administrative step that allows for the overall distribution of teachers to be monitored.
- At the same, the process of being certified can also be a means of selecting the right candidates by implementing additional requirements for certification. This may be motivated by a desire to select only the strongest candidates from a large pool of ITE graduates, such as by using competitive examinations or standardised tests that successful graduates from ITE have to pass to be formally certified and allowed to teach (see Box 2.1).
- Alternatively, additional requirements may also be a response to concerns around the effectiveness of all ITE programmes to fully prepare candidates for the classroom (see Lever 2 below). Additional requirements may be a means of monitoring the quality of ITE graduates being developed, as well as encouraging a degree of competition between providers.
- Additional requirements such as standardised tests are, of course, associated with additional resource costs too. In particular considering how it can be challenging to gauge the quality of a potential teacher through single standalone measures. It may also create specific incentives for ITE providers who might focus more on the demands of the additional requirements than the classroom itself.

Notably, this form of additional requirements and selection may occur in the latter stages of an ITE programme as part of its successful completion.

- For instance, a standardised test could be a requirement to graduate from an ITE programme.
- Alternatively, in Finland, oral exams are drawn upon in the ITE programme to gauge motivation (see Annex B).

Finally, a second connected question here is if teachers have a probationary period or not when they enter the classroom.

- For instance, teachers may be able to start teaching, but still have a certain duration of probation where they are monitored. One example of this is Japan (see Box 2.1).
- Some have pointed to the potential value of probationary measures due to their ability to allow for the effectiveness of teachers to be better gauged. Hence, probation can allow for teachers to be retained based on a better measure of their proven effectiveness than, say, competitive examinations (Rockoff et al., 2011^[3]; Staiger and Rockoff, 2010^[4]).
- At the same time, these have to be balanced with resource constraints; evaluation processes of teachers are complex and demanding (see Chapter 3, Lever 6).
- A further consideration is that additional barriers may demotivate certain candidates and depress overall supply (Hanushek and Pace, 1995^[5]; Vegas and Ganimian, 2013^[6]).

Define pathways into teaching

One of the most common routes into teaching is through ITE programmes. Even within these, typically two models dominate.

- **Concurrent (or parallel) approaches** see pedagogical and subject matter studies occur at the same, alongside any practical training features too like practicums.
- For instance, this has historically been a common approach to preparing teachers at the primary level. In Japan, teachers at the primary and lower secondary level are prepared through a concurrent model that covers a range of content related to academic subjects, pedagogical studies, educational science, child adolescent development studies, research skills development, as well as a 20-30 practical experience (see Annex B).
- **Consecutive approaches** see a focus on subject matter initially. These are then followed by courses in pedagogy and more practical training.
- For instance, this model may be characterised by individuals completing a bachelor's degree in a specific subject and then a master's in pedagogy. In England (United Kingdom), teachers can pursue a consecutive model where they can qualify for entry onto a one-year postgraduate certificate of education with at least a bachelor's degree (3 years) (see Box 2.1). Their route may be shaped by the particular subject of that degree.

These models have ramifications for the coherence of preparation and how different types of knowledge and skills may be integrated. They intersect with the content of preparation, which is discussed more below too (see Lever 2).

A further important consideration is that of so-called alternative pathways into teaching. A 2018 Eurydice report (2018^[7]) found that approximately one-third of European countries have some alternative pathway to a teaching qualification.

In particular, a common feature has been the development of accelerated, so-called 'fast-track' routes into the classroom. These have grown in popularity in many systems (OECD, 2025^[8]; OECD, 2024^[9]).

- One key objective of these is to offer a means of bringing different skills and experiences into schools.
 - For instance, alternative pathways may allow individuals to undertake career changes into classrooms, reflecting the shifting attitudes and norms in society around working life (OECD, 2025^[10]).
 - The prestigious, competitive nature of certain courses may be significant for attracting certain high-achieving candidates, as well as the opportunity such routes can present for moving quickly into the classroom and developing one's career in schools.

- Such alternative pathways may also reflect the fact that features such as a teacher's ability to craft strong relationships matter greatly for student learning but are hard to measure. As previously mentioned, questions have been increasingly raised about the predictive power of certain requirements (Rockoff et al., 2011^[3]; Staiger and Rockoff, 2010^[4]). Alternative pathways may place more emphasis on entry to teaching being about what teachers know and can actually do, rather than passing through a specific route.
- Similarly, they may also be a response to concerns about the overly-theoretical approach to preparation of some ITE programmes and seek to increase emphasis on a more practical-focussed training model.
- Finally, alternative pathways may be driven more broadly by a need to increase the supply of candidates by maximising routes into the classroom.

Alternative pathways may combine several different features, including:

- Offering teachers the opportunity to earn a salary and essentially practice as an uncertified teacher while working towards their formal certification.
- Adopting a more apprenticeship-style training approach, as may be typically used in certain crafts or vocational fields. This may be combined with more emphasis on practical assessment too for certification.
- In England (United Kingdom), several pathways similar to these above exist alongside each other, as set out in Box 2.1.

Overall, the evidence on the effectiveness of such approaches for teachers is largely inconclusive thus far (OECD, 2019^[11]).

- In general, the evidence base is hindered by methodological constraints, such as a reliance on correlational results or the sheer challenge of measuring quality teaching. There is also the fact that long-term effects (such as those who later return to teaching, or who move into adjacent areas) can be hard to capture.
- Studies in the United Kingdom (Allen and Allnutt, 2017^[12]) and the United States (Clark et al., 2017^[13]) tend to find either no difference or very marginal benefits to alternative pathway teachers, with unknown general equilibrium effects due to their tendency to leave teaching at more rapid rates.
- The OECD's Teaching and Learning International Survey (TALIS) 2024 notes that comparing perceived quality across different pathways is challenging due to the small sample size of teachers who completed non-regular teacher training. Holding this in mind, it is suggested that there does not seem to be large differences in perceived quality of initial teacher education between fast-track routes and regular routes (OECD, 2025^[8]). There is variation among countries and data suggest that routes may have more emphasis on certain features (e.g. subject understanding, practical elements) than others, which can be both advantageous and a drawback.
- More theoretical arguments have also been raised; one notable concern is that with their strong focus on more 'on-the-go' learning in the classroom, they may minimise the need for mastery of knowledge and theory, which arguably undermines the intellectual status of the profession.
- That said, at the same time, the 'fast-track' nature may also serve as a significant trajectory for individuals to grow into teaching, school leadership and other related areas – such as policy making and social enterprise – which may still have a positive contribution to schools and systems (Schleicher, 2018^[14]). In particular, the impacts in wider roles may be harder to fully measure, but significant, nonetheless.

Incentivise and attract candidates

Building on the topic of pathways that incentivise, countries may also turn to other mechanisms to further incentivise and attract candidates. For instance, the use of financial incentives (e.g. scholarships, subsidies, stipends) may be drawn upon as levers to enact change.

- These may serve to raise the profile of ITE and to attract particularly highly skilled candidates. For instance, as in Estonia (see Box 2.1).
- These may also be a means of addressing specific shortages or needs. For instance, in England (United Kingdom) (see Box 2.1), the Netherlands and Slovak Republic there has been use of scholarships and subsidies for students in so-called shortage subjects, such as the sciences (OECD, 2019^[1]).
- Notably, these may also focus not only on recent graduates from secondary or higher education, but also those seeking a career move; in Chile, there are scholarships for individuals with experience outside education who would like to teach, which is coupled with an alternative pathway similar to those set out above.

Incentives may be attached to certain criteria, such as a certain number of years of service.

- For instance, in Chile, the Teacher Vocation Scholarship (“Beca Vocación de Profesor” [BVP]) targets academically talented secondary education graduates.
- It offers a scholarship (among other benefits) if they choose teacher education as a higher education degree and commit to teaching in a publicly-funded school for at least three years (Santiago et al., 2017^[15]).

Alternatively, incentives may serve as a lever for also attracting beginning teachers to particular contexts, such as more disadvantaged or remote ones.

- Some studies have pointed to the promise of monetary incentives to help distribute teachers where they are needed the most.
- That said, such policies will work different depending on how the size of incentives interacts with the wider framework of career progression and employment.

There may also be efforts to concentrate incentives earlier in the ‘pipeline’ of attracting individuals to the profession. For instance, in England (United Kingdom), a system of ‘teaching internships’ are available for those interested in the profession to understand what it is like in the classroom. The internships last three weeks in June or July at the end of the school year. Interns are paid, receiving £425 (~€500) per week. Notably, these paid internships are targeted; paid internships are available to those studying for an undergraduate or master’s degree in chemistry, computing, languages, maths, physics. They thus seek to address particular subject shortages. Those interested in teaching from other subjects can visit schools to get unpaid experience in the classroom.

Finally, it is also worth noting that the attractiveness of the profession does also extend to wider financial incentives such as the pay structure (see Chapter 3), as well as beyond. The intellectual attractiveness of the profession, as well as the opportunities for learning, growth and agency in their career, are also significant (OECD, 2025^[10]). This highlights the interconnectedness of the four ambitions.

Box 2.1. International examples of approaches to selecting candidates

England (United Kingdom)

'Qualified Teacher Status' (QTS) is a minimum requirement needed to teach in publicly-funded primary, secondary and special schools in England (United Kingdom) and is awarded as part of Initial Teacher Training (ITT) courses.

There are entry requirements to be achieved prior to the Initial Teacher Training course commencing and separate requirements to be awarded QTS that will be met over the course of the Initial Teacher Training course.

- To be awarded QTS, candidates must have a bachelor's degree – or equivalent qualification – in any subject.
- For entry into primary and secondary teaching courses, candidates must have passed Maths and English at the upper secondary level at the equivalent minimum grade (GCSE grade 4, or recognised equivalent).
- In addition, primary school teachers must also hold a minimum qualification (GCSE grade 4) in a science subject.

There are three routes to QTS:

- The undergraduate route, which normally runs over three or four years and includes teacher training via an undergraduate degree route and the new Teacher Degree Apprenticeship pilot which is a salaried route (and is accepting applications from 2024/25 onwards).
- The postgraduate fee-funded route, which normally runs for one year full-time:
 - Combines theoretical learning with at least two classroom placements in schools.
 - Courses have to be paid for (approx. 9 500 GBP per year [~11 000 EUR]), though there are some bursaries or scholarships to incentivise participation in certain subjects. A system of loans also exists.
- The postgraduate salaried route, which also normally runs for one year full-time:
 - These allow individuals to earn a salary while they work towards QTS.
 - They follow a more apprenticeship-based approach with most of the time spent in school placements over theoretical learning.
 - This includes the School Direct salaried programme, the High Potential ITT programme (HPITT) and, since 2018/19, the Postgraduate Teaching Apprenticeship (PGTA).
 - These are highly competitive and in high demand.

An example of a salaried route into teaching is via Teach First. Teach First is a charitable organisation that also delivers so-called 'high potential initial teacher training'.

- Teach First is a charitable organisation that also delivers so-called 'high potential initial teacher training'.
- This is an employment-based route into teaching for high-performing graduates and those who are looking to change careers.
- Similar to the above, participants on the programme also earn a salary while they work towards qualified teacher status (QTS) with a postgraduate certificate in education (PGCE).
- Teach First is part of the Teach for All network, which offers similar alternative pathways into teaching in a range of countries.

- For instance, Teach for Ukraine is a non-governmental organisation that is also part of the Teach for All network, and which offers an alternative pathway into the teaching profession in Ukraine. Teach for Ukraine focusses on engaging talented graduates and young professionals to teach in rural settings in order to focus on learning losses in these communities. This includes offering stipends as financial incentives. Like other providers in the Teach for All network, they follow a similar competitive multi-step selection method and intensive summer training institute.

Following gaining QTS, individuals can apply to teaching positions to complete a mandatory two-year induction period (see Annex B) as a so-called ‘Early Career Teacher (ECT)’.

Estonia

In recent years Estonia has concentrated considerable efforts on enhancing the quality of its teacher workforce. New teacher standards were introduced in 2013, which were developed through an interactive process representing a considerable range of stakeholders. New standards have been introduced since these with the latest entering force as of 1 September 2025.

The standards are an important tool; institutions need to demonstrate that their curriculum prepares teacher candidates for all the requirements stated in the professional standards, with an explicit matching occurring between the learning outcomes of the curriculum, modules and courses on the one hand, and the teacher standards on the other (Révai, 2018^[16]).

There are two major initial teacher education providers in Estonia: Tartu and Tallinn University. There are also two other institutions that only focus on preparing teachers of certain areas (music, arts and theatre).

For those entering at the primary level, programmes of ITE consist of mostly concurrent studies. At the lower or upper secondary level, it generally consists of a consecutive programme.

On the back of concerns that a teaching career was not a very popular choice among upper secondary graduates, Estonia has developed a series of programmes to further incentivise talented candidates to join the teaching profession. This includes:

- A scholarship programme: This seeks to motivate students with talent for teaching to apply to teacher education specialisations and to acquire a teaching qualification. Students who enter the teaching profession within 18 months of completing their education receive an additional allowance. The programme was also extended to professional support specialists.
- The “Youth to School” programme (“Noored Kooli”), part of the “Teach for All” programme: This seeks to raise interest in teaching and education by awarding scholarships to a select group of university students. Recipients teach at a school for two years, while at the same time participating in teaching and leadership training. After the programme, students can choose to keep working at school, return to university or work elsewhere.

Japan

In Japan, ITE is provided by approved universities or departments of universities. There are hundreds of universities and departments that have met the necessary requirements to provide ITE programmes. They must undertake self-evaluations and undergo accreditation processes by evaluation and accreditation organisations certified by the Ministry of Education, Culture, Sports, Science and Technology (MEXT) at least every seven years.

In order to enter ITE programmes, individuals need to pass entrance examinations and be enrolled as students at universities with ITE programmes. In order to be able to become a teacher, candidates need to complete all required teacher-preparation courses and a practicum, in addition to an associate or a bachelor’s degree.

A key actor here is the of the local Boards of Education (BOE). Following completion of an ITE programme, teacher candidates must apply to the local BOE. The BOEs of prefectures and ordinance-designated cities are responsible for certificating and hiring teachers to work in schools in their jurisdiction. They administer their own competitive employment examinations for assigning teachers as permanent teachers at public schools (see Annex B).

More broadly, there have been efforts in recent decades to build more collaboration between universities and BOEs to help raise the quality of initial teacher education programmes, as well as that of ongoing professional learning.

Notes: England (United Kingdom) shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2019^[11]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>; (Santiago et al., 2016^[17]), *OECD Reviews of School Resources: Estonia 2016*, <https://doi.org/10.1787/9789264251731-en>; (Révai, 2018^[16]), *What difference do standards make to educating teachers?: A review with case studies on Australia, Estonia and Singapore*, <https://doi.org/10.1787/f1cb24d5-en>; (OECD, 2018^[18]), *Education Policy in Japan: Building Bridges towards 2030*, <https://doi.org/10.1787/9789264302402-en>

Lever 2: Nature of preparation for the classroom and profession

Evidence on initial teacher education and what makes it effective is growing, but it remains a somewhat incomplete picture (OECD, 2019^[11]). This means that there is a need for policymakers as well as those engaged in delivering initial teacher education to remain flexible with a view to being responsive to an evolving evidence base.

At the same time, research has demonstrated how important good preparation is. For instance, the importance of carefully considering how teachers are prepared for the classroom is exemplified in the fact that the better prepared and early career teachers feel, the longer they tend to stay in the profession. This relevance is heightened in a context of shortages as faced by many systems (OECD, 2024^[19]).

Ensure quality in the provision of ITE

The question of who provides ITE is a foundational one to many education systems. Their quality shapes the effectiveness of teacher preparation and, ultimately, the strength of the entire education system.

Policymakers are often confronted by several overarching challenges to navigate.

- As previously mentioned, in general there can be high degrees of autonomy among higher education universities. This can be important for spurring innovation – particularly considering the limited evidence base on ITE as well as the changing demands – and responding more flexibly to different contextual needs (OECD, 2019^[11]). But it also means that trust and collaboration may be needed to navigate this balance between autonomy and assurance.
- ITE programmes may be directly accredited on a programme-by-programme basis, or it may be that institutions are accredited and then have the autonomy for quality assurance of its own programmes.
- Different time frames also present a problem to consider. There can be multiple, interlocking cycles of evaluation, preparation and accreditation occurring simultaneously. For instance, programmes may be evaluated on historic data for a renewal of their accreditation, while still continuing to implement ITE programmes with current students.
- The example of Ireland has carefully balanced these practical challenges with rigour and thoroughness, with all institutions reviewed and accredited during a ‘cycle’ lasting approximately two years. An independent review panel is appointed, with each review of an institution taking approximately eight months (see Annex B).

In systems where the delivery of ITE is more directly accredited to higher education institutions or specific programmes, one key lever of quality assurance is the criteria for accreditation. Common features may include:

- The quality of the academic staff, such as in terms of their research outputs (e.g. impact metrics, research expertise).
- How ITE providers integrate the latest research evidence into their curricula in a responsive yet rigorous way.
- How ITE providers facilitate quality grounded practice, that is also sufficiently substantial for exposing teachers to the different experiences that await them (OECD, 2019^[11]).

Further sources of information that may be drawn upon include:

- Evaluation by teacher candidates of ITE providers, both in general as well as in a more granular form in terms of certain courses.
- Evaluation by schools of ITE providers based on the quality of candidates that a provider develops.
- Self-evaluations or independent audits may also be undertaken. There may be requirements for these to be transparently communicated as in Finland (see Box 2.2).
- Notably this information may be used in a summative way to inform accreditation, but also used formatively to support ongoing improvement of ITE programmes.

Where there is less direct accreditation of ITE or concerns around the quality of provision, alternative means may be drawn upon to support quality assurance.

- Teaching standards – or more bespoke ‘graduate standards’ focussing exclusively on beginning teachers – can be a key tool. They can serve to concretely define what graduates from ITE programmes should know and be able to do.
- As set out in the next section, such standards can be a particularly valuable lever for shaping the design of actual content in an ITE programme. They may also serve as an opportunity for engagement by different stakeholders, who can have a say in determining the type of graduates they seek to be developed (e.g. school leaders).
- As mentioned, additional requirements after an ITE programme for certification and starting to teach can also be a key means of quality control. In particular, standardised examinations may be a means of assessing the quality of preparation by ITE programmes. Lever 1 above has addressed this in more detail.

Develop a coherent body of theory and knowledge

One of the most fundamental functions of preparation is to introduce teacher candidates to the body of theory and knowledge that underpins education. While this body of theory and knowledge is less well-defined in education than in some other professions (e.g. medicine), and also under ongoing construction, there remains nevertheless a large body of important knowledge that has been developed over time (OECD, 2025^[20]; OECD, 2019^[11]). This can be considered as the minimum ‘content’ that ITE should impart to teacher candidates.

More specifically, some of the particular knowledge that may be concentrated on includes subject knowledge and general pedagogical knowledge.

- These can be seen as two key bodies of knowledge that teachers draw upon alongside their more locally-situated pedagogical content knowledge.
- The latter essentially sees the translation of more abstract theory and knowledge around a subject or teaching into actual actions that support student learning. It is primarily developed through

focussed effort in the classroom, which is why there is a need for both theory and practice in ITE, as addressed below.

Striking the right balance between breadth and depth in the content of ITE is no easy task.

- The nature of concurrent or consecutive models can have a particular influence here.
- Similarly, particular attention may be needed to subject knowledge depending on the route to specialisation in a subject. For instance, in some systems one can teach a particular subject based on a similar degree (e.g. teaching chemistry with a physics degree).
- Alternatively, at certain levels where there is not specialisation like primary, some systems may still dedicate particular attention to certain subjects (e.g. mathematics) to respond to wider priorities or system-level needs.
- It has been argued that there has been little shift over the past decades in how prepared teachers feel for the classroom (OECD, 2019^[11]). This suggests that preparing teachers with the more grounded, pedagogical content knowledge for the actual classroom remains a challenge.

One tool for developing coherence in this content is clear, well-defined teacher standards.

- Standards can help to define what is needed in terms of theory and knowledge. It can also help to provide coherence with what teachers need to be able to do in terms of their practice and how they actually use any theory and knowledge.
- Standards may be particularly relevant for prioritisation. It is, after all, more common to add to the expectations of teachers than to take away from them.
- Exercises to map standards to the content of ITE programmes may be a fruitful exercise for identifying gaps and a shared understanding across different stakeholders (Révai, 2018^[16]).
- Such mapping exercises may also help to expose the types of experiences and more practical knowledge that is needed. After all, standards cannot encompass everything due to the challenge of capturing the more tacit pedagogical content knowledge of the classroom.
- One further advantage may be that clear standards can also underpin career progression beyond the initial stages as a teacher, as outlined below (see Chapter 3). They can thus help to ensure alignment between ITE and future progression and learning opportunities.

Another tool may be thoroughly attending to the coordination of content across ITE programmes.

- One particular limitation in the teaching of this fundamental content has been its typically episodic nature; small chunks of content are taught in isolation often by different individuals, with little attention to how they may align with or build on each other.
- This may be further exacerbated by the separation of teaching and research in ITE institutions (e.g. universities). It may mean that the latest research findings are not always easily integrated into the curricula of ITE institutions.
- Some systems such as Finland have sought to bridge this gap through the integration of more subject-specific studies alongside pedagogical studies (see Box 2.2).

Move beyond theory to build an understanding of practice

Both theory and practice are important in the content of ITE programmes. Theory helps to prepare teachers for a life of learning, becoming research-informed and aware of why certain actions are taken. Practice is about ensuring that implementation is effective and the actions and decisions are translating this theory effectively. To this end, ITE should also seek to incorporate quality time and experiences in school.

Two particular forms of exposure may be especially relevant: observing practice in action, or actually engaging in practice themselves in the classroom – what is often referred to as grounded practice.

- **Observing colleagues** may help candidate teachers see practices in action and understand better their different components. In particular, observation of colleagues should be about exposure to high-quality practice.
- Moreover, the reflection on this type of experience is particularly relevant. This can support specific learnings to be derived from observation and potentially mobilised into a teacher candidate's own practice or knowledge base.
- Observation may be especially relevant when it comes to exposing teachers to more innovative, student-led practices which candidates may have had little exposure to in their own educational experiences.
- **Grounded practice** is particularly important for the process of transferring one's theoretical knowledge into actual action.
- For instance, subject/content knowledge can be relatively easily developed, but as mentioned beginning teachers tend to find it much harder to develop the aforementioned 'pedagogical content knowledge' that relates to mobilising this knowledge of a subject to actually support student learning. This type of knowledge is grounded in the actions and decisions of teaching, and thus locally-situated in the classroom.
- It means that opportunities to actually experience the process of mobilising and transferring knowledge into teaching actions is particularly important.
- Again, opportunities to reflect on grounded practice is particularly important to support candidate teachers to identify areas for further development, as well as better identifying the rationale of certain decisions or actions.

Notably, there is variation across countries in how school-based grounded practice occurs (OECD, 2019^[1]; OECD, 2019^[11]). This is partly shaped, again, by whether a more concurrent or consecutive approach is adopted to ITE, or whether alternative pathways are pursued.

- Broadly speaking, grounded practice can be seen as either an extensive or an intensive model. There is limited research on the effectiveness of different approaches.
- The former sees grounded practice spread across a prolonged period of time and with normally a breadth of experiences.
- The intensive model sees a more concentrated period in schools during a shorter time frame, but with more immersion in the sustained or interconnected nature of experiences.
- There are also several important implementation considerations that inform the design process (see below).

Notably, in some systems, such as Ireland, increased emphasis in recent years has been placed on the time that is spent in schools in ITE. The 'school placement' as defined in the underpinning 'Standards for Initial Teacher Education' includes direct teaching of 200 hours, as well as beginning teachers experiencing a variety of placement settings so they have a wider range of experiences to draw on. Notably, ITE providers are also expected to schedule an 'extended period' school placement. This should constitute a minimum of 10 weeks in one school in the second half of the programme, meaning that beginning teachers experience more immersion in a school community (see Annex B).

Attend to the wider knowledge, skills, values and attitudes for their career ahead

ITE not only prepares teachers for the classroom, but for a career as a teacher and their active contributions to the profession throughout this career. The fast-paced nature of the 21st century has heightened the need and challenge of nurturing a future-proof profession.

In this context, one key area of consideration is the wider knowledge demands that are placed on teachers, and thus ITE programmes too.

- An increasingly important area of content for countries is supporting teachers to broaden their knowledge from expertise in only standalone subjects to also accommodate knowledge of transversal themes grounded in global trends. The aforementioned OECD Teaching Compass reflects these changing knowledge needs (OECD, 2025^[21]).
- There are examples of content coming to increasingly incorporate knowledge of digitalisation, including AI and digital literacy; student's social-emotional development, including well-being and health; and the environment, including sustainability and civic values (see Box 2.2).
- Similarly, as the example of Singapore in Box 2.2 shows, this has also included teaching this content through new methods to reflect the type of learning environments future teachers may seek to nurture.

A second key area is how ITE develops and nurtures the disposition of teachers.

- A key aspect of this is their disposition towards ongoing learning. For instance, it is in ITE that a critical stance towards one's own practice can be first developed, with early career teachers learning how to self-inquire into their practice and adapt.
- Coupled with this process of self-inquiry is a willingness to and understanding of how to update their practice as new evidence or needs emerge. ITE can build the foundations of research literacy in education and engender the habits of research-use that can develop evidence-informed practitioners long into the future.

A third key area is how ITE lays the foundations of teachers' identity as a teacher and professional.

- Establishing a solid and evolving sense of self is crucial for teachers' professional identity, development and effectiveness, as exemplified in the role that teacher's sense of self plays in anchoring teachers in the OECD Teaching Compass (OECD, 2025^[21]).
- In particular, working with experts and policymakers from a number of countries, the OECD's Initial Teacher Preparation Study outlined some of the most important qualities for beginning teachers to endeavour to develop (OECD, 2019^[11]), such as strong communication and relationship skills, confidence, resilience and proactivity, and values such as a sense of mission, responsibility and ethics.

Box 2.2. International examples of preparing teachers for the classroom and profession

Singapore

In Singapore, there is one single provider of initial teacher education to prepare teachers for the classroom and profession: the National Institute of Education. Preparation seeks to develop the knowledge and skills related to teaching, as well as imbuing candidates with the purpose and values of teaching and the education system.

In 2023, the Enhanced TE²¹: Empowering Teachers for the Future Model was launched to help develop teachers that are ready for the future, insofar as they themselves can develop students' 21st-century competencies so they are also future-ready. The new model builds on the original model but with a particular focus on preparing teachers and the school systems for the fluctuating changes of the future. As outlined in PISA 2022 Volume V (OECD, 2024^[22]), the new model is rooted in:

- Three core values:
 - Commitment to the learner: believing that all children can learn, nurturing each learner holistically, and valuing diversity.
 - Commitment to the teaching profession: engaging in lifelong learning not only for one's personal and professional growth but as a role model to students.

- Commitment to the community: being cognisant of their role in the ecosystem and contribute to society by impacting the next generation of learners.
- Knowledge of the self as a teacher, the learners, the subject content and pedagogy. In particular, there has been a focus in the new model of expanding teachers' knowledge of topics such as sustainability, global and environmental issues, and health and mental health.
- Skills for teachers of 21st-century learners such as promoting reflection and metacognition, digital and data literacy and cross-cultural literacy.
- Three core competency dimensions: professional practice, personal growth and development, and leadership and agency. These aims to help teachers to enable teachers to perform five roles, namely, shapers of character, creators of knowledge, facilitators of learning, architects of learning environments and agents of educational change.

In terms of delivery, there has been a focus in the past decade or so on promoting a learning environment in ITE that is more centred on and driven by teacher candidates rather than teacher educators. This was grounded in seeking to develop more self-directed, inquisitive learners. Simultaneously, there was also a focus on strengthening the link between theory and practice. A manifestation of these shifts is that candidates were, for instance, increasingly asked to engage in inquiry and reflection in the classroom, such as through using real-life school scenarios as a basis for this. More problem-based, blended and flipped learning have become more important.

Finland

There are eight universities and five universities of applied sciences with responsibility for teacher education in Finland. Teacher education is organised by different groups depending on one's future career, such as class teacher (primary education), subject teachers (lower secondary education or general upper secondary education), or special needs teachers, among others.

Finland's preparation of teachers has a strong focus on the development research skills. Student teachers learn to produce and consume research-based knowledge.

- ITE often includes courses on qualitative and quantitative research methods.
- Teachers are typically required to complete a master's degree, including a master's thesis.
- Similarly, there is often use of inquiring teaching methods in ITE, such as case-based approaches or project-based learning.
- Notably teacher educators are immersed also in this research-based model of ITE. Teacher educators do research in their own teaching and draw upon this.

Overall, this seeks to develop the type of reflective thinking that is necessary in the profession and ongoing, self-driven learning. This also seeks to foster a strong level of research literacy that prepares teachers with the capacity to undertake their own research and to critically evaluate their work. It leads to an overall competence in research methods, including broad knowledge of a range of methods and more detailed, specialised knowledge in at least one.

ITE in Finland is also characterised by concerted efforts to foster robust pedagogical content knowledge. Teacher education is a shared responsibility between the teacher education faculty and the academic subject faculty. This combination means that there is substantial attention to the type of more subject-specific pedagogy necessary for robust pedagogical content knowledge. It avoids treating pedagogy as generic and devoid of particular subject contexts (e.g. questioning approaches in different subjects). This notably applies to the preparation of upper-grade teachers, but there are also some opportunities for classroom teachers as the primary level to complete more studies in a certain subject.

A focus on more theoretical pedagogical studies is coupled with guided teaching practice. Candidates to the profession practice in specific teacher training schools during their studies. These schools provide

a space not only for candidates to the profession to practice but also support experiments, research and continuous professional learning too. Such practice is not an additional 'extra', but an essential part of achieving the credits to complete specific programmes.

Underlying this is a blend of both autonomy but also evaluation, transparency and cooperation. There is no single 'curriculum' for ITE programmes and universities have autonomy in determining this. At the same time, universities are required to evaluate their education and research work and its effectiveness, as well as to engage in external evaluation activities too. The outcomes of evaluations are published.

There is also a notable degree of co-operation, with the Teacher Education Forum funded by the Ministry of Education and Culture supporting this. It has been particularly significant in supporting providers to respond to emerging needs and challenges. For instance, the Teacher Education 2050 Vision has been recently developed, which invites greater consideration of the challenges to be increasingly accounted for such as eco-crisis, digitalisation and well-being.

Estonia

There is a strong focus on linking theory and practice in ITE in Estonia. In particular, ITE seeks to prepare teacher candidates for the realities of the classroom through a range of school-based experiences. These occur in the two years of the master's programme, and include:

- Observations to help teacher candidates relate their theoretical knowledge to practice.
- Teaching practice of 6-10 weeks in a school, organised through a range of partnerships with schools.
- A university-based supervisor and a school-based mentor to support reflection on practice, and to provide general oversight or guidance.
- In some cases, simulated pedagogical practice also occurs in a university setting (e.g. the University of Tartu).

There is also a notable focus on research skills, grounded in bridging theory and practice. This is evidenced by the writing of a master's thesis, for instance. Or, in some settings, building teacher candidates' research capacity through involvement in research projects (e.g. the University of Tartu).

There is no standardised curriculum for ITE, but institutions use the aforementioned teaching standards to inform the content of ITE programmes. The standards define a broad range of knowledge areas teachers need (e.g. development psychology, basics of research, basics of reflection, age-related peculiarities) and competencies, as well as particular tasks. Analysis has suggested that the breadth of tasks teachers undertake demands a wide knowledge base that is both theoretical – stretching across general pedagogical knowledge and more specific content-based pedagogical knowledge – and including knowledge of research methods (Révai, 2018^[16]).

Notes: Finland and Singapore shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2024^[22]), *PISA 2022 Results (Volume V): Learning Strategies and Attitudes for Life*, <https://doi.org/10.1787/c2e44201-en>; (Révai, 2018^[16]), *What difference do standards make to educating teachers?: A review with case studies on Australia, Estonia and Singapore*, <https://doi.org/10.1787/f1cb24d5-en>

Lever 3: Attending to the transition into the classroom and school

No matter the quality and extensiveness of a teacher's initial preparation, the transition to the classroom will always be a key moment. The nature of teaching and the rhythm of school life means that it can be hard to ever fully capture these realities for candidates to the profession.

Accordingly, the transition process of teachers truly stepping into the classroom for the first time as a responsible teacher – whether they have already been certified, are on probation, or preparing for a final certification – is an important foundation in building an effective and meaningful career.

Research has documented how beginning teachers can experience a very acute ‘shock’ in encountering the realities of teaching and schools.

- One worrying consequence is an elevated risk of burnout.
- A second is the tendency to resort to so-called ‘survival strategies’ in the face of initial challenges, which is unlikely to expose students to quality teaching (OECD, 2019^[11]).

Establish induction programmes with coaching and mentoring

As noted in TALIS 2024 (OECD, 2025^[8]), the literature has noted that induction programmes have been found to have positive effects on teacher quality (as perceived by students) (Maulana, Helms-Lorenz and van de Grift, 2015^[23]) and student achievement (Rockoff, 2008^[24]). They may also help reduce teacher attrition (Smith and Ingersoll, 2004^[25]; Helms-Lorenz, van de Grift and Maulana, 2015^[26]). TALIS 2024 also points to a positive relationship between induction programmes and job satisfaction, and in some systems with well-being for teachers (OECD, 2025^[8]).

In terms of specific activities for induction, there is no clearcut understanding of what it should constitute, with research still relatively nascent (OECD, 2019^[11]), though providing multiple types of support seems to be particularly significant (OECD, 2025^[8]). Broadly speaking, induction may consist of mentoring and coaching, as well as the integration of teachers into professional learning opportunities with other colleagues (e.g. department teams):

- **Mentoring** can be understood as a means of providing more general support to beginning teachers, such as in relation to their well-being and developing their reflective thinking. This may also focus on some of the acute early challenges beginning teachers can frequently encounter, such as managing workload. Mentoring is the most visible, implemented and researched of induction practices (Valenčič Zuljan and Marentič Požarnik, 2014^[27]). TALIS 2024 reports that in about one-third of education systems, novice teachers with an assigned mentor are more likely to report fulfilling their lessons’ aims and enjoying higher job satisfaction (OECD, 2025^[8]).
- **Coaching** can be understood as being more orientated to improving specific practices of beginning teachers. This may see more directed guidance from a coach through the identification of bespoke strategies for particular challenges or weaknesses. Again, an initial focus of coaches may be on acute early challenges, such as behaviour.
- **Integration into the school community** may consist of how beginning teachers are provided with opportunities to collaborate and learn from new colleagues, such as in department or age-group teams. It may also consist of specific onboarding opportunities that share the more tacit ‘ways of working’ in the school and its institutional knowledge.

In recent years, rates of participation in induction activities have seemingly increased. TALIS 2024 notes that participation in any induction programme (whether formal or informal) increased by 31 percentage points, on average, when considering education systems that participated in both TALIS 2018 and TALIS 2024 (OECD, 2025^[8]). This is an indication of the potential that induction may hold.

In particular, due to the potentially high positive impact of initiatives such as coaching (Kraft, Blazar and Hogan, 2018^[28]), there may be a need to attend to induction being systematic across schools if it is to fairly support all teachers and their pupils. For instance, TALIS 2024 noted that in almost one-third of education systems, public school teachers who have started working in their current schools in the five years before the survey have lower chances to benefit from induction activities – be it formal or informal activities (OECD, 2025^[8]).

Align induction as a coherent continuum with professional learning

One identified limitation with induction has been how it is still typically viewed in a very narrow way.

- Specifically, induction has been primarily seen as a standalone one-year feature. This is despite the fact that it has been documented that certain practices, such as adapting to student thinking, can be particularly challenging for beginning teachers (van de Grift, van der Wal and Torenbeek, 2011^[29]).
- Moreover, this rigidity does not meet the varied needs of teachers, and that some may need more sustained support. This may be particularly relevant for more challenging contexts.

There has been, accordingly, arguments for induction to be viewed more broadly (OECD, 2019^[11]):

- For instance, induction can be viewed in terms of a two-to-seven-year duration form of support. This can be adapted to individual teachers or schools.
- Coupled with this may be a more ‘faded’ approach, where induction supports are steadily removed. For instance, the regularity of coaching and mentoring can be reduced. Naturally, features such as orientation largely become irrelevant, too.
- There may also be scope for not only fading induction but also strategically reorientating it; for instance, concentrating on more complex, relational practices once certain fundamentals of the classroom (e.g. behaviour management) have been first addressed (OECD, 2025^[20]).

One example of this more open-ended support can be seen in the example Singapore (see Annex B). Beginning teachers receive a mentor in the school who then supports them with their eventual identification of a particular ‘track’ in its career structure. This identification of a track comes when there is agreement with the mentor that a teacher is ready for this step.

Lastly, the case of Singapore also serves as a reminder of the need to consider alignment between induction and the wider career structure and professional learning.

- Induction may be combined with a probationary period that is part of the certification process, as set out above.
- Induction may also be orientated towards preparing teachers for specific standards and certain evaluation processes (see Chapter 3).
- Induction may also be fused with efforts to support ongoing professional learning, such as the use of teacher development plans or wider school development plans.
- Indeed, induction can also be conceptualised as a cycle; there are examples of approaches that transition those who received induction into eventually becoming the mentors of the next generation of beginning teachers (see Box 5.6 in OECD, 2025^[11], p.159).

Facilitate feedback loops between schools, ITE providers and stakeholders

A fundamental tool for ensuring the transition process goes well is that there is alignment between the needs of schools and ITE institutions. Is the content that ITE preparing teachers with what schools need? And are schools able to provide ITE institutions with what they need if their content is to be sufficiently supported and built upon through practical experiences hosted by schools and induction as schools assume more responsibility for teachers?

Accordingly, opportunities for meaningful dialogue between schools and ITE providers are important.

- Loops can help to promote quality and efficiency, and that resources from different stakeholders are being channelled in the best way.
- As the OECD’s Flying Start study noted (OECD, 2019^[11]), the use of ‘meaningful’ is particularly important; dialogue should not be purely informative and logistical based (e.g. orchestrating the

times when schools will host teachers for practical experiences) but engage both actors in really designing the content to ensure different needs are effectively met.

- Moreover, these feedback loops may also be significant for ensuring more localised needs of schools are met. For instance, those needs of rural schools or those with particularly high proportions of second language learners, or with special education needs.

Box 2.3. International examples of attending to the transition

Estonia

Teachers beginning their careers in a school have the opportunity to voluntarily participate in a 12-month induction programme. This is funded by the Ministry of Education and Research and the induction programme is provided by both of the public comprehensive universities that also provide ITE.

The induction programme hinges on the allocation of a mentor to supervise the work of the beginning teacher. The mentor is a teacher at the receiving school and is appointed by the school principal. There are certain criteria that the mentor must fulfil too, namely having at least three years of experience in pedagogical work and having passed a specific training in supervision.

In terms of tasks, the mentor provides a support for beginning teachers as well as supervising their work. The beginning teacher will, in particular, prepare an individual development chart which contains self-reflections of their experience. The mentor is also required to provide feedback to the ITE provider from which the new teacher received their qualification. In this respect, there is a feedback loop between schools and institutions.

The beginning teacher also has access to tailored workshops that address the frequent challenges of the first years in a school. These seek to address experiences and explore solutions, be that individually or jointly with other teachers.

Japan

Boards of Education (BOE) in Japan are required by law to provide induction training for one year for all beginning teachers with a regular-term contract. Candidates who have been successful with the examinations of their BOE to be hired, are on a probationary status in their first year of employment.

The induction training process includes assigning a mentor teacher for every new teacher, who are commonly selected by the principal of the school from among their pool of vice principals or senior teachers in the school. It often includes in-school professional learning opportunities too.

More broadly, Japan has seen more concerted efforts in recent years to foster collaboration and closer connections between different stakeholders. For instance, the system of affiliated schools in Japan are an example of strong university-school partnerships that seek to support practical insights into the classroom. More than 250 affiliated schools provide opportunities for on-site innovative teaching experiences for teacher candidates, as well as serving as a space for collaborative research between the school and university. Such experiences can mean that induction is not beginning teachers' first exposure to the realities of the classroom.

Chile

Chile's Induction and Mentoring Program provides dedicated support to new teachers and their integration into the education community. It reflects their ongoing efforts to recognise and attend to the importance of the first years of teaching.

Induction is a right of all beginning teachers who are appointed or contracted to a state-funded institution. It lasts for 10 months during the school year, and beginning teachers receive an allowance for the time spent on induction.

Beginning teachers have an assigned mentor who is an experienced teacher (at the upper stages of the career structure: Advanced, Expert I or Expert II) and who has received specific training to be able to lead and support effective induction.

- Mentors may be from the same institution or another.
- Mentors may oversee up to three beginning teachers.
- Mentors are also entitled to receive allowances for each beginning teacher they mentor and support.
- Mentors must design, implement and evaluate a mentoring plan based on the beginning teachers' needs. Guidance is published by the Ministry to support this, including a rubric for self-assessment with beginning teachers.

The induction process requires a commitment of between four and six hours a week for the beginning teacher and mentor to meet and reflect on practice and challenges, as well preparing classroom work, analysing learning outcomes, and considering integration into the education community. Notably, the emphasis is on mentoring and formative development, rather than supervision and evaluation.

High-performing schools have the right to implement and manage their own induction processes.

A commissioned evaluation of the system of professional learning noted that induction was highly valued, but has yet to fully permeate the school culture. It also did not have wide coverage yet. This has been a valuable evaluation for further refinement of the system of professional learning, including induction. For instance, now there are now ongoing efforts to enhance the relevance of induction to specific, more localised contextual needs. This includes trying to better match beginning teachers with mentor teachers from that particular local context.

Notes: Chile shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2019^[1]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>; (Santiago et al., 2016^[17]), *OECD Reviews of School Resources: Estonia 2016*, <https://doi.org/10.1787/9789264251731-en>; (Révai, 2018^[16]), *What difference do standards make to educating teachers?: A review with case studies on Australia, Estonia and Singapore*, <https://doi.org/10.1787/f1cb24d5-en>; (OECD, 2018^[18]), *Education Policy in Japan: Building Bridges towards 2030*, <https://doi.org/10.1787/9789264302402-en>

Key implementation considerations

Building the initial foundations of an effective and meaningful career is complex. It demands the co-ordinated effort of a range of different stakeholders and the alignment of different structures and resources. Some key implementation considerations that surface from international examples and best practice include:

- **Coordinating across multiple actors:** Initial teacher education is a complex system that depends on multiple actors. In many systems, it is increasingly a case that initial teacher education depends on multiple ITE institutions, as well as a range of key partners. For instance, schools are especially key partners in a system seeking to balance theory with practice. Similarly, if ITE is to serve as a foundation for a continuum of learning, then in-service providers of continuous professional learning are also important actors if they are to appropriately build on the foundations that are laid. More broadly, the actors that build the evidence base any initial teacher education and professional

learning occurs on are important, be that researchers, knowledge brokers and networks of practitioners. There is a need for systems to understand, first, who is involved, and, second, how they are and how they should be connected if there is to be alignment and coherence.

- **Facilitating buy-in of different actors:** Coordinating across multiple actors may also constitute facilitating their collective ownership of the ITE approaches. Some systems have sought to reflect the multi-stakeholder and complex nature of ITE by engaging different stakeholders in its design and development. For instance, in Norway, collective guidelines were developed with different stakeholders (OECD, 2019^[1]). In Ireland, the Teaching Council is very much a partnership-based organisation and as part of the review of the policy on initial teacher education undertook four rounds of consultations, constituting some 12 meetings, with different stakeholders. This ranged from teacher unions, parent bodies, principals' bodies to student unions, to name but a few. A similar picture unfolded in Ireland with the professional induction programme for new qualified teachers ("Droichead"). This was also developed by the Teaching Council in Ireland in consultation with the profession. Moreover, it was also launched first as a pilot, in 2013, and, following a positive evaluation of the pilot in 2016, expanded to all newly qualified teachers (OECD, 2024^[30]).
- **Delivering quality school-based experiences:** Ensuring grounded practice or quality observation through school-based experiences demands considering resources:
 - One aspect is who enables a school's teachers to engage in supporting beginning teachers, such as watching their lessons and providing feedback.
 - Moreover, as it is about exposure to good practice, the most appropriate schools for teacher candidates to undertake grounded practice also need to be identified. This often demands a central oversight. It may also mean trying to expose teacher candidates to good practice in similar contexts to where they will eventually work as an independent teacher to support their transition.
 - There may also be some training needs for preparing those teachers who are supporting new colleagues. For instance, in how to best be observed and share the more tacit expertise of their decision-making. This is also a significant point in relation to mentoring and induction, as outlined below.
- **Enabling impactful induction through sufficient resources:** Mentoring and coaching need to be both enabled and recognised. Enabling may be providing the time for mentoring or coaching meetings to occur in the school day by freeing up time in teachers' timetables. Recognising it may include providing opportunities to develop as a mentor or coach through specific professional learning, or it being a recognised feature of career progression. For instance, mentoring is far from being universally available for teachers. TALIS 2024 finds that, on average across OECD education systems, about one in four novice teachers (teachers with up to five years of teaching experience) have an assigned mentor (OECD, 2025^[8]). This share of novice teachers with an assigned mentor increased in about one-third of education systems since the previous TALIS cycle (2018). However, considering the value of mentoring that has been documented in the research, there remains scope for further growth still. There is also substantial cross-country variations in the prevalence of mentoring programmes for novice teachers; for instance, in Shanghai (China) almost 79% of novice teachers report having an assigned mentor.
- **Concentrating sufficient attention and resources on teacher educators:** For some time, it seems the role of teacher educators has been ill-defined and not properly considered (OECD, 2019^[11]). This is despite their important role in shaping the profession. After all, teacher educators serve as role models for their student teachers. Their approach to the classroom as well as their disposition (e.g. towards research) are significant. Indeed, the motivation and skills of teacher educators in ITE institutions are important for shaping what is imparted. For instance, some teacher educators may be primarily researchers who are obliged to engage in supporting teacher candidates, and this may shape their motivation for this function and its subsequent quality.

Alternatively, it is rare that teacher educators get induction or support in their role, as well as bespoke capacity enhancing measures such as professional learning.

- **Look outwards and build on others' experiences:** Many challenges in teacher professional learning are not unique to individual systems. There may be inspiration and lessons to be drawn from the example of other systems. In developing its Procedures for the Professional Accreditation of Programmes of ITE, the Teaching Council in Ireland undertook a range of consultations. One particularly valuable step was their dialogue with representatives from Ontario (Canada) and Scotland. These international case studies helped to inform their policymaking process.

Box 2.4. Key reflection questions for system leaders

To build the initial foundations of an effective and meaningful career, system leaders may reflect on the following:

Process for selecting candidates for the profession

- How can entry requirements to ITE and, in connection, certification, best drive quality standards forward?
- How do the pathways to the classroom foster quality as well as a diversity of candidates?
- What could be the role of incentives to further attract quality candidates?
- How can the overall selection process send positive signals about the teaching profession?

Nature of the preparation for the classroom and profession

- What blend of mechanisms and criteria could best promote quality provision of initial teacher education?
- How is information on the quality of provision used to support further learning and development among ITE providers?
- What could be the balance of quality control with supporting the need for innovation and flexibility?
- How can teachers be supported to develop a coherent, foundational body of theory and knowledge?
- How might teachers be supported to go beyond theory to develop their actual classroom practice through quality experiences?
- How can coherence across the content taught by teacher educators, as well as other actors like schools, be ensured?
- How can the knowledge base and content of ITE programmes be updated in tune with evolving research and evidence?
- How can teachers be supported early on to develop a critical, reflective stance towards their own practice to help its ongoing development?
- How might foundational research and inquiry skills be developed to support a teacher's lifelong learning?
- How might a teacher's self-concept and sense of purpose as part of the profession also be attended to in ITE?

Attending to the transition into the classroom and school

- How can resources and capacity-building be used to promote quality coaching and mentoring experiences?

- What can support all teachers to have access to quality induction supports?
- What structures could enable meaningful dialogue between ITE providers and schools to ensure that different needs are being appropriately met?
- How might induction contribute to wider learning in the school by introducing new ideas or questioning received wisdom for greater clarity?

Implementation considerations

- How might the capacity of teacher educators be supported and developed to increase their impact?
- How can the different roles of stakeholders in ITE be defined and coordinated?
- How might the voices of different stakeholders be heard and considered to foster buy-in?

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3

Ambition Two: Develop an infrastructure that incentivises and recognises excellence

This chapter presents the second of the four ambitions that orientate this report. It concentrates on the career structure of the teaching profession in a system, including both the opportunities for vertical progression and for horizontal diversification through specialisations. A third consideration is how trustworthy, reliable evaluation mechanisms support the effective implementation of this career structure. In discussing these, the chapter presents key design considerations from research literature, insights on implementation, and examples from systems around the world.

Career structures can be thought of as a set or sequence of recognised professional positions with associated tasks and responsibilities, as well as the rules that govern individuals' progression across these positions (OECD, 2019^[1]). Two core benefits underlie opportunities for promotion or specialisation:

- One, they are important for teachers' motivation. For instance, differentiated levels allow public acknowledgement of teachers' skills and professional growth. Accordingly, they also communicate important wider messages about the profession that shape both retention and attraction of teachers.
- Two, they also matter for ensuring effective allocation of resources. Differentiated levels can support efforts to align between teachers' competencies and particular tasks. For instance, highly effective teachers may be able to be assigned certain classes or roles such as coaching.

Key highlights

This ambition focusses on the goals of rewarding excellence and expertise through career progression; recognising the variety of roles of teachers; and ensuring fair promotions while identifying development needs at different levels. Some of the considerations it draws particular attention to include how:

- A key tool for differentiating and communicating stages can be through alignment with specific competency frameworks.
- Opportunities for progression or specialisation can play a distinct role in teachers' long-term motivation, even in the absence of end-of-year bonuses or other incentives.
- In determining any financial incentives, the positioning of the 'floor' and 'ceiling' can have profound effects on teacher supply and retention.
- Maintaining the 'external' nature of high-stakes evaluation can help support trust and consistency in the evaluation mechanisms.
- As measuring the quality of teaching is challenging, one option can be to increase the quantity and diversity of measures that are drawn upon for a more rounded picture; the more varied the information is that is generated, the more likely are teachers to trust it and the less likely it is that teachers will try to manipulate it.
- Focussing evaluation and the criteria for career advancement on the intended outputs of professional learning, such as growth in practice, competencies and knowledge, may help to ensure that professional learning remains impactful and relevant, rather than a 'box-ticking exercise'.
- Formative evaluation to improve practice is also an important feature, and it can be helpful to consider this as being a more localised, school-led form of evaluation.

Lever 4: A clear and attractive vertical career structure

One way of differentiating teachers' careers is in terms of vertical progression. Like a ladder, teachers may have opportunities to progress through a succession of formal positions or roles. These are organised hierarchically with distinct task profiles and increasing levels of responsibility, that in turn demand increasing levels of skills or experience.

The clear articulation of these stages and the associated expectations can allow teachers to have clarity over what is necessary to progress. It is also important for any associated certification and evaluation process.

Define stages to be clear, coherent and comprehensible

When designing multi-stage career structures, the number of stages and their associated roles and responsibilities is a foundational consideration for policymakers:

- Stages need to be specific and clearly articulated.
- At the same time, they also need to be distinct to ensure that they can effectively motivate teachers and facilitate fair progression.
- Ideally each stage of a teacher's career should be associated with a clear set of progressively increasing responsibilities and competencies (see below).

A review of 20 OECD systems in 2019 found that a little less than half provide their teachers with multi-stage vertical career structures that offer opportunities for promotion based on a succession of formal positions with distinct task profiles in the classroom (OECD, 2019^[1]). Among these, structures comprise between two and six distinct stages, which gives an indication of the number of stages that may allow meaningful differentiation and motivation.

Often, the articulation of career progression typically begins by considering teacher's entry point as a novice teacher. This forms the first stage in teacher's journey, and from which they build. This can also help support alignment with initial teacher education and any forms of induction that are in place to support teachers' early development.

Align with competency frameworks

A key tool for differentiating stages can be through alignment with specific competency frameworks. These frameworks can be considered as the articulated 'Teacher Standards' for what is expected from those in the profession at each level (Hooze, 2016^[2]).

- Differentiated competency frameworks that specify what is expected of teachers at different stages of their career can help to build a shared understanding of good teaching.
- This shared understanding can serve as a valuable focal point for connecting how professional learning and career progression interlink; the standards articulate the competencies that teachers should endeavour to develop and which they will eventually be assessed on for their progression.
- In this respect, competency frameworks can give coherence across learning opportunities, evaluation mechanisms and career progression decisions.

In some systems, such as Estonia (see Box 3.1) and the Flemish Community of Belgium, connections between competency frameworks and teachers' career structures have been clearly established (OECD, 2019^[1]). The latter is a valuable example; differentiated professional standards exist, providing teachers with a list of basic teacher competencies adapted for different levels and covering a range of teachers' expected functions (e.g. content expert, innovator and researcher, partner of parents etc.). There are also a series of professional profiles designed to guide teachers' development over their career. Notably, the standards are not linked with a corresponding vertically differentiated career structure. The result is limited awareness and use of the standards to plan professional learning needs or guide their appraisal processes (Nusche et al., 2015^[14], pp. 148, 164).

Clarify connections between stages

In thinking about the nature of the stages in a vertical career structure, a further design consideration is the type of connections between stages. There is a need to define the types of criteria that are necessary for advancement.

- On the one hand, there is the consideration of progression being voluntary or not.
- While vertical career advancement is often voluntary, some countries require teachers to apply for higher career stages after a certain amount of time (OECD, 2019^[1]). This is a means of monitoring continued effectiveness.
- Voluntary and meritocratic access to higher career stages can be important though for effectively motivating and rewarding professional growth.

A related consideration is that of the duration of progression. Is progressing to a higher stage permanent or temporary?

- Again, a temporary progression that sees the need to demonstrate the appropriate competencies for that higher level can be a means of monitoring continued effectiveness.
- At the same time, it may have ramifications for teachers' motivation, potentially reducing the attractiveness of certain progression steps.
- There are also administration burdens and resource costs attached to re-evaluating teachers' progression.

Underpinning the above considerations is the question of evaluation. Evaluation can inform appraisal processes and a teacher moving up the structure. In defining stages and ensuring that what is expected in terms of roles and responsibilities is clear, so too the 'step' between stages must be defined. These can be considered as the criteria for advancement, and is considered in more detail below in relation to Lever 6: Evaluation mechanisms.

Leverage appropriate financial incentives

Opportunities for progression can play a distinct role in teachers' long-term motivation, even in the absence of end-of-year bonuses or other incentives that are more common in on-teaching professions (Crehan, 2016^[3]; Natale et al., 2013^[4]). That said, financial incentives may be a key feature of progression. This is an extremely complex matter as it demands policymakers consider teacher salaries and their wider fiscal implications.

In particular, one tension that is navigated is between the positioning of the 'floor' and 'ceiling' of the salary structure.

- On the one hand, systems may compress the salary levels between the stages of a career progression and have a higher entry salary – a high 'floor' – for beginning teachers (e.g. Denmark). This may have particular implications for the attractiveness of the profession.
- On the other hand, teachers may have a lower entry salary but a steeper progression in salaries, including a higher salary for reaching more advanced stages of the progression – a higher 'ceiling' (e.g. Korea). This may have particular implications for retention and the way that progression is incentivised (OECD, 2019^[1]).

Box 3.1. International examples of multi-stage career structures

Estonia

Estonia introduced a new vertical career structure in 2013, alongside a reformed system of teacher professional qualifications. It comprises four distinct stages reflecting different levels of professional skills and experience. The four stages are Teacher (Level 6), Teacher (Level 7.1), Senior teacher (Level 7.2), Master teacher (Level 8). Level 7.1 is awarded following completion of a master's degree level initial teacher education programme or equivalent qualifications. Senior teachers are awarded for, in addition to regular teaching activities, aspects such as supporting the development of the school or other teachers, and Master teachers also work closely with a higher education institution and participate in development and creative activities in and outside of their school.

Level 6 and Level 7.1 are awarded indefinitely, while Level 7.2 and 8 are awarded for a five-year period after which the teacher must reapply. The stages are not formally linked to salaries and financial incentives remain variable (see below), and access to higher stages is voluntary. New responsibilities are given to teachers as a result of progression, rather than being a criterion for progression.

With regard to the financial incentives, career progression decisions are made independently of financial decisions. Promotions are accompanied by a guaranteed salary increase, but the exact amount depends on the number of teachers at each level; local authorities share out salaries from a lump salary sum. This means that there is not an upper limit on promotions but at the same time salary expenditure remains constant and predictable for the authorities.

Notably, the absence of financial incentives has meant that most teachers show little interest in attaining the certifications and engaging in the professional learning that allow access to higher levels. A recent review of the system suggested that the roles of senior and master teacher are not universally understood (Cordingley et al., 2022^[5]). Moreover, the review suggested that there was a need to better leverage the two more senior roles of senior and master teacher for fostering more collaboration.

Slovak Republic

The Slovak Republic has four stages: beginning teacher, independent teacher, teacher with first certification and teacher with second certification. In progressing through these stages teachers can undertake different specialisations in the Slovak Republic's horizontal structure (see Box 3.2). For instance, beginning teachers can only serve as class teachers, and independent teachers can undertake more roles but not yet serve as mentors to other teachers. Meanwhile, those with second certification – the fourth stage – can become members of national and international expert committees and sponsor professional development programmes.

Progress across the four stages is based on teachers' growing professional competencies, their experience and the accumulation of credits through the continuing professional development opportunities that are provided. For instance, one way to become an independent teacher is by accumulating 60 professional development credits.

Source: (Santiago et al., 2016^[6]), *OECD Reviews of School Resources: Slovak Republic 2015*, <https://doi.org/10.1787/9789264247567-en>; (OECD, 2019^[11]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>; (Crehan, 2016^[3]), *Exploring the impact of career models on teacher motivation*, <http://unesdoc.unesco.org/images/0024/002462/246252e.pdf> (accessed 21 February 2025); (Cordingley et al., 2022^[5]), *Policy Recommendations for the Professional Development of Teachers and School Leaders in Estonia*, https://www.britishcouncil.ee/sites/default/files/policy_recommendations_for_the_professional_development_of_teachers_and_school_leaders_in_estonia_eng.pdf (accessed 15 February 2025)

Lever 5: Opportunities to specialise through horizontal diversification

Teachers may also have opportunities to undertake horizontal diversification. This is typically a way of recognising and encouraging specialisation and can be thought of as lateral career moves. These lateral moves can be combined with vertical career progression opportunities too (see Box 3.2).

Teachers naturally enjoy and excel in some roles more than others. It is therefore important to acknowledge and support their capacity for leadership in some areas, just as it is to recognise their need for further development in others. Opportunities for horizontal diversification can give teachers autonomy in shaping their careers (Crehan, 2016^[3]), as well as potentially supporting more efficient use of teachers' time in relation to the alignment of their knowledge and skills with tasks. Moreover, it is noteworthy that data from TALIS 2024 finds that mentoring others is positively associated with job satisfaction in about half of education systems (OECD, 2025^[7]), which points to the satisfaction that teachers may derive from certain horizontal roles.

It is worth noting too that new roles in a horizontal structure may be effective responses to new and emerging areas of need, which is particularly relevant in an era of rapid change (e.g. technology specialists, social-emotional specialists).

Define specialisations

As with vertical progression, it is important that specialist roles are clearly defined.

- This should encompass the nature of its roles and responsibilities. For instance, specialisations such as mentors or in digital tools may demand specific knowledge, skills or competencies, and they may also have expectations in terms of what teachers do or who they work with.
- This also encompasses outlining what is the relationship with vertical structures. For instance, some systems have developed both horizontal and vertical structures, but this demands clear articulation of the type of roles and responsibilities, as well as competencies expected (see below) that these different structures demand (see Box 3.2).

Accredit and recognise specialisations

A connected question to defining specialisations is how these roles are obtained and retained. Specialisations tend to rely on specific knowledge, skills and competencies and it is therefore important to verify these for the specialisation to be accredited.

- This may take the form of proxy measures, such as experience levels for assigning the specialisation of mentor to a teacher.
- Alternatively, it may draw upon outputs from summative or formative evaluations as part of the vertical structure; again, taking the example of specialisations as a mentor, this may be assigned based on high performance in evaluations (see Lever 6 below).

Another approach may see instead bespoke evaluations of the skillsets of specialisations. In this respect, the space of micro-credentials may be particularly significant. Micro-credentials are a form of credential that individuals can earn upon the completion of organised learning activities that are “smaller, more targeted and more flexible” than traditional education and training programmes (OECD, 2023^[8]). This touches upon some of the identified potential advantages of micro-credentials with regard to professional learning:

- micro-credentials allow for the division of particular skills and competences for a more granular recognition of these,
- are learner-centred by allowing teachers to choose areas of focus relevant to their needs,

- enable provision to responsively evolve in step with potential new demands on competences (Minea-Pic, 2020^[9]).

Nevertheless, their implementation demands careful consideration.

- One consideration is attending to the quality assurance mechanisms that allow for formal accreditation of the micro-credential (e.g. gathering evidence of skill mastery through student work and videos that are then evaluated by a reviewer).
- A further consideration is how there is coherence across micro-credentials appear particularly relevant. After all, the very name suggests the existence of a connected, wider credential, with micro-credentials typically seen as a tool to complement more conventional programmes (Kato, Galán-Muros and Weko, 2020^[10]).
- These challenges are complicated by the considerable lack of evidence with which to make decisions. While there has been particular attention on micro-credentials at the tertiary level (OECD, 2023^[8]), in teacher professional learning their use remains in its infancy, due partly to the sheer novelty of the approach. That means that research on the effectiveness of micro-credentials and, crucially, what makes them successful in implementation is nascent.

Finally, as with vertical progression, the question of duration in horizontal specialisations is also significant. How long is a specialisation assigned for? This is connected to the question of evaluation mechanisms and the resources that may be demanded, and balancing this with the degree of quality control that having mechanisms for teachers reapplying to specialisations may bring.

Enable and incentive specialisations

One consideration is how specialisation is incentivised. This may be purely based on teachers' intrinsic motivation.

- The opportunity to assume more responsibility in terms of an area teachers are passionate about may be a primary means of encouraging teachers to seek out lateral career moves.
- This depends on understanding the demand for particular roles and ensuring provision and demand align.

Other mechanisms that systems may make use of includes reduced teaching hours.

- A common approach is to see teachers who adopt a specialisation compensated with fewer hours in the classroom.
- This also provides teachers with time for actually enacting these specialised roles, thus attending to teacher workload and well-being. Indeed, the reduction in hours may actually mean that the most effective teachers stay in the classroom as they do not need to move to administrative or leadership roles due to the opportunities the specialisation facilitates (e.g. see Austria in OECD, 2019^[12]).

There may also be the availability of certain learning opportunities with particular horizontal tracks that serve as an incentive and reward. Financial incentives can also be a factor too (e.g. see Uruguay in Santiago et al., 2016^[17], p.229). This may be tied to the related career progression pay structure, or it may be the case that regular or occasional additional payments are in place.

Box 3.2. International examples drawing upon horizontal diversification

Singapore

In Singapore, vertical career progression is combined with opportunities for specialisation in a particular role through horizontal diversification. In terms of the latter, three parallel streams exist:

- The teaching track is designed to cultivate experts in the classroom, with teachers staying in the classroom and supporting younger teachers in the school. Those at the top of this scale earn the same as school principals.
- The leadership track provides an additional pathway for teachers to develop their leadership competencies – both within a school and across schools. Hence, teachers can progress to Subject Head, Head of Department, Vice Principal and Principal, before progressing beyond a school to roles such as Cluster Superintendent up to the Director-General of Education. Some 25% of educators within Singapore's schools are on this track.
- The specialist track caters to those wishing to become researchers and to develop specific subject matter knowledge. The specialist track is relatively small, with an estimated less than 1% of Singapore's 33 000 employees in this track. Specialists are expected to have or be pursuing a doctorate.

Each of these comprises at least four stages of career advancement (vertical progression). Each level on the three tracks does not correspond directly to a set job description or school-based function, with school principals determining the kinds of responsibilities and experiences associated with each level based on the needs of the school.

Teachers' pay increases as they progress. In the first three years, pay increases independent of performance. Following their specialisation into one of the three tracks, pay increases as teachers progress. Retention bonuses are also available every 3 to 5 years, as well as one-off, performance-based bonuses every year.

Appointment to a higher level is performance-based (see Box 3.3) and can depend on teachers' or school leaders' completion of certain forms of professional learning. For instance, some of these can be time-intensive preparation courses of six months that teachers are recommended to (e.g. The Leaders in Education Programme).

There is a degree of central coordination across the career ladder. There is a quota for how many teachers can reach each level. Progression can be viewed more as an 'appointment' than 'certification'. Teachers can preference a certain track, with teachers being supported by a mentor in their first three years to choose a track, but the Ministry could reassign teachers to certain tracks if they demonstrated they are more suited to that track.

England (United Kingdom)

England (United Kingdom) has a series of 'National Professional Qualifications' (NPQ) for teachers and school leaders. The qualifications aim to:

- Improve career progression
- Improve quality and consistency of leadership
- Improve retention of teachers and leaders
- Improve the quality of teaching

These were reformed in 2021 to ensure they were underpinned by the best available evidence of effective teaching and leadership practice. Examples of qualifications include "Leading Behaviour and

Culture”, “Leading Literacy” and “Leading Teacher Development”. There are also qualifications aimed at the leadership level (e.g. “Senior Leadership”, “Headship”).

Qualifications are formally taught through a blend of face-to-face, online and independent learning with a defined curriculum, the qualifications are both a learning opportunity as well as recognition of the teacher or school leader’s knowledge and competence in a specific area. To successfully complete the NPQ there is an assessment, which was reformed in 2021.

Qualifications typically last for 12 months with a minimum of 50 hours of engagement. Leadership qualifications are longer at 18 months and most demand a minimum of 75 hours of engagement. Provision and delivery is outsourced to an approved set of providers, with the selection of providers rigorous based on specific criteria. Moreover, for each NPQ, the Department for Education has developed and published a framework to support organisations developing and delivering NPQs. The content of the frameworks for NPQs and the supporting evidence are independently assessed and endorsed by the Education Endowment Foundation (EEF) (see Annex B for more information).

Slovak Republic

In the Slovak Republic, a range of horizontal specialised positions are available. Different specialisations become available at certain points in teachers’ vertical progression. Although these positions are not organised hierarchically, they allow teachers to develop areas of expertise and deepen certain aspects of their profession.

Specialised positions include roles such as mentor teacher, educational advisor, head of subject committee (takes responsibility for a school’s pedagogical work on a given subject or study area), career advisor, ICT co-ordinator or a co-ordinator overseeing a specific area (e.g. special needs children) in a school.

There can be some variability as school leaders decide on the definition and assignment to certain vertical positions in their school.

Notes: England (United Kingdom) and Singapore shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (Crehan, 2016^[3]), *Exploring the impact of career models on teacher motivation*, <http://unesdoc.unesco.org/images/0024/002462/246252e.pdf> (accessed 21 February 2025); (Santiago et al., 2016^[6]), *OECD Reviews of School Resources: Slovak Republic 2015*, <https://doi.org/10.1787/9789264247567-en> ; (National Institute for School Leadership, 2019^[11]), *Singapore’s Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025)

Lever 6: Trusted evaluation mechanisms

An attractive and effective career structure goes hand-in-hand with the evaluation mechanisms that allow progress to happen fairly and appropriately. Evaluation mechanisms serve as a mechanism for determining career progression and thus facilitating reward and acknowledgement. They are also a form of quality control and thus a way of promoting growth, providing valuable information that can be used to support individual teachers as well as the system as a whole to improve.

Protect the role of the ‘external’ in evaluation for certification and promotion

One central goal of evaluation is to reliably and fairly support career progression. It is helpful to consider this as a distinct objective in terms of how it can be best approached, even if it may also furnish valuable information for formative growth as set out below.

To support trust and consistency in the evaluation mechanisms, it can be helpful to consider this form of evaluation as consisting of a distinct *external* aspect.

- That is, the evaluation is overseen by external actors to the teacher and the school. It may be that school leaders – and even teachers' own self-evaluations – play a role, but there is a degree of external oversight that operates across the evaluation of teachers.
- After all, for career progression to effectively function it needs to account for teachers across multiple schools, and this is naturally something that stretches beyond the remit of a school leader.
- There is also a question of the quality of judgements and potentially the need for specialised skills or multiple perspectives; research has found that leaders could distinguish between those teachers that added the most and least value in terms of student standardised achievement gains, but that they struggled to distinguish between teachers in the middle (Jacob and Lefgren, 2008^[12]).
- Furthermore, the external aspect also avoids the personal relationships between teachers and leaders affecting evaluations (OECD, 2019^[11]).

Defining the nature of this external body is important and its constituent inputs.

- There should be a clear process to how they work in relation to teachers, as well as the measures that they draw upon to actually conduct evaluations.
- The evaluation system of Singapore captures this blend of more detailed internal knowledge from leaders with external oversight (see Box 3.3).
- In some systems, such as those of Austria and Poland, an external perspective is only sought out in cases of complaint or appeal (OECD, 2019^[13]).

Draw upon well-defined, multiple measures

A range of measures can be drawn upon to determine teachers' competency levels and their career progression.

- More direct methods include observation of practice such as by school leaders or inspectorates, other more indirect methods include student performance such as in external assessments.
- Other indirect measures of competency may be participation in professional learning opportunities, such as the obtention of a certain amount of 'credits', or successful completion of qualifications.

The internal validity of measures is a key question.

- For instance, measuring performance of teachers is difficult; one way is through student performance, but this is confounded by the composition of the class and student starting points, while it may also mean that teachers prioritise – as mentioned – the mechanism that measures student performance (e.g. examinations over holistic development).
- Alternatively, rubrics for observing practice can be resource-intensive; for instance, observers need to be trained with specific rubrics.
- Accordingly, one option can be to increase the quantity of measures that are drawn upon to provide a more rounded picture, though this also incurs time and resource costs and increases the burden that evaluation carries on both evaluators and those being evaluated.

The measures that are used to evaluate a career structure will shape how teachers engage with that career structure. If the incentives function, the measures can receive increased attention from teachers – as the example of student examinations illustrates above. Accordingly, an overarching focus on competency in measures seems particularly important.

- This is connected to the question of the provision of professional learning opportunities. If engagement with professional learning is part of the evaluation criteria for career advancement this may impact incentives when it comes to engagement with professional learning opportunities.
- Participation in professional learning opportunities may become the primary incentive or objective, rather than the actual growth that such opportunities can yield (Santiago et al., 2016^[6]). Thus, professional learning may become a 'box-ticking' exercise (Crehan, 2016^[3]).
- Focussing evaluation and the criteria for career advancement on the intended outputs of professional learning, such as growth in practice, competencies and knowledge, may help to ensure that professional learning remains impactful and relevant.

Calibrate and audit measures

Confidence and trust in evaluation measures are important for the measures to effectively function.

- Transparency in how evaluation measures are conducted is a fundamental of this.
- This is connected to the specific standards that are associated with different career progressions; teachers should understand what clearly is expected of them.
- They should also understand how these expectations are to be measured and what success looks like on the evaluation mechanisms.

Efforts can also be made to foster greater confidence in how the measures are applied.

- For instance, if observation is used, it may be that multiple observers or observations are conducted to increase reliability.
- Alternatively, it may be that there are trained, specialist observers.
- There may also be auditing of measures, such as the consistency of how observation ratings are provided.

Again, this points to the value of the external nature of evaluation for progression. This process of building confidence is, however, shaped by constraints in terms of time and resources.

Use formative evaluation for development

A second central goal of evaluation is to facilitate learning. Its importance can be considered as tantamount to that of evaluation for certification and promotion, but it is again helpful to consider it as distinct, even if connections between the two require careful consideration (see below).

Considering formative evaluation as being a more localised, school-led form of evaluation can be helpful.

- The closer relationship between school leaders and teachers, as well as school leaders' knowledge of the school context, may facilitate more effective identification of a teacher's strengths and weaknesses and, in particular, targeted ongoing support for the latter.
- Teachers' self-evaluation may be less honest and, thus, less effective if the stakes are especially high for this evaluation, but more localised formative evaluation may reduce this risk.
- Moreover, one limitation of summative evaluations for progression are that they may not occur very regularly, with intervals of a year or more common. More regular evaluation and support at the school-level may better facilitate growth.

Formative evaluation closer to the grassroots level can see collaboration between school management and teachers to create individualised development plans, defining activities aimed at improving specific aspects of teaching practice (Maghnouj et al., 2020^[14]). For instance, in Colombia, teachers are formally appraised by leaders every year. This consists of an assessment of skills and qualifications, with a

summative rating out of 100 applied. However, evaluation for appraisal occurs only every three years and is managed externally by the national testing agency and the sub-national governments (OECD, 2019^[1]).

Facilitate connections between the summative and formative to identify learning needs

While distinct, considering the mechanisms of summative and formative evaluation in total isolation from one another may be inefficient. One central question remains how to align evaluation for certification and promotion with professional improvement.

- Recent data from TALIS 2024 (Teaching and Learning International Survey) suggested while teacher appraisal is widespread, there is a lot of variation in what happens after appraisal. On average across OECD education systems, 65% of teachers are engaged in post-appraisal discussions about how to improve their teaching. Less than half of teachers (46%) are offered development or training plans, on average, with this varying considerably across systems from less than 15% to over 90% of teachers (OECD, 2025^[7]).
- Questions have been raised historically by teachers around how appraisal processes can substantially impact on their classroom practice (OECD, 2014^[15]; OECD, 2019^[1]). This is significant when considering that evaluation for appraisal purposes is demanding, both in the sense of resources for a system but also often emotionally for teachers.

Carefully considering what connections exist between evaluation for certification and promotion and formative evaluation for development can be meaningful.

- For instance, how can shortcomings in the former feed into the more day-to-day support provided by schools to their teachers?
- This is where school leaders taking on more oversight for formative development can be meaningful, as they may understand the expected standards for progression better than individual teachers.
- Similarly, system leaders may choose to better align the provision of professional learning based on identified shortcomings from looking across summative evaluations.

Box 3.3. International examples on unlocking the fuller potential of evaluation

Singapore

Singapore's *Enhanced Performance Management System (EPMS)* guides evaluation and advancement in the career progression. Teachers (and school leaders) are evaluated every year based on three sets of outcomes: student outcomes (e.g. learning outcomes, holistic development), professional outcomes (e.g. collaboration, sharing resources), and organisational outcomes (e.g. contributions to the school). Teachers are assigned a specific grade, which determines progression and annual bonuses.

The evaluation is multi-step and multi-stakeholder. Teachers are assigned a specific supervisor to oversee the process. There is a close relationship between goal-setting and evaluation, with teachers meeting at the beginning of the year to set goals with their supervisor and to co-design an evaluation plan. Observation and monitoring unfold throughout the year, with a mid-year check-in designed to monitor ongoing progress towards goals and discuss what further support may be needed. A final evaluation sees all members of the school leadership team meet to decide on a final evaluation.

The EPMS does not specify exactly how supervisors carry out an assessment but provides a broad set of competencies (e.g. individual attributes, professional mastery, organisational excellence and effective collaboration). There is a high level of consistency in the shared understanding among

principals of the evaluation process due to the professional learning opportunities they undertake and the clear, consistent communication around the EPMS.

The evaluation process is notably time-intensive with its expectation of thorough evaluations every year involving a range of individuals. This has a cost for both evaluators, as well as teachers in terms of preparation for evaluations. At the same time, this process also can contribute to trust and consistency across a system due to the engagement of a range of individuals.

Estonia

In Estonia, the more resource-intensive forms of evaluation with external personal are reserved for more high-stakes assessment. For instance, Level 6 and Level 7.1 teachers (see Box 3.1) are granted by the head of the institution, while the head of the institution appoints a commission for those being attested as senior teachers, and the Ministry of Education establishes a commission for those being appointed as master teachers. The external aspects not only accounts for the greater weight of these roles but that there may also not be sufficient expertise within a school to make certain judgements on more senior promotions.

Teachers submit a written application for attestation at the next level. This includes documents that can help document their fulfilment of the requirements. There is a sense that the attestation is a verification of fulfilment, as opposed to a full evaluation for promotion (Crehan, 2016^[3]).

Estonia has, interestingly, also established a 360-degree feedback tool for the directors and leaders of schools. This is designed to support their self-evaluation and own leadership on their development, with anonymous commentaries from teachers, students, parents, colleagues and other parties.

Chile

Chile introduced wholesale changes to its system of professional learning and teacher career structure in 2016. Progression across its five-stage career model hinges upon a certification process that is competency-based and regulated by an external component, as well as teachers' years of experience. Interestingly, this was recently further reformed to try to create more efficiency and reduce the burden on teachers.

Competencies are specified in the national teaching standards. Evaluation consists of two external components: the teacher completes a standardised written assessment on their pedagogical knowledge and prepares a professional portfolio. The latter seeks to evaluate different aspects of the pedagogical practice of classroom teachers, based on multiple direct evidence of the work inside and outside it, including samples of work (e.g. lesson planning), a recording of their teaching, and proof of collaborative work with colleagues.

Teachers are expected to move from the first stage ('Initial') to the second stage ('Early') and, under the new reform, onto the third stage ('Advanced'). Teachers have the opportunity to engage in the evaluation process between these stages twice. This means that evaluation serves as a mechanism to remove underperforming teachers from the profession if they fail the examination more than twice (Santiago et al., 2017^[21] pp. 240, 253). In some cases, teachers can re-enter the profession after a two-year break and try the evaluations again.

Progression to the two highest stages ('Expert I' and 'Expert II') are voluntary. Thus, those at the 'Advanced' or higher levels do not need to engage in evaluation processes but, instead, they must participate in the 'Deepening Cycle'. This cycle lasts for four years and seeks to actively promote professional learning. During this, teachers must complete one of four actions, which can include taking and passing at least one of the evaluation mechanisms; mentoring colleagues; participating and passing a certified programme, course or peer-action activity; or actively engaging in the programme of the Teachers' Network of Teachers.

Under their new salary and career re-structuring in 2016, all beginning teachers were assigned to the lowest stage while current teachers were designated a step based on their years of experience and their previous results in multiple teacher evaluation processes (Santiago et al., 2017^[21] p. 240). In the more recent reforms, an additional provisional category was created for teachers who did not have evaluations from recent years, who have several years to complete an evaluation with which they would essentially 're-join' the new progression ladder.

The reforms centred on creating a high-stakes evaluation system that, as mentioned, could see the removal of underperforming teachers. The evaluation was also intensive in terms of measures. Central to its successful implementation was it being a multi-stakeholder affair, with the government creating a tripartite technical committee that provided a platform for central and local authorities alongside the country's main teacher union to develop consensus positions.

Notes: Chile and Singapore shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2019^[11]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>; (Crehan, 2016^[3]), *Exploring the impact of career models on teacher motivation*, <http://unesdoc.unesco.org/images/0024/002462/246252e.pdf> (accessed 21 February 2025); (Santiago et al., 2017^[16]), *OECD Reviews of School Resources: Chile 2017*, <https://doi.org/10.1787/9789264285637-en>; (National Institute for School Leadership, 2019^[11]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025); (CPEIP, 2025^[17]), *El centro de perfeccionamiento, experimentación e investigaciones pedagógicas del ministerio de Educación*, <https://www.cpeip.cl/> (accessed 25 February 2025)

Key implementation considerations

Developing an infrastructure that incentives and recognises excellence is complex, with multiple considerations at play and many of these interconnecting. Further complexity comes from the fact that there are also historic, existing structures to typically engage with. Some key implementation considerations that surface from international examples and best practice include:

- **Translating existing structures to new structures:** Considering how the transition process will unfold is an important element of change, with adjustments to career structures naturally inviting uncertainty for a range of stakeholders. Re-assigning teachers to certain levels needs a clear process and transparency. In particular, it demands balancing between placing too many teachers at a certain higher level, which can impact the credibility of the new structure and potentially have financial implications, with too few teachers at a certain higher level which may lead to frustration and reduced motivation. The example of Chile is particularly relevant here (see Box 3.3). Designing and evaluating pilot projects may be a way of allaying some concerns, as well as supporting consensus-building (see below).
- **Communicating changes and building ownership:** A lack of clarity around changes and the sense that one's voice is not heard in these changes may be significant barriers. Participative processes may be important. For instance, the French Community of Belgium's vertical expansion of the career structure saw an emphasis on early involvement of stakeholders and the clear communication of the reform package's goals to build support. There was a participatory consultative process that encompassed teachers, school leaders, parents and students alongside the economic, social and cultural sectors (OECD, 2019^[11]). Again, this is echoed in the case of Chile too (see Box 3.3).
- **Understanding the role of cultures:** The question of buy-in mentioned above is connected to a further challenge around understanding how to navigate existing tacit norms in the system. There may be historic ways of working or thinking when it comes to careers and processes such as

evaluation that may need to be attended to. For instance, formative, learning-orientated appraisals may take time to adapt to when cultures of more summative appraisal are historically the norm in a system. Alternatively, it may take time to understand and adapt to the language of underpinning teacher standards, particularly if they demand new knowledge or practices from teachers in a changing world. These more ‘tacit’ ways of working and interacting may demand sustained effort and attention during implementation if they are to steadily become ingrained and to foster a new culture around the career infrastructure.

- **Alignment with ITE provision:** It is limiting to view the career structure for teachers in isolation without considering its close connections with the wider Initial Teacher Education (ITE) and professional learning provision (see Ambition 2). ITE can be considered as the entry point to the first stage in the vertical teacher career. It thus builds the foundations of the career and the competencies it develops should have some alignment with what is expected in future progressions. Meanwhile, in its most fundamental form the nature of professional learning is a vehicle for teachers to grow and progress. If certain roles, be that vertical or horizontal, demand certain competencies, then the opportunity for these to be developed is important.
- **Coherence with leaders’ career structures:** It is also important to attend to how teachers’ career structure intersects with that of leaders. For instance, assuming that a final stage of a particular teacher track is becoming a ‘leader’ is insufficient. School leadership demands particular competencies that should be duly recognised, nurtured and rewarded. While leadership lies beyond the remit of this paper, a view to its structure – from the entry requirements to leadership positions to its progression – alongside processes of change in teachers’ career progression may be fruitful. A range of systems have explored ways of establishing preparatory support and ongoing support for leaders (see 4.3 “Developing leadership for inquiry, dialogue and learning” in OECD, 2019^[5]).

Box 3.4. Key reflection questions for system leaders

To develop an infrastructure that incentivises and recognises excellence, system leaders may reflect on the following:

A clear and attractive vertical career structure

- How can each stage of the career, as well as the connections between them, be clearly defined and communicated?
- How could teacher standards with their expected knowledge, skills, values and attitudes underpin the career stages?
- How can ownership and shared understanding of teaching standards and career structure be developed?
- What may be the role of financial incentives in career progression?

Opportunities to specialise through horizontal diversification

- How can specialist roles be clearly defined, from tasks to expectations for success?
- What features of the career structure may incentivise teachers’ engagement with specialist roles?
- How can effective fulfilment in specialist roles be ensured through sufficient resources, from time to financial resources to the learning opportunities that may be included?
- How are specialist roles accredited to teachers to ensure they are suitable for this role?

Trusted evaluation mechanisms

- How can evaluation positively shape career progression?
- What is the potential to draw upon multiple measures, including the perspectives of different stakeholders, to address the complexity of teaching in evaluating?
- What mechanisms can support the reliability and validity of measures? (e.g. capacity of evaluators, auditing, calibrating)?
- What can be the role of more regular and ongoing internal formative evaluation in supporting professional learning?
- How can learning needs from summative evaluations be identified and harnessed for positive formative experiences?
- How can both individual needs of teachers be identified, but also wider school and system needs by school or system leaders?

Implementation considerations

- How are the distinct needs of leaders met and considered in the career structure, while retaining coherence?
- How does the career structure build upon the foundations of the system of Initial Teacher Education?
- How have different stakeholders' perspectives been incorporated to ensure ownership of the career structure?

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4

Ambition Three: Facilitate quality, meaningful ongoing learning that enables the profession to grow and face future challenges

This chapter presents the third of the four ambitions that orientate this report. It focusses on the opportunities for ongoing professional learning for teachers, encompassing the type of provision of relevant learning opportunities, features of their delivery to maximise impact, and the incentives for teachers to engage with these opportunities. In discussing these, the chapter presents key design considerations from research literature, insights on implementation, and examples from systems around the world.

The work of teachers is remarkably complex. Teaching is highly relational, with teachers making decisions contingent on both real-time, 'live' information, as well as the historic experiences of the class and wider long-term learning goals. Moreover, teachers are not only processing and responding to the signals and needs of one student, but multiple students.

Accordingly, professional learning opportunities for teachers are not only necessary but essential. There is a need for teachers to continue to craft their existing practices. At the same time, they also need to respond to changes in research as well as changes in what knowledge, skills, values and attitudes societies demand education systems to develop.

Key highlights

This ambition focusses on the goals of ensuring learning opportunities respond to needs at the individual, school and system levels; delivering impactful learning experiences for teachers; and fostering a culture of effective self-regulated engagement. Some of the considerations it draws particular attention to include how:

- In determining the content focusses of professional learning opportunities, these need to balance the needs of individual teachers with that of the school and wider system.
- While research remains ongoing, a shift has occurred in the delivery of professional learning in recent years to more structured communities of learning that are grounded in particular contextual issues.
- Coaching and mentoring have shown particular promise to positively impact teachers.
- There has been increased interest and attention on the potential of observation of practice and accompanying feedback discussions to support professional learning.
- In general, delivery of quality professional learning is challenging and a distinct skill and knowledge-base, which demands both further research and concerted capacity-building.
- Mechanisms for quality control are important in professional learning provision. A characteristic of strong systems is the use of a combination of approaches to this end.
- Time constraints and limited financial support may present barriers when it comes to accessing professional learning, and accordingly need addressing.

Lever 7: Provision of relevant learning opportunities

Formal professional learning opportunities are important structured occasions for teachers to engage in understanding and reflecting on new approaches and practices (Cordingley et al., 2015^[1]). They serve as a key means of supporting teachers to improve in their work and develop as professionals. In particular, their formal nature can be a means of responding to emerging trends in a systematic way (e.g. using digital tools or teaching social-emotional learning) or addressing specific gaps in terms of teachers' competencies (e.g. fostering students' metacognitive thinking).

Establish who are the providers

One design consideration of quality provision is the infrastructure of formal learning opportunities in terms of who provides such opportunities and how they occur.

- Clearly establishing who is providing the professional learning is important in a whole-system approach.
- Similarly, it is also important that the formal channels for procuring and participating in professional learning are clear.

Broadly speaking, this may range from a highly centralised approach to a more decentralised one.

- It may be that systems directly develop and provide formal learning opportunities through an internal formal body.
- Alternatively, it may be slightly less centralised with the provision outsourced to a set of selected providers, with the latter applying to fulfil certain provision needs.
- There may, in contrast, be a highly decentralised model that sees essentially a ‘market’ of professional learning opportunities, where providers are free to propose and provide opportunities. Schools or teachers may directly choose the appropriate provider that meets their needs.

Related to this is the question of funding. In a more decentralised model, schools or teachers may have funding to directly pay for particular opportunities. In more centralised funding, system-level actors may oversee the funding of opportunities.

Align provision with needs and competency frameworks

A second key consideration is that of the content of what is being provided. Professional learning opportunities are, in essence, about learning *on something*. Defining this ‘something’ needs to balance multiple vantage points:

- At one level there are the needs of the individual teacher. Teachers’ competencies can be highly variable, and they have different areas of practice for potential improvement. Teachers may also have different competencies that they want to develop as part of their intrinsic motivations.
- There are also the needs of the school. This takes into account the collective strengths and weaknesses of the teachers in the school. By considering data such as student performance on external assessment, as well as other sources of information such as student surveys, school leaders may identify particular areas of need for the school to prioritise (e.g. student literacy, student well-being).
- A third level is the needs of a broader system. This may also respond to particular patterns in assessment data, while also attending to emerging needs as mentioned in terms of more long-term objectives for an education system.

In determining the content focusses of professional learning opportunities, these three factors are in dialogue with each other. Different emphases can be placed on the three levels.

- There may be a reliance on teacher-driven alignment. This could take the form of teachers choosing directly what to engage with; essentially, a broad ‘market’ of provision is provided and, based on teacher engagement, this is adjusted in line with demand. Alternatively means of surveying teachers for their particular needs with which provision is adapted.
- The needs of school and system leaders may be sought out to inform the provision too. Provision is also important for the strategic growth of teachers collectively. There may be needs in a school that school leaders feel are a pressing priority. Alternatively, system leaders might have to respond to new, emerging needs and they may therefore establish a more universal provision.

This alignment is also closely connected to the nature of incentives (see Lever 9 below).

Box 4.1. International examples of managing the provision of quality learning opportunities

Singapore

Throughout their career, teachers have access to a wide range of professional learning opportunities. This includes through seminars, courses and qualifications. These are centrally coordinated by the National Institute of Education or the Academy of Singapore Teachers.

Teachers may engage in as much as 100 hours of professional learning a year. These learning opportunities are largely free of charge or, where required, paid for by a teacher's school. For instance, a bespoke 'Management and Leadership Studies Programme' offers 17 weeks of funded training (alongside participants' salary) for department, grade or subject group heads. This includes a residential aspect with international visits too.

Learning opportunities often play a role in teachers' targets for advancing to the next level of their career track or improving their practice to make them eligible for a performance-related bonus.

Japan

In 2019 Japan launched a new era ("Reiwa") of education focussed on redefining schooling as providing individually optimised and collaborative learning that draws out the potential of all children. To this end, Japan seeks to further improve the quality of school-based collaboration and professional learning – despite it already being a system with well-engrained traditions of collaboration through peer lesson observation and feedback, notably through the "lesson study" format (see Annex B).

This is founded on a set of national indicators for improving teacher quality across five areas (e.g. classroom instruction, supporting children who require special consideration and support) and supported by a new digital 'Platform for Teachers and Staff Development'. The latter aims to support efficiency and effectiveness:

- Teachers are required to record all professional learning activities in the platform.
- There are specific tests to check knowledge and skill acquisition in professional learning activities, as well as reflective tools to support contextual translation.
- Teachers and school leaders can select from a catalogue of development opportunities those that best fit their own and their school's needs.
- A library of video on-demand teacher development programmes developed by universities, boards of education and public-interest corporations, among others.
- Tools for school-based collaboration (e.g. sharing content) and lesson study.

Notably, the platform will also enable teachers to evaluate the quality of their professional learning and its impact on their practice. Through its centralised nature, this can give policymakers a view of needs across the system as well as the effectiveness of the current provision and delivery.

Notes: Singapore shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2019^[2]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>; (OECD, 2024^[3]), *Education Policy Outlook 2024: Reshaping Teaching into a Thriving Profession from ABCs to AI*, <https://doi.org/10.1787/dd5140e4-en>; (Boeskens, Nusche and Yurita, 2020^[4]), *Policies to support teachers' continuing professional learning: A conceptual framework and mapping of OECD data*, <https://doi.org/10.1787/247b7c4d-en>

Lever 8: Delivery of quality learning opportunities

There is no shortage of research on the types of delivery that may be most effective for professional learning opportunities. Despite this, concerns remain around the exact features that constitute high-quality professional learning, with it proving deceptively difficult to identify features that consistently produce improved instruction and student outcomes (Kennedy, 2019^[5]; Kennedy, 2016^[6]; Sims and Fletcher-Wood, 2020^[7]). Indeed, the latest data from the OECD's Teaching and Learning International Survey (TALIS) suggests that impactful professional learning is a challenge; while nearly all teachers participate in continuous professional learning, just 55% of teachers report that the professional learning activities they participated in during the 12 months preceding the survey had a positive impact on their teaching, on average across OECD education systems (OECD, 2025^[8]).

Rather than there being a simple definitive set of 'ingredients' for effective professional learning, it seems more appropriate to view these as design considerations or mechanisms that interact with the objectives and context of professional learning (Boeskens, Nusche and Yurita, 2020^[4]). These design considerations may help produce more effective professional learning when appropriately attended to:

- Keeping learning tied to teachers' localised needs and particular contexts.
- At the same time, making use of quality external input to introduce fresh perspectives and challenge existing orthodoxies.
- Designing activities that facilitate their active engagement as learners and iterative knowledge-builders.
- Ensuring that the duration and intensity is appropriate for the type of the learning goal.

In particular, these considerations point to a need to move away from thinking about formal professional learning opportunities in terms of passive, standardised one-off seminar-style courses, and to consider more active and context-based learning that can better engage with the realities and complexities of schools.

Support structured communities of learning

One manifestation of the shift away from more seminar-style 'training' courses has been increased focus on how formal learning opportunities can be formulated more as structured collaborative communities that draw upon the aforementioned design principles. More concretely, typical manifestations include:

- Going beyond simply the imparting of theory and treating teachers as active participants with expertise to share. Space for discussion and actually 'doing' or developing specific outputs. Even if a whole-school approach is adopted on a mandatory topic, teachers work in smaller collaborative groups.
- Clear opportunities to relate learning to the particular contextual needs, be it a school need or individual problem of practice.
- Making use of external expertise (e.g. learning specialists, social workers) and concrete inputs such as vignettes, case studies or videos (see below for more) that can tie learning to the classroom.
- Revisiting learning in an iterative way over a set period of time.

Such communities may be particularly relevant when considering approaches to build teacher capacity around specific focusses at scale, say across a whole school or region. These may take both an in-person format but also, with appropriate tools, occur in the digital space too (Mineea-Pic, 2020^[9]). There may also be overlap with how a wider culture of inquiry is fostered within a school, or across schools more widely (see Chapter 5).

Utilise coaching and mentoring

Considering the range and complexity of skills that teachers need, practical and regular coaching or mentoring has particular promise. It has the capacity to promote iterative thinking and refinement through its blend of a focus on reflecting on specific issues, sustained duration, and external input.

- Research has found that assigning individualised, structured instructional coaching to teachers, either with designated positions (Blazar and Kraft, 2015^[10]; Kraft and Blazar, 2016^[11]) or matching effective teachers with less effective ones (Papay et al., 2016^[12]) has shown promise in improving instruction and students' learning.
- That said, large-scale coaching programmes present particular challenges. A meta-analysis of 62 studies found that the benefits of coaching were substantially reduced in larger coaching programmes serving over 100 teachers at time (Kraft, Blazar and Hogan, 2018^[13]).
- One particular challenge appears the identification of the quality external input needed in the form of high-quality coaches. This suggests that the design of structures around schools to provide external input need to carefully attend to their scale and size to ensure their effectiveness.

Enable peer observation and feedback

Much of teachers' work is often in isolation and rarely seen. This is despite the fact that the quality of teaching is the most impactful factor that can be influenced at the school level for shaping student learning outcomes (OECD, 2020^[14]). Yet classrooms are primarily closed spaces that are rarely accessible. For instance, TALIS 2024 (2025^[8]) found that 9% of lower-secondary teachers observe other teachers' classes and provide feedback at least once a month, on average.

- There has been increased interest and attention on the potential of observation of practice and accompanying feedback discussions to support professional learning (Stoll et al., 2006^[15]; Kersting et al., 2012^[16]; Fischer and Neumann, 2012^[17]).
- On the one hand, observation of practice and feedback can provide valuable information for the teacher being observed on the teaching process with which to refine it.
- At the same time, observation is also a way of potentially building new knowledge; it may be that in observing a colleague that a teacher 'learns' something new, or that the observation serves as a common, tangible resource for further discussion (Gamoran Sherin and van Es, 2008^[18]).

Of note is the particular potential of video observation.

- Video has the potential to make observation more accessible through its digital nature, as well as providing potentially relevant external input through the showcasing of high-quality videos.
- A range of studies on so-called 'video clubs' have found teachers can improve their ability to discern and interpret factors that are conducive to student learning in the classroom (Gaudin and Chaliès, 2015^[19]).
- Observation may be particularly relevant in contexts of change where there are new expectations of teachers and their practice. For instance, in Shanghai (China), observation is an important arm of supporting the teaching of new topics by teachers (see Box 4.2).

Implement quality assurance

Because teacher professional learning opportunities are shaped by the context and learning goals at hand, there is a need for quality assurance. The design principles outlined above are helpful guidelines, but they do not guarantee effectiveness alone. Indeed, the relevant quality external input, or the duration and intensity of professional learning, are shaped by the problem of practice being considered.

Mechanisms for quality assurance are thus important for not only promoting high standards, but also for supporting formative learning on delivery methods.

- Understanding the effectiveness of certain approaches may help these to be further refined, or scaled in other settings.
- Understanding the effectiveness of certain opportunities is especially significant for individual teachers or schools that have agency over choosing particular providers if they are to make informed and empowered choices.

Nevertheless, in general monitoring the quality of professional learning at a system-level and its respective effectiveness is challenging. Defining the mechanisms through which evaluation takes place requires careful consideration of the information they yield as well as their feasibility.

- An instinctive choice is that of student outcomes in terms of standardised tests (Darling-Hammond, Hyler and Gardner, 2017^[20]), though these may be too narrow to capture the full range of desirable effects – including those on teachers (e.g. self-efficacy) – or offer only a delayed perspective on effectiveness.
- Teacher evaluation of professional learning opportunities may be one low-cost and straightforward solution to monitoring effectiveness. This could also be coupled with self-reported data on impact in teachers' practice. Notably, evaluation may be susceptible to certain biases, such as teachers' motivation for such learning opportunities.
- Alternatively, self- or peer-audits conducted by providers themselves may be a reasonably efficient means of gauging an indication of the quality of professional learning. Again, this may be shaped by the incentives of providers in the system (e.g. market forces, teacher choice).
- External evaluations may be a more cost- and resource-intensive approach. Approaches that examine the causal effect of specific provisions can be insightful but are also particularly expensive and demanding. Moreover, more long-term effects on teachers' practice and student outcomes may be missed.
- Another form of external evaluation may come in the form of auditing by independent actors, for instance appointed overseeing authorities or inspectorates. These may, for instance, periodically audit a certain sample of professional learning activities (see Ireland below).

Accordingly, a characteristic of strong systems is the use of a combination of approaches for gauging the quality of professional learning.

- This can avoid the one-dimensional perspective that only one approach may yield, as well as the risk of an undue focus on one single aspect creating a distorted picture of the quality.
- More varied information can yield a more robust understanding of quality, and greater trust in it too, which may actually create a more efficient, quality delivery of professional learning.
- For instance, in Ireland, the central professional support provider ("*Oide*") makes regular use of teacher evaluation as well as reviewing themselves the quality of samples of in-person and online professional learning courses too. A third perspective comes from the central inspectorate in Ireland, who evaluates the design and facilitation of a sample of online and in-person professional learning courses each year (see Box 4.2).

Finally, strong channels of communication and feedback loops between relevant stakeholders seem particularly important for monitoring the quality of learning opportunities. It is important that the quality assurance feeds back into the question of provision; there may need to be a re-evaluation of certain providers if delivery of quality learning is to be assured.

Box 4.2. International examples of delivering quality learning opportunities

Finland

Online learning is widely available to Finnish teachers for them to engage with. The alignment with teachers' needs is led by teachers, who have agency in their engagement with this.

The high potential of tutoring is also drawn upon. Between 2016 and 2020, earmarked government funding supported the development of regional networks of 'Tutor Teachers'. The initiative aimed to address contemporary and future challenges in education by supporting teachers through peer mentoring and training. For instance, benefits were particularly noted in relation to teachers' use of digital tools in their practice – one of the goals of the initiative – while tutors themselves also benefited from the opportunity and helped mobilise good practice more widely (Pennanen et al., 2021^[21]).

The availability of funding and the allocation of time in teachers' schedules for their engagement was particularly key. Indeed, by 2020, many providers had taken steps to ensure that the tutor teachers could continue following the end of allocated funding (Pennanen et al., 2021^[21]).

In Finland, there are also tentative means of tracking the contributions of professional learning to teaching and student learning. The data system "Koski" has been exploring real-time tracking of student progress, and how this could be analysed alongside that of the professional learning engaged in by teachers. This could allow for an evaluation of the contributions of learning opportunities (Mikkonen, n.d.^[22]). Such information may be relevant for quality insurance.

Ireland

In 2023, *Oide* (an Irish word meaning "teacher" or "tutor") was introduced to consolidate professional support services for teachers and school leaders. Its overarching purpose is to support the professional learning of teachers and school leaders. This is primarily achieved through 21 full-time and 9 part-time 'Education Support Centres'. Each Centre is managed by an annually elected voluntary Management Committee and serves the local education community. Some 60 million EUR is provided in annual funding by the Irish Department of Education, including covering:

- the cost of over 360 seconded teachers providing continuing professional learning (CPL),
- and funding for substitute teacher cover in schools that can enable participation in professional learning.

A range of professional learning opportunities are available. Notably, there is a concerted effort from *Oide* to promote the formation of communities of practice. There are also efforts around building capacity in data and research-inform school planning.

Significantly, there are several mechanisms in place to support monitoring of the quality of professional learning. This helps to build a rounded picture of the quality of provision and its relevance to schools:

- *Oide* make regular use of teacher evaluation. All summer courses include a teacher evaluation. *Oide* also regularly reviews the quality of in-person training courses at the primary and post-primary levels, as well as a sample of online courses.
- The Irish Department of Education Inspectorate evaluates the design and facilitation of a sample of online and in-person professional learning courses each year. The Inspectorate also helps during the scoping, design and review of new professional learning programmes, providing valuable insights on particular needs that have been observed in the field.
- The Educational Research Centre published in 2023 a framework intended to support the evaluation of different forms of formal professional learning, either by providers or external evaluators.

- The Teaching Council plays an important role in shaping professional learning opportunities. In addition to its role in licensing teachers and accrediting initial teacher education (ITE) programmes, the Council is responsible for advising the Minister for Education on teachers' CPL, promoting engagement in CPL, conducting research and raising awareness of the benefits of teachers' professional learning.

Shanghai (China)

Professional learning is a central feature of the education system in Shanghai (China). Participation in comprehensive and regular professional learning activities is expected of teachers.

In particular, there is a long history of observation within and among schools. This includes senior teachers being observed by more junior colleagues, and vice-versa to support mentoring conversations. In particular, demonstration lessons are not uncommon, where a teacher delivers an example lesson for a large number of colleagues to observe and comment upon. In general, a culture of 'open classroom doors' pervades the system.

Close collaboration between practitioners is also intentionally structured into teachers' time. 'Teaching-study groups' are a subject-based structure for collaboration between teachers. Timetabled time in teachers' daily schedules allow the group to meet, discuss and refine specific lessons. Indeed, the average teacher in Shanghai (China) teaches for only 10-12 hours per week (Schleicher, 2016^[23]), with ample time for learning and structured collaboration. Teaching-study groups produce a general lesson scheme that teachers then follow with their classes, developing more detailed lesson plans based on this scheme.

Incidentally, teachers' career progression depends on being able to demonstrate their ability to work collaboratively, reiterating the importance of this skill in the system.

Notes: Finland and Ireland shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2024^[3]), *Education Policy Outlook 2024: Reshaping Teaching into a Thriving Profession from ABCs to AI*, <https://doi.org/10.1787/dd5140e4-en>; (Schleicher, 2011^[24]), *Building a High-Quality Teaching Profession: Lessons from around the World*, <https://doi.org/10.1787/9789264113046-en>; (Schleicher, 2016^[23]), *Teaching Excellence through Professional Learning and Policy Reform: Lessons from around the World*, <https://doi.org/10.1787/9789264252059-en>; (OECD, 2024^[25]), *OECD Review of Resourcing Schools to Address Educational Disadvantage in Ireland*, <https://doi.org/10.1787/3433784c-en>; (OECD, 2012^[26]), *Lessons from PISA for Japan*, <https://doi.org/10.1787/9789264118539-en>

Lever 9: Incentives to engage in professional learning

What incentivises teachers to engage in professional learning is dynamically shaped by a multiplicity of forces in a system. The career structure, both horizontally and vertically, is particularly foundational, as well as the accompanying evaluation mechanisms and how they operationalise teachers' professional learning opportunities. The quality of learning opportunities is similarly essential; when teachers see professional learning as benefiting them and supporting their work, the incentives to participate increase too.

Incentives may thus arise and diminish in relation to other ambitions and levers. There are though some particular design considerations around incentives that stand out for system leaders.

Balance mandatory and voluntary choices

A key tension is the balance between mandatory and voluntary choices. Systems may be guided by a primary focus on one of these, or a blend of both.

Voluntary choices may be particularly incentivising for teachers.

- This gives them agency and ownership, which is important for both their sense of professional identity and well-being. Indeed, the OECD's Teaching Compass notes the importance of teachers feeling a degree of ownership as a professional (OECD, 2025^[27]).
- Teachers may be able to pursue learning opportunities based on their passions and interests and thus more actively engage with these opportunities.
- Similarly, many teachers join the profession out of a deep sense of purpose, and this intrinsic motivation means can drive their efforts to constantly improve.

At the same time, teachers may not be perfect judges of their professional learning needs.

- Teachers – like anyone – can be susceptible to unconscious biases and this may leave 'blind spots' in relation to their needs.
- Learning can also be challenging and, at times, uncomfortable, and some teachers may prefer to focus on consolidating their existing strengths rather than addressing their struggles, particularly if they have limited time or bandwidth to engage in professional learning.

In this respect, the more mandatory assignment of participation may be relevant, be it by school or system leaders. Their oversight across bodies of teachers may, as mentioned, help support collective efficacy, with more mandatory approaches being potentially important for strategic development of the teaching profession.

The voluntary and mandatory choices may also be nested within each other.

- There may be mandatory areas of growth that are prioritised (e.g. use of digital tools), based on school or system needs.
- Yet, within this, teachers can voluntarily choose a particular provider or more granular content focus.
- Processes of ongoing formative evaluation between school leaders and teachers may also facilitate harmonising both school and teacher professional development plans, with a blend of school objectives and teacher objectives being pursued.

Facilitate the means of participation

Time constraints and limited financial support may present barriers when it comes to accessing professional learning. According to TALIS 2024, lack of time due to other commitments or responsibilities (63%) is the factor that teachers are most likely to report as a barrier to their participation in professional learning, followed by conflicts with the work schedule (60%) and the cost of learning activities (46%) (OECD, 2025^[8]).

The distribution of teaching time is particularly significant here.

- In a review of 24 school systems with available data, only five saw teachers spend more than 50% of their total statutory working time teaching (OECD, 2019^[21]). Alongside the expected responsibilities of teachers, this may be time that can be used to facilitate professional learning.
- To support professional learning, it may be that school closures are enacted (e.g. French Community of Belgium). Alternative models may see teachers have a certain entitlement of annual hours to use for professional learning at their own discretion. This may depend on the aforementioned degree of teacher choice.
- There are signs that there are associated challenges with some approaches, such as covering absent teachers (Nusche et al., 2015^[28]).

Recognise and track participation

A focus on professional learning in evaluation mechanisms may, as mentioned, create perverse incentives, meaning that the actual growth such learning opportunities are designed to facilitate is not appropriately monitored or incentivised. In general, mechanisms to monitor teachers' engagement in professional learning are not widespread (OECD, 2019^[2]).

That said, systems may find value in recognising and tracking professional learning participation.

- Portfolios that track participation can support teachers' self-regulation of their learning. It may provide a means of revisiting previous learning, or identifying potential gaps not yet addressed. Moreover, it should not be overlooked that the ability to survey one's growth over time may be important for some teachers' sense of self-efficacy and motivation.
- It can be helpful for supporting actors, such as school leaders or assigned mentors, to be able to understand what teachers have and have not engaged with. In fluid systems where teachers move jobs between schools, understanding the history of professional learning may facilitate more efficiency for school leaders in continuing to offer more sustained support.
- Furthermore, where there is a high amount of teacher choice in choosing participation, means of tracking participation across the profession over time may enable policymakers to identify levels of demand and adjust provision accordingly.

Certain systems, such as Wales (United Kingdom) have made use of centralised digital 'passport' systems that track participation across a teacher's career. For instance, Welsh teachers, school leaders and teaching assistants have access to a digital 'Professional Learning Passport'. This allows them to self-assess their needs, record their participation in different learning opportunities, and self-evaluate effectiveness (OECD, 2024^[3]).

The use of digital tools has been particularly significant when it comes to recognising and tracking learning. Digitalisation has enabled the emergence of new forms of skill certification, such as the aforementioned micro-credentials (see Chapter 3, Lever 5).

Box 4.3. International examples of incentives to engage in professional learning

United States

Digital Promise, an independent organisation authorised by the United States Congress as the National Centre for Research in Advanced Information and Digital Technologies, has developed an eco-system of more than 400 micro-credentials for educator skills. These micro-credentials are on skills that are supported by research evidence, for instance teachers' ability to engage students in deeper learning such as concept mapping or constructively critiquing arguments from their peers.

The delivery of micro-credentials is outsourced with micro-credentials provided by more than 50 organisations (e.g. higher education institutions, non-profit organisations, school districts). Digital Promise also collaborates with schools and districts on the design of bespoke broader learning structures based on micro-credentials (Minea-Pic, 2020^[9]).

In terms of wider policies in the United States, 29 states have micro-credential policies through legislation of their Department of Education (Digital Promise, n.d.). In 2017, eight states enabled teachers to rely on micro-credentials for fulfilling their continuing education requirements (DeMonte, 2017^[29]). There has been experimentation in some states too regarding licensure (e.g. Tennessee, Louisiana) and teachers completing new training in areas experiencing teacher shortages (e.g. Virginia with regard to science, technology, engineering, and mathematics).

Spain

The National Institute of Educational Technologies and Teacher Training (INTEF), part of the Ministry of Education and Vocational Training, provides a range of online training.

This includes online tutored courses which encompass around 60-70 hours of training, last two months and rely on expert tutors and co-ordinators who guide participants in the course. The catalogue of courses is updated yearly, and teachers who complete a course receive a certification for the training hours undertaken (INTEF, n.d.^[30]).

A range of massive open online course (MOOCs) and Nano-MOOCs (NOOCs) are also available. Training hours are not certified, but digital micro-credentials are issued to recognise learning upon completion and stored in an “INTEF Badges” backpack. Moreover, more sustained learning can be recognised too with the issuing of open digital meta-credentials when teachers collect several micro-credentials (INTEF, n.d.^[30]).

Source: (Minea-Pic, 2020^[9]), *Innovating teachers’ professional learning through digital technologies*, <https://doi.org/10.1787/3329fae9-en>; (Campbell, Lieberman and Yaskina, 2013^[31]), *Teacher Learning & Leadership Programme*, https://www.otffeo.on.ca/en/wp-content/uploads/sites/2/2013/09/tllp_full_report-.pdf (accessed 20 February 2025)

Key implementation considerations

In surveying international examples and best practice, the following implementation considerations appear particularly relevant when enacting processes of change:

- Coherence with ITE:** The competencies that teachers enter the classroom with are shaped by how they are prepared for the classroom. This accordingly shapes the agenda for the professional learning opportunities too. Sizeable proportions of teachers report feeling unprepared for the classroom’s realities (OECD, 2025^[8]; OECD, 2019^[12], p. 129). While the complexity of teaching means that it takes years to master – and indeed even then ongoing refinement – such that perfect preparation cannot exist, the better prepared teachers are the more their ongoing professional learning opportunities can focus on particularly challenging, relational practices (OECD, 2025^[32]). In particular, teachers often enter the profession having experienced minimal classroom-based learning opportunities. This may have ramifications for the type of professional learning opportunities they need, but also means they may be less habituated to particular forms of learning – notably those more collaborative and inquiry-orientated forms (OECD, 2019^[33]). In considering the coherence with ITE it is thus also an important consideration of how ITE connects to ongoing professional learning as a formal teacher, such as the transition mechanisms like induction that can support this. Systems such as Estonia, the French Community of Belgium and the Netherlands have sought to give particular focus to more practical learning experiences in schools (OECD, 2019^[2]).
- Nurturing the skills that delivering high-quality professional learning demands:** Facilitating effective professional learning or providing impactful mentoring is difficult. It draws upon distinct skills that are not necessarily automatic, nor simply an extension of being an effective teacher (Coles, 2019^[34]). Bespoke capacity-building efforts for these skills may be necessary to help support their effectiveness. For instance, in Estonia, there is support for mentors working with beginning teachers to develop their mentoring skills through bespoke seminars (OECD, 2019^[2]).
- Understanding the distinction between teacher professional learning and that for school leaders:** School leaders have distinct needs due to the role they play in creating a supportive school environment and facilitating teachers’ growth. For instance, they play a particularly

important role in fostering teacher collaboration, but large-scale surveys have suggested that school leaders' competencies to support collaboration may be an area of need (OECD, 2019^[35]). Crafting a high-quality professional learning system for teachers may not meet the needs of school leaders and should not be considered a direct substitute. Specific capacity-building efforts for leaders require careful consideration too. For instance, in the Slovak Republic principals are required to complete a course in specific management competencies ("functional training"), while in the French Community of Belgium have 180 hours of training, coaching provision, and a three-year mandatory entry phase (OECD, 2019^[21]). The aforementioned 'Management and Leadership Studies Programme' in Singapore is a further example (see Box 4.1).

- **Remaining sensitive to market forces:** Market forces can help self-regulate the demand for certain learning opportunities from individual teachers and the provision from different independent actors. This can support efficient matching of needs and provision, while also potentially driving up standards if there is a degree of competition among providers. These market forces may also act in unforeseen ways that need sensitive monitoring. For instance, external quality control can still be valuable to help differentiate between higher-quality providers, and to allow schools and teachers to look beyond just very well marketed providers. Similarly, monitoring of provision may avoid distortion towards particular 'trends' and allow wider strategic goals to be retained.
- **Attending to equality of access to learning opportunities:** The challenge of market forces is a reminder of the importance of attention to how accessible different learning opportunities are to different schools and teachers. For instance, market forces may see on-site provision concentrated around urban areas where there is higher demand, meaning some regions are left behind. Alternatively, it may be that specific infrastructure is needed to facilitate certain learning opportunities and that this needs to be targeted to certain schools. Ambition 4 (see Chapter 5), where digital tools are particularly prevalent, is also a reminder of this. The same may be true of specific capacity-building needs, such as in the most disadvantaged settings; in Ireland, for instance, disadvantaged schools receive additional teaching and leadership resources and priority access to training aimed at supporting students with the highest levels of needs. It is thus a pertinent consideration during implementation of how there will be an equality of opportunity to engage with quality learning opportunities.
- **Troubleshooting potential challenges through pilot programmes:** Pilot programmes may be valuable for identifying the types of common barriers that exist and allow for adjustments of initiatives before wider deployment. They may also help to gradually build buy-in by providing tangible evidence of the impact of changes and develop a cohort of 'early adopters' whose expertise can be relied on. For instance, in the context of micro-credentials, evidence from the United States and states that have adopted this approach suggests that starting with pilot programmes can be valuable for successful implementation (DeMonte, 2017^[29]).
- **Gather an external perspective:** Independent evaluations can be a means of confidently understanding what is working well and what are ongoing challenges. For instance, in a commitment to the constant improvement of its offer, an evaluation report of the system of professional development was commissioned by the Chilean Ministry of Education. This identified certain accomplishments of the system (e.g. promoting the social valorisation of the teaching profession, promoting the improvement of professional performance), as well as some of the ongoing challenges that required further consideration (e.g. a school leadership career, the processes of induction and mentorship – despite their high levels of appreciation – needing further coverage).

Box 4.4. Key reflection questions for system leaders

To facilitate quality, meaningful ongoing learning that enables the profession to grow and face future challenges, system leaders may consider the following:

Provision of relevant learning opportunities

- How can the responsibility for the provision of professional learning opportunities be best addressed for high-quality learning?
- What could be the criteria and mechanisms for promoting high-quality offers of professional learning?
- What might be the right balance of agency and choice with more mandatory learning opportunities?
- How can the provision be designed with evolving needs of teachers, schools and the wider system?

Delivery of quality learning opportunities

- What resources support structured communities of learning? (e.g. time, physical resources, capacity-building)
- How are data and research provided to support structured communities of learning?
- How is a culture of learning embodied in the system and across schools?
- What opportunities exist for quality coaching and mentoring in the system?
- What enablers in terms of resources (e.g. time) and capacity-building support effective coaching and mentoring?

Incentives to engage in professional learning

- How can potential barriers to teachers' participation in professional learning be addressed? (e.g. cost, time)
- What could be the approach to recognising and tracking professional learning participation?
- How could recognising and tracking support sustained reflection on professional learning?
- What measures can be used to gather teachers' and school leaders' views on professional learning opportunities and drive further improvement?
- How can the overall quality across the provision and delivery of professional learning be monitored and used?

Implementation considerations

- How can professional learning build on the foundations of initial teacher education coherently?
- How could the capacity of professional learning providers to deliver high-quality learning be further enhanced?
- How may the potential effects of market forces on provision of professional learning be best understood and responded to?
- What could be the role of pilot programmes in supporting potential change?

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5

Ambition Four: Create an ecosystem for collective learning and sharing to ensure opportunities for excellence for all

This chapter presents the fourth and final ambition that have orientated this report. It focusses on the question of ensuring that rich professional learning characterises an entire system. To this end, the chapter considers how collaborative inquiry can stretch across schools, fostering rich, quality research engagement, and some of the ways that new knowledge can be collectively developed, systematised and disseminated by schools. In discussing these, the chapter presents key design considerations from research literature, insights on implementation, and examples from systems around the world.

Fostering an ecosystem of collective learning and sharing can help drive excellence across a system. Schools are communities of dynamic interaction between a multitude of unique individuals. This is an environment that presents unforeseen opportunities and challenges, and an inquiry stance can help schools navigate these. But schools also share many similar objectives and accompanying barriers. The connections of an ecosystem can help build greater efficiency in overcoming these barriers to realise objectives.

Moreover, these objectives are not necessarily static. Schools do not exist in a vacuum but are shaped by the wider forces of change in the world. Schools also have to adapt, learn and build new knowledge in step with the evolving needs of societies. An ecosystem of learning can also help build robust school systems that can respond to new challenges long into the future.

Underlying this is a recognition of the expertise that exists in the teaching profession. As the OECD Teaching Compass notes, a sense of agency and ownership for teachers will only become increasingly important in education systems (OECD, 2025^[1]). Not only is it motivating and empowering for the profession, but essential for the quality and relevance of its work (Suarez and McGrath, 2022^[2]).

Key highlights

This ambition focusses on the goals of encouraging inquisitive and reflective practitioners; nurturing a culture of evidence-informed work; and ensuring the identification of and learning from best practices. Some of the considerations it draws particular attention to include how:

- Fostering a culture of collaborative inquiry can nurture the types of informal interactions of teachers that are important for sharing and building knowledge around their day-to-day work in that particular context.
- Initially, developing these inquisitive, growth-orientated cultures in schools typically hinges upon clear structures, anchored around tangible problems of practice in the local context. These structures can allow the routines and habits of inquiry to become more embedded and, eventually, self-sustaining as a culture.
- Bespoke resources may be needed for collaboration, such as capacity building in terms of leaders' or facilitators' collaborative skills; providing dedicated time and financial support; or furnishing resources such as data.
- Collaboration may look outwards too, with external actors in the community or beyond offering additional expertise.
- There should be trusted sources or avenues for accessing quality, relevant research, coupled with developing practitioners' skills for engaging with and using this research effectively in their context.
- A key feature of an ecosystem of learning is the accessibility to knowledge it presents to all. Technology can be a crucial tool for supporting schools' accessibility to codified knowledge, and allowing knowledge-development to be driven from the bottom-up too.

Lever 10: A culture of collaborative inquiry within and across schools

While the provision of structured formal learning opportunities is an important means of supporting strategic teacher growth, the complexity of teachers' work means that the informal exchanges and networks of support within a school remain essential. These more informal interactions of teachers are important for

sharing and building knowledge around their day-to-day work in that particular context (Akinyemi, Rembe and Nkonki, 2020^[3]). According to the OECD's latest Teaching and Learning International Survey (TALIS) teachers who report collaborating more tend to be more likely to fulfil their lesson aims, and teachers who collaborate more frequently tend to be in the top quartile of self-efficacy in their country (OECD, 2025^[4]).

Working together can take many forms, varying greatly in intensity and frequency. One area of particular interest has been around cultures of collaborative inquiry that focus on instructional improvement and fostering collective learning in a school. Such cultures appear particularly significant when considering how challenging it is to change behaviours in teaching and schools, as reflected in the ineffectiveness of much formal professional learning when it comes to changing teachers' actual day-to-day practice (Copur-Gencturk and Papakonstantinou, 2016^[5]; Webb and Sheeran, 2006^[6]). The work of teachers and schools is, quite simply, very demanding and accordingly routines and habits are important to its effective functioning. When it comes to enacting changes, the culture within a school can help sustain attention on and provide support with new initiatives or approaches. Growing research on implementation science has suggested that change in schools, and more broadly, requires not only careful planning and preparation, but ongoing attention and dynamic refinement (Education Endowment Foundation, 2024^[7]).

Furthermore, it should not be forgotten that, as the OECD Teaching Compass highlights, teachers' own well-being is essential to the effectiveness of their work (OECD, 2025^[1]). A supportive and collaborative community may be particularly significant here too. Indeed, some research has suggested that opportunities to collaborate and build professional relationships could be linked to teacher well-being (Soini, Pyhältö and Pietarinen, 2010^[8]).

Developing these inquisitive, growth-orientated cultures in schools typically hinges upon clear structures that allow the routines and habits of inquiry to become more embedded and, eventually, self-sustaining. They also demand investment of time from all participants, and orientation around clear goals with a shared purpose (OECD, 2025^[4]).

Promote tangible and specific focusses

Collaboration is not an end in itself but a means to achieving a particular goal. Thus, the ultimate goal is promoting collaboration among colleagues about a particular problem of practice. The processes of identifying the problem of practice are thus vital.

- One aspect is that collaboration focusses on problems of practice that are highly relevant. This is significant for the motivation and investment from teachers.
- It may be that participatory processes of identifying problems of practice, or a degree of autonomy exists within a school for teachers to pursue issues they see as highly relevant.
- A second aspect is ensuring that the problem is concrete and feasible to maximise success. One challenge when it comes to implementing changes that address specific problems is a focus on too many things that may divide attention and lead to inefficiency (Education Endowment Foundation, 2024^[7]). Strategic prioritisation may be of value.
- For instance, one way of keeping collaboration concrete and results-driven comes from promising initiatives to support collaboration around school data (Poortman and Schildkamp, 2016^[9]). Data can help schools identify particular areas in need of attention. With this, schools can develop hypothesis for addressing the problem of practice and then use data to monitor its effectiveness (see Box 5.1).
- The provision of relevant data and supporting capacity to engage in decision-making with data may be particularly impactful.

Delineate clear structures and roles

Connected to the challenge of time is ensuring efficiency and coordination across multiple participants. Most often, collaborative opportunities seem to emerge systematically through specific structures within schools (OECD, 2019^[10]; Kools and Stoll, 2016^[11]), such as subject committees and teams, or specific groups united by broader transversal themes.

Part of the question of structuring is the question of who is responsible for what in collaboration.

- Specific roles and responsibilities relating to collaboration can be an effective means of ensuring that collaboration does occur and is facilitated.
- Moreover, these roles can provide a means of oversight with a dedicated person responsible for how collaboration unfolds and any agreed outputs.
- There may be multiple roles, with tasks split between several members of staff, or they may be more centralised around specific individuals within a school or across schools.
- This may relate to the extent that collaboration is a relatively new process or not; more directed collaboration may be necessary if a particular culture of collaborative inquiry has not developed.

These roles may also be reflected as a specific responsibility of a certain part of the career structure. For instance, in Singapore and Shanghai (China) the engagement of the teacher in their school and their fostering of horizontal collaboration is significant in its evaluation process (Schleicher, 2016^[12]).

Alternatively, signals may be sent by the system with regard to the value of collaboration; schools in the French Community of Belgium are expected to develop strategies for collaborative work as part of their school development plans (OECD, 2019^[10]).

Dedicate resources to support collaboration

Connected to the idea of providing data to support collaboration is that of providing the means that can enable effective collaboration.

- While collaboration depends on there being a sense of shared ownership, it nevertheless faces the challenge that collaboration demands coordinated time for it to occur.
- Scheduling dedicated time for collaboration is a common feature in a range of systems (e.g. Chile, Austria). For instance, the weekly working time for teachers in Portugal includes dedicated time for collaboration (OECD, 2019^[10]).
- An additional challenge is the physical or human resources that inquiry processes may demand. For instance, in New Zealand, the Teacher-led Innovation Fund is an initiative of the Ministry of Education that provides teachers not only with time in their schedules but also additional expert support and the ability to apply for specific funds (OECD, 2019^[10]).
- Resources may range from direct sources of funding to bespoke resources that facilitate collaboration to external expertise.

Create structures for school-to-school collaboration

Notably, collaborative inquiry may stretch beyond the walls of an individual school.

- Much like the role of external input in supporting effective professional learning opportunities, school-to-school collaboration can also be a means of fostering fresh perspectives and challenging established orthodoxies.

- It can also be a means of developing more localised knowledge that may be of particular relevance to specific communities and contexts, as the Finnish system has focussed on in particular with its Lighthouse network (see Box 5.1).

This demands careful consideration of the design principles that underpin within school collaboration afresh.

- Identifying a specific, concrete shared focus that is of meaning to multiple schools may be challenging, running the risk of becoming more abstract as it seeks to accommodate multiple perspectives and problems of practice.
- Roles and structures become particularly important too when coordinating multiple actors that are not in a shared physical location.
- Creating specific structures to start such collaboration may be particularly significant. This may take the form of more direct pairing. For instance, in Portugal clusters unite between 2 and 29 schools under a single administrative team, enabling a certain degree of knowledge sharing.
- Alternatively, it may be approached through the creation of intermediary actors; in Denmark, a national body of learning consultants support schools in their improvement efforts but also facilitate peer exchange through this.
- It may also draw upon digital connections. In Austria, a virtual networking and learning space allows leaders to connect among new secondary schools in its Centre for Learning Schools.

Tap into the role of additional actors

Collaboration may also stretch beyond schools to include other actors that can bring relevant expertise. Indeed, there has been considerable increase in recent years in the number and type of actors engaged in the education space (OECD, 2022^[13]).

- This may include local actors that school leaders identify as being able to support the school and the learning of its students (e.g. local museums, art galleries, sports facilities).
- It may also include wider national actors with more formal structures established to provide support and input to schools. For instance, higher education institutions may function as satellite institutions that can coordinate and convene different schools (OECD, 2023^[14]) (see Lever 11: Developing engagement with research too).

Box 5.1. International examples of nurturing a culture of collaborative inquiry

Finland

A range of networks support knowledge-building and sharing in Finland. For instance, the National Agency for Education oversees the “Lighthouse network of leadership” to support school-to-school collaboration. It includes typically monthly webinars, including a presentation and subsequent discussion, as well as occasional in-person events or seminars. Numerous similar examples could be included, such as the National Network for Bilingual Education (“Kaksari”), coordinated by the City of Helsinki and funded by the National Agency for Education, or the Early Childhood Education Development Network (“Loisto”), coordinated by the Finnish National Agency for Education.

Indeed, an orientation towards collaborative inquiry permeates the entire system.

- An example of this can be seen in the National Teacher Education Forum, which seeks to meet the new needs of teacher education in the context of the 2020s through broad cooperation based on research and knowledge. It brings together all higher education institutions

responsible of providing teacher education, as well as all other relevant stakeholders (e.g. representatives of teachers, leaders, students, municipalities etc.). There is also a small international advisory board.

- The Forum was first coordinated by the Ministry for Education and Culture (2016-2023), with the University of Helsinki since responsible with the Ministry participating and financing the Forum.
- Notable outputs include research-based development projects and reports, the collaboratively developed national guidance (“The Teacher Education Development Programme”), and a range of seminars and workshops. In particular, in March 2025, the Forum launched the Teacher Education 2050 vision, the first time such a vision has been created in Finland.
- Since 2013, the “LUMA Centre Network” has operated in Finland. This is a network of 11 universities and which focusses on the subjects of science, maths, technology and environmental studies (“LUMA” subjects), and in particular approaching them in a multi-disciplinary way. It aims to foster a high level of knowledge in these subjects, as well as a high number of professionals across Finland in fields associated with these subjects. To this end, it works to:
 - Develop research-based solutions for teaching LUMA subjects.
 - Support the work of teachers and their continuous professional learning.
 - Inspire children and youth towards studying these fields and having them as hobbies.
- The LUMA Centre Network has built a collaborative ‘ecosystem’ around these goals, including ranging from museums and libraries to the business sector and science centres. During 2021 and 2024, some 185 000 people participated in its activities each year and the network provided in-service training to some 4 600 professionals.

Estonia

Lifelong learning has a strong historical importance in Estonia. The Estonian Education Strategy 2021-2035 emphasises the development of principals and teachers to prepare the population for a changing world.

This emphasis on learning is also coupled with high levels of agency. Estonian schools have considerable autonomy. School leaders and teachers are seen as key agents of change at the classroom and school level, as well as local and national level.

A series of initiatives supported at the national level are in place to support teachers to develop communities of practice focussed on student learning and well-being. This ranges from:

- The availability of specific training on principles of professional learning communities for potential leaders of communities or training to teams in schools on leading innovation and school improvement.
- The ability to apply for financing for initiatives that focus on improving teaching, such as engagement in seminars, training sessions or international conferences.
- Regional and national subject associations with a strong focus on leveraging networks to enhance student learning at greater scale.

Notes: Finland shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (Crehan, 2016^[15]), *Exploring the impact of career models on teacher motivation*, <http://unesdoc.unesco.org/images/0024/002462/246252e.pdf> (accessed 21 February 2025); (Cordingley et al., 2022^[16]); *Recommendations for the Professional Development of Teachers and School Leaders in Estonia*, https://www.britishcouncil.ee/sites/default/files/policy_recommendations_for_the_professional_development_of_teachers_and_school_leaders_in_estonia_eng.pdf (accessed 15 February 2025); (OECD, 2019^[10]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>

Lever 11: Developing engagement with research

One important source of information that teachers can draw upon to inform their practice is scientific research. There has been considerable growth in the quantity of research in recent decades (Torres and Steponavičius, 2022^[17]). While challenges remain (see (OECD, 2025^[18]) for a summary), this wealth of research is a valuable source of information for teachers. Moreover, building the infrastructure for engagement is not only a case of enabling this research to impact classrooms, but for classrooms to also potentially inform and strengthen the wider world of research going forward (OECD, 2024^[19]).

Supply high-quality, relevant and accessible research

Quality research is a foundation of its effective use. What counts as quality has long been contested, but in recent years there has been a movement towards more plurality in handling evidence to reflect that there is no single best method or type of evidence. Rather the most appropriate methodological approach depends on the question that is being investigated (Nutley, Powell and Davies, 2013^[20]).

This invites consideration of access to and engaging with this research:

- There should be trusted sources or avenues for accessing quality research. For instance, centralised hubs, such as those developed by many knowledge brokers, may act as gatekeepers for ensuring research is of a sufficiently rigorous nature.
- Research should also be relevant to teachers and their particular needs. Teachers are busy and research that does not speak to their own experience may reduce their engagement with research. The co-development of research agendas may be significant in this respect.
- This supply from trusted sources may also be supported by a process of translation to ensure research is accessible. For instance, summaries of key findings can still retain rigour and communicate transparently the nuances of research while making it digestible for often busy teachers.

Underlying consideration of quality and relevancy is the question of context: quality, relevant evidence for who? Building generalisable knowledge in education that works across contexts is very challenging. Certain methodologies are better at enabling this type of inference, but they are expensive and time-intensive (e.g. randomised control trials). Even then, contexts can be highly variable and there has accordingly been a range of efforts in recent years to better contextualise findings from one context to particular national or regional contexts (e.g. the Laboratory of Education Research and Innovation for Latin America and the Caribbean [SUMMA], Leerpunt) (OECD, 2023^[21]).

Build research engagement skills

Supporting engagement with research is also about building the skills and capacity of teachers to engage with research. As mentioned, a degree of translation is necessary in terms of applying the insights from scientific research to a particular localised context, and this is why research skills are necessary too.

However, this remains a challenge for many systems; in 2021, the OECD Strengthening the Impact of Education Research policy survey found that just over half (53%) of surveyed education systems reported that “low levels of skills and capacity to use research” were a barrier to the use of research in school practice (OECD, 2022^[13]).

Opportunities to develop research engagement skills can be significant here.

- This can relate to both developing the knowledge with which to interpret certain methodologies and their associated findings, as well as knowledge of how to conduct research such as through collaborative research projects.

- For instance, there has been particular attention in action research models, which can allow teachers to address challenges in specific contexts, by questioning their own practices, exploring relevant theory, trialling new approaches, and generating new knowledge (McKenney and Reeves, 2020^[22]).
- This may also extend to explicit opportunities for teachers to step formally into research roles. For instance, since 2010 in the Netherlands, the Dutch Research Council has been organising calls for their Doctoral Grant for Teachers programme. This is aimed at teachers in primary, secondary, vocational, higher vocational and special education, with some 500 projects having been awarded. Significantly, the grant seeks to serve as an enabler by including the costs of a replacement teacher up to the equivalent of a maximum of 0.4 full time for a maximum of five years (OECD, 2022^[23]).
- Notably, there is an important connection with initial teacher education (ITE) here (see Chapter 2). Some systems have started to build the development of research skills into ITE programmes with a view to this hopefully fostering longer term engagement with research (see Box 5.2).

Create system-supported approaches for research engagement

In recent years it has become clearer that active efforts are needed to develop the structures and cultures for effective research use. System leaders play an important role in building particular connections, with it helpful to think of a research ‘ecosystem’ being fostered (OECD, 2023^[21]).

- For instance, this may be the development of brokerage agencies that can offer a centralised place to access research (e.g. the Education Endowment Foundation in England (United Kingdom) – see Box 5.2). Such agencies may be created without any initial foundations, or they may draw upon existing infrastructure and organisations by expanding their responsibilities (e.g. Higher Education Institutions or research institutions).
- Alternatively, it may be the establishment of partnerships between specific universities and schools (e.g. Netherlands Initiative for Education Research [NRO] in the Netherlands). Overall, such brokerage agencies and partnerships may also coordinate research efforts and identify particular gaps to inform research agendas, as is the case of the aforementioned NRO in the Netherlands.
- Alternatively, it may overlap with aforementioned approaches to connecting schools and fostering collaborative research networks (Rickinson et al., 2023^[24]).
- This may be coupled with bespoke resources such as the provision of funding for collaborative research projects that require multi-stakeholder engagement.

There may also be particular incentives. For instance, engagement with research may be rewarded through specific roles or as part of career progression. Alternatively, there may be expectations around research engagement that are communicated. Evaluation mechanisms could draw upon evidence of research engagement, though this still invites consideration of the potential unforeseen perversion of certain measures as set out above.

Box 5.2. International examples of developing engagement with research

Education Endowment Foundation in England (United Kingdom)

In 2011, the Education Endowment Foundation (EEF) was established as the ‘what works’ centre for education by the Sutton Trust, with an endowment from the UK Department for Education. It was set up as an independent charity with the primary aim of closing the attainment gap between disadvantaged pupils and their peers. The EEF was created to promote the use of evidence-based approaches in education, by funding and evaluating research into effective teaching and learning strategies – what is known as evidence generation – and to synthesise available research evidence and to mobilise the use of evidence to the sector.

One key output is its Teaching Learning and Toolkit. The Toolkit is an online resource developed in collaboration with academic researchers. It provides accessible summaries of educational research to help teachers and school leaders make evidence-informed decisions about how to improve learning outcomes, especially for disadvantaged pupils.

The Toolkit presents strategies and interventions (e.g. feedback, homework, small group tuition, and metacognition) providing evidence on three key factors:

- Impact on pupil attainment (measured in months of additional progress)
- Strength of the supporting evidence
- Cost to implement

It is designed to be practical and easy-to-use, offering a quick way to compare different approaches and understand more about their effectiveness. The Toolkit is regularly updated with new research findings.

The EEF has also funded a range of empirical research studies, in particular randomised control trials. It has experienced considerable growth from its original primary focus on randomised control trials and meta-analyses, such as in relation to its extensive funding of qualitative research too.

To better support schools to access, understand and use evidence, the EEF set up a partner network (“Research Schools Network (RSN)”) in 2016. The RSN has grown to a collaboration of 33 schools across seven regions in England. Research schools serve as evidence advocates in their local and regional networks and develop strategic partnerships through a blend of training, exemplification and school-to-school support.

The RSN has collectively engaged with more than 40% schools across England and provided training to over 6 000 schools, with its role in fostering stronger cultures of evidence-use reflected in the wider fact that 70% of senior leaders in England cite use of the EEF’s Toolkit when making decisions about school spending.

As outlined above (see Chapter 3) the content of the frameworks for National Professional Qualifications (NPQs) in England (United Kingdom) and the supporting evidence are independently assessed and endorsed by the Education Endowment Foundation (EEF) too.

Ireland

The Teaching Council in Ireland is the professional standards body for teachers in Ireland. It was established on 28 March 2006 on a statutory, and it is governed by the Teaching Council Acts. The Council is funded by the teaching profession and has an independent regulatory function. It also has a mandate to promote the teaching profession.

The Teaching Council provides a range of resources to support teachers and schools to engage in and with research.

- **Online Library:** The Teaching Council provides all registered teachers with free access to an online library of research journals and articles and a collection of eBooks.
- **Teachers Research Exchange (T-REX):** T-REX is an online network for teachers and other educational researchers in accessing and sharing research. It supports conversations, collaborations and stronger connections between research policy, practice and between schools and higher education institutions. It also actively supports the development of research skills.
- **Research Support Framework:** The John Coolahan Research Support Framework offers support to teachers and others conducting new research, or exploring existing research in their practice, either independently or in collaboration with others.
- **Commissioned Research:** The Teaching Council also commissions research. For instance, a recent study commissioned and jointly funded by the Teaching Council and the Department of Education is a longitudinal study exploring teachers' early career experiences.

Connected to this but also overlapping with efforts to foster collaborative professional learning is the Félite Bursary Scheme. This provides bursaries of 2 000, 3 000 or 5 000 EUR with the goal of supporting collaboration between schools and across sectors (e.g. colleges of further education, universities, education support centres) to enhance teachers' professional learning.

- Some of the types of activities supported include forming professional learning communities, hosting workshops or events, inviting speakers and organising a lesson study.
- Some of the themes that have been considered by projects include outdoor learning, artificial intelligence, anti-bullying and well-being, and English as an Additional Language.

In 2024, the Félite Bursary Scheme saw a total value of 100 000 EUR awarded in bursaries, with 27 projects consisting of 98 collaborating schools. Digital assets are created by each project for dissemination, such as video clips.

Norway

Norway has developed an infrastructure to foster effective research engagement. A specific special unit – the 'Section for Policy Analysis' – has been developed in the Norwegian Ministry of Education and Research with a mandate to provide educational research and data to support policymaking. It cooperates with all departments of the Ministry, serving as a bridging knowledge broker by disseminating relevant research as well as by offering analytical support to the departments and to the political and ministry leaders.

The section also endeavours to support evidence production; it works to develop and increase the evidence base across education levels, from kindergarten to schools to higher education. To this end, it follows international research and facilitates strategic discussions in the field of education research. It also coordinates research initiatives across a range of different stakeholders, with one task being the allocation of budget means to educational research being one of these tasks.

A further actor is the Norwegian Knowledge Centre. This is funded by the Ministry but is an autonomous institution, and in 2019 it moved from Oslo to the University of Stavanger as part of a more general decentralisation effort by the government of Norway.

The Programme for Research and Innovation in the Educational Sector (FINNUT) has sought to support high-quality research aimed at spurring innovation and informing practice and policy development at all

levels of the education sector. Running from 2014 to 2023, it was carried out by the Research Council of Norway and provided funding for research projects across the education sector focussed on four main areas: teaching and learning; professional education and practice; management, leadership and organisation; and education and work. Notably, the Programme attended to monitoring the impact of the projects, as well as supporting their targeted dissemination to particular target groups through communication and dissemination plans.

Present alongside the infrastructure for effective research engagement are also important initiatives to foster a complementary culture too. Hence, it is worth noting that efforts to encourage research engagement starts early too; initial teacher preparation in Norway includes an academic dimension that sees teachers learning about research disciplines and carrying out action research.

Notes: England (United Kingdom) and Ireland shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2022^[13]), *Who Cares about Using Education Research in Policy and Practice?: Strengthening Research Engagement*, <https://doi.org/10.1787/d7ff793d-en>; (OECD, 2012^[25]), *Lessons from PISA for Japan*, <https://doi.org/10.1787/9789264118539-en>; (OECD, 2019^[10]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>

Lever 12: Systematising and disseminating knowledge

As teachers and leaders engage in different types of professional learning, they are also potentially generating new knowledge, in relation to their particular context and ways of working. Similarly, as the array of system actors engage in their work of supporting professional learning – such as around knowledge mobilisation – they too may be developing relevant insights on their way of working and what can best maximise their impact and effectiveness.

Codifying the knowledge that schools and systems develop is a critical step of an effective system of professional learning. It can help improve efficiency and quality; for instance, it may mean that ineffective processes or initiatives are not adopted, or that those that are highly effective are adopted at greater scale without teachers needing to invest considerable resources again. Sharing can reduce duplication and the efficient allocation of resources (OECD, 2025^[26]).

Codifying knowledge can also help the more systematic building of new knowledge. A codified knowledge base can be carefully built upon with outstanding questions specifically targeted and addressed.

It has proven challenging to develop coherent approaches to codification, despite calls from both practitioners and researchers (Goldstein, 2012^[27]; Kane, 2018^[28]). There has been remarkably less attention on the processes of learning in schools, or the perspective and insights of schools (OECD, 2025^[18]). The relevance of this codification is particularly relevant in an era of high teacher turnover which may reduce the degree of ‘institutional wisdom’ being held inside a system or schools.

Capture new knowledge in systematic, coherent ways

One key consideration is the type of knowledge from schools that is shared. Generalisable knowledge is often understood through the lens of more expensive and intensive causal research designs. These lie beyond the capacity and resources of many schools and systems. Systems may attend to how they draw upon other sources of knowledge:

- More intermediate levels of evidence may be gathered through processes such as collective insights across multiple teachers or schools on the effectiveness of particular strategies and approaches. These may make use of teacher self-reported data on particular approaches.

- Alternative approaches include that of detailed case studies. Researchers have long argued for the benefits of ‘case studies’ with their rich description of particular contextualised learning as yielding a different type of generalisation that relies on the user as a critical reader (Merriam and Tisdell, 2016^[29]).
- Case studies do not provide generalised knowledge in the statistical probabilistic sense, but rather detailed accounts from a specific setting, which the user can then reflect on and evaluate for themselves. Sometimes referred to as ‘hypothesis testing’, this approach essentially sees the user test their own context-specific needs against the knowledge such case studies share.
- Such case studies may capture the complexity of processes of change and learning which can be particularly valuable for leaders or teachers considering similar changes themselves.

In both cases, the establishment of more standardised and consistent ways of capturing knowledge from schools, and wider actors, is a significant question for a system.

- For instance, shared templates for case studies of practice or an agreed set of items for self-reported data can help to facilitate comparison between different sources of knowledge that are collected.
- Moreover, transparency and detail tend to be essential; because the knowledge has not been generated through a causal design, it may be that multiple methods are particularly valuable – such as that of parents, students, leaders and teachers.
- It also means that transparent communication of limitations can be valuable, particularly for the process of understanding if such a case study may apply to one’s own situation.

Curate new emerging knowledge

Knowledge needs to be well-organised and accessible to be useable. It has been argued that one of the challenges teachers and schools face today is the sheer quantity of information that they are presented with in the digital age (Trust and Horrocks, 2016^[30]). This sits in tension with the fact that diverse sources of information can be also relevant for responding to diverse challenges, sparking inspiration, and teachers’ process of critically interpreting how information is relevant to their context.

One particular consideration is what is prioritised.

- Knowledge may be curated to focus on particular goals and objectives, such as the priority objectives of a system (e.g. fostering engagement in Science, Technology, Engineering and Mathematics (STEM), social-emotional learning).
- Alternatively, teachers may have more agency to browse different knowledges. These can still be enabled through comprehensive systems such as digital platforms that use tagging systems for schools to find the knowledge they seek.

A second consideration is how trust is built.

- Transparency on both the success and the limitations or challenges of particular approaches, acknowledging that so-called ‘silver bullets’ in education do not exist, can support teachers’ critical engagement with knowledge.
- Similarly, clarity on the contexts that knowledge comes from may be relevant here too and the process of interpretation.

Ensure effective tools for dissemination at scale

A key feature of an ecosystem is the accessibility to knowledge it presents to all. Technology can be a crucial tool for supporting schools’ accessibility to codified knowledge.

- Digital platforms and digital networks can be a means of teachers and schools in more rural settings of having access to the same knowledge as those that exist in more physically-connected geographies (Minea-Pic, 2020^[31]). Of course, this is dependent on the digital infrastructure that exists.
- Digital also holds promise insofar as helping to bring the tangible nature of teaching to life. Lesson resources and materials are essential features of many initiatives and the day-to-day action of classrooms. Digital tools can provide access to these at greater scale.
- Furthermore, videos of practice may help illustrate new approaches from teachers more concretely than purely descriptive accounts (Fischer and Neumann, 2012^[32]).

Box 5.3. International examples of systematising and disseminating knowledge

Singapore

Singapore has sought to create an infrastructure that can better allow for networked learning. This seeks to support the co-construction and production of new knowledge as a community.

Singapore's One Portal All Learners platform allows teachers to set up online collaboration groups, facilitating asynchronous knowledge construction that complements face-to-face interactions. A series of tools are also available for the knowledge-building processes:

- Blogs and forums allow for the sharing of resources (e.g. artefacts, literature reviews) and commenting or discussion around these.
- A chat feature allows for real-time synchronous conversations or quick exchanges, while more formal webinars can also be hosted for virtual convenings.
- Multi-media such as audio and video clips can be shared through podcast and vodcast features.
- Wiki features allow for collaborative creation and co-editing.
- Survey/poll features allow for surveys to be created to facilitate data collection.

While the digital realm is essential in its function, it is also worth noting the importance of simplicity too. Users appreciate the platform's compatibility with mobile technologies and its unified log-in with existing email systems (Lee et al., 2020^[33]). Researchers have also found that facilitation of collaboration in the digital space, potentially drawing upon assigned roles for members, may be essential to successful collaborative knowledge-building (Lee et al., 2020^[33]).

Shanghai (China)

Shanghai has developed a sophisticated digital platform to share lesson plans that draws upon the wisdom of the crowd for curation through reputational metrics. The more other teachers download, or critique or improve lessons, the greater the reputation of the teacher who had shared them.

This, in turns, feeds into the wider professional progression of the teacher. Hence, the ability to work collaboratively is one of the expectations of the system. In evaluations of staff, school leaders can enquire into the contribution of the teacher to improving the teaching profession and the wider education system.

In this respect, Shanghai (China) has sought to tap into teachers' desire of people to contribute, collaborate and be recognised for their efforts. It has helped to nurture creativity and quality control simultaneously through its giant open-source community.

Source: (Lee et al., 2020^[33]), *The development of an implementation framework to support knowledge construction in online networked learning*, <https://www.tandfonline.com/doi/full/10.1080/19415257.2020.1763430>; (Campbell, Lieberman and Yaskina, 2013^[34]), *Teacher Learning & Leadership Programme*, https://www.offeo.on.ca/en/wp-content/uploads/sites/2/2013/09/tllp_full_report-.pdf (accessed 20 February 2025); (OECD, 2023^[35]), *Teaching for the Future: Global Engagement, Sustainability and Digital Skills*, <https://doi.org/10.1787/d6b3d234-en>

Key implementation considerations

- **Supporting leaders:** School leaders are key actors in fostering collaboration. As mentioned, this may mean appreciating that nurturing collaboration demands a particular skillset and building this capacity. Leaders' time can be limited for engaging in more instructional work with colleagues (OECD, 2019^[36]), and thus this time constraint for any skill development as well as enacting such practices demands consideration too.
- **Identifying 'earlier adopters':** To support the steady, targeted development of new initiatives, identifying those in schools that can serve as 'champions' and serve as the first adopters of the new approaches the initiative proposes can be valuable (Education Endowment Foundation, 2024^[7]). For instance, these may be relevant for the use of new digital tools (OECD, 2024^[19]). Such champions may help to mobilise others and provide them with sustained support. Of particular relevance is the question of fostering truly 'whole' school cultures; with TALIS 2018 results suggesting that teachers tend to collaborate with only a few colleagues, while others in the same school may not collaborate at all. A key implementation consideration may be how early adopters with the support of school leaders steadily build wider connections for more inclusive collaborative cultures.
- **Appreciating the time needed for new cultures to embed:** Moving to more collaborative ways of working may take time. In particular, it may be jarring when transitioning from cultures of competition. The skills that underpin collaboration need steady nurturing within the structures that enable these to manifest, so that they become more habitual and refined. Indeed, this type of sustained support may be particularly vital in the earlier phases where resistance may be most likely.
- **Attend to reform fatigue across actors:** Connected to the previous point about patience is also the need to be mindful of the negative effects of constant change (OECD, 2025^[26]). Reform or change fatigue can be demotivating and lose buy-in, particularly if there are frequent U-turns or overhauls in the approach. Similarly, too many changes may be overwhelming, and a more gradual, targeted approach may avoid this sensation of fatigue among stakeholders.
- **Celebrate success:** One way of navigating potential feelings of fatigue or uncertainty can be clear monitoring of the change process and, in particular, highlighting its progress. The positive identification of successes – however small – can help to highlight the steps being achieved and may support engagement (OECD, 2025^[26]). Indeed, much like positive praise from teachers to students in the classroom, this too can be important for the adults in the system as well.

Box 5.4. Key reflection questions for system leaders

To create an ecosystem for collective learning and sharing to ensure opportunities for excellence for all, system leaders may consider the following:

A culture of collaborative inquiry within and across schools

- How can structures (e.g. networks) and roles (e.g. facilitators) be used to support collaborative inquiry within and across schools?
- What resources can be provided to support collaborative inquiry within and across schools? (e.g. time, physical resources [e.g. videos, digital tools], capacity-building)
- How can practitioners enact agency over defining tangible focusses for collaboration?
- What role can data, research and artefacts play in grounding collaboration?
- How can additional actors be drawn upon to enhance collaborative processes?

- How could best practice around collaboration be more widely shared?

Developing engagement with research

- What mechanisms can support access to high-quality, relevant research?
- How could teachers and school leaders be supported to develop their research literacy?
- What system-level infrastructure would enable a culture of effective, sustainable research engagement?
- What role could teachers and school leaders play in shaping the research agenda?

Systematising and disseminating knowledge

- How could new knowledge from school-level actors be coherently and systematically captured?
- What could be the role of different actors in generating and curating new knowledge?
- What tools could enable an evolving knowledge base to be accessible to a wide range of actors?

Implementation considerations

- How can leaders be supported to become enablers of strong collaborative cultures in their school and more widely?
- What could be the potential of identifying and supporting early adopters when implementing change?
- What might be the role of patience and incremental steps in the face of a time-consuming process of changing cultures?

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6

Teacher initial education and ongoing professional learning in Ukraine: Strengths, challenges and reflections for further development

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The implementation of the New Ukrainian School (also “NUS”) reform alongside the challenges posed by the pandemic and the full-scale war following Russia’s invasion, create new demands for the initial and ongoing professional learning of Ukrainian teachers. Current circumstances require teachers to acquire new professional competencies to equip the next generation with the skills needed to rebuild the country. The basis for these urgent changes in professional learning is provided by progressive educational legislation, which reflects modern expectations of the teaching profession, as well as the Strategic Plan of the Ministry of Education and Science of Ukraine up to 2027 which outlines a renewal of approaches to initial teacher education and ongoing professional learning, as well as a new teacher remuneration system.

To explore potential directions for reforming the system of professional learning, state institutions and civil society organisations in Ukraine, as well as the international expert community, were engaged. Specifically, at the request of the Ministry of Education and Science of Ukraine (MESU), the Organisation for Economic Co-operation and Development (OECD) analysed approaches to teacher learning from both national and sub-national education systems around the world. This analytical work was summarised and presented for review by the Ukrainian educational community, and is presented here in Chapters 2 to 5. It was discussed during the seminar “International Pre-service and In-service Teacher Learning Cases – for Ukraine”, held on 13 and 14 May 2025 in a peer-to-peer format. Over the course of two days, Ukrainian educators engaged with the analytical work of the OECD, attended detailed presentations by international experts on their respective national experiences, and took part in group discussions aligned by shared professional interests and expertise. Based on these discussions, the participating groups identified practices with strong relevance and potential for adaptation within the Ukrainian teacher learning system.

Building on this the seminar, this chapter was created. It outlines Ukraine’s experience in teacher learning and professional development, identifies key issues and challenges, and provides recommendations for updating and enhancing this sector. It reflects the perspectives and assessment of the authors, a group of experienced education analysts with in-depth knowledge of the Ukrainian education system. The chapter outlines some of the current strengths and challenges in Ukraine’s approach to professional learning. For ease of cross-referencing, the structure of this chapter mirrors that of the preceding chapters and the overarching structure of four ambitions and corresponding levers (see Chapter 1, Box 1.1). For each of these levers, the chapter also tentatively presents some reflections on potential directions that could be explored for strengthening professional learning in Ukraine, informed by international experiences and the aforementioned seminar’s discussions.

Key highlights

- An underlying challenge is the low status of the profession and its subsequently declining appeal among prospective students. Stakeholders may consider re-examining remuneration alongside addressing the intellectual attractiveness of the profession, such as through clear opportunities for professional growth and reliable recognition of this through the career structure.
- Initial teacher education can better align to contemporary school realities, particularly in terms of the implementation of the New Ukrainian School reform's more competency-based pedagogy and in responding to wartime challenges.
- Valuable mechanisms are in place to support new teachers in their transition to the classroom and professional life, but there may be scope to enhance the actual day-to-day implementation of these mechanisms through more specialised training efforts and peer-support networks.
- Teacher appraisal remains a demanding yet highly formulaic process that does not necessarily reflect a teacher's actual professional competence, and which does not serve as a sufficient motivational incentive for professional learning.
- Recent development of professional learning platforms in Ukraine have helped to centralise access to learning opportunities while the financing system for professional learning has also been updated and diversified. There remains scope for greater co-ordination to create a more unified system with clearer mechanisms of quality control in terms of provision and delivery.
- Developing stronger cultures of inquiry and research-use within and across schools demands capacity-building among school leaders and sustained support through networks.

Current state of teacher initial education and ongoing professional learning in Ukraine

The Revolution of Dignity (2013–2014) and the signing of the Association Agreement with the European Union in 2014 marked a turning point in the reform of Ukrainian education. These events provided significant momentum for embedding democratic principles within the governance of the education system. Ukraine has committed to European integration as a strategic priority. Accordingly, Ukrainian education legislation has been fully revised to align with EU policies, particularly in the area of general secondary education.

Since 2017, the New Ukrainian School (NUS) reform has been in progress, introducing a shift to competency-based learning, the integration of socio-emotional skills into learning outcomes, and the extension of specialised upper secondary education to three years to support more personalised and in-depth learning. The success of these and other elements of the reform is contingent on a well-prepared teaching workforce (Box 6.1).

Box 6.1. Foundations of reforming teachers professional learning

New approaches to the professional development of modern teachers in Ukraine are outlined in key regulatory documents adopted as part of the New Ukrainian School reform. These include: the Law of Ukraine “On Education” (Verkhovna Rada of Ukraine, 2017^[1]), the Law of Ukraine “On Complete General Secondary Education” (Verkhovna Rada of Ukraine, 2020^[2]), Resolution of the Cabinet of Ministers of Ukraine No. 800, dated 21 August 2019, “Some Issues of Improving the Qualifications of Pedagogical and Scientific-Pedagogical Workers” (Cabinet of Ministers of Ukraine, 2019^[3]), Order of the Cabinet of Ministers of Ukraine No. 988-p, dated 14 December 2016, “On Approval of the Concept for the Implementation of State Policy in the Field of Reforming General Secondary Education: The ‘New Ukrainian School’ for the Period Until 2029” (Cabinet of the Ministers of Ukraine, 2016^[4]).

In 2020, the requirements for teachers’ professional competencies were formalised through the professional standards for the positions of Primary Teacher of a General Secondary Education Institution and Teacher of a General Secondary Education Institution (Ministry of Economic Development Trade and Agriculture of Ukraine, 2020^[5]). These standards were updated in 2024, and since then, the revised professional standard Teacher of a General Secondary Education Institution has been in effect (Ministry of Education and Science of Ukraine, 2024^[6]).

At the same time, the implementation of the New Ukrainian School reform is taking place under the pressure of significant challenges. Following the 2013-2014 ‘Revolution of Dignity’, Russia illegally annexed Crimea and temporarily occupied regions in Eastern Ukraine, with Russia fomenting armed conflict and instability. The COVID-19 pandemic brought about a radical shift in the organisation of the educational process in Ukrainian schools, which were forced to transition to distance and blended learning formats. This shift highlighted the urgent need to enhance teachers’ skills in using information technology and digital tools, as well as in diagnosing and addressing learning losses.

Subsequently, the outbreak of full-scale war following Russia’s invasion further compounded these difficulties, resulting in the widespread destruction of Ukraine’s educational infrastructure. As of early 2024, according to a report on direct damage to infrastructure, at least 380 facilities had been destroyed and 3 429 damaged as a result of Russia’s military aggression. Among these are 1 888 schools across 21 regions of Ukraine, with the Donetsk, Kharkiv, Kherson, Mykolaiv, Dnipropetrovsk, and Zaporizhzhia regions suffering the greatest losses (Kyiv School of Economics, 2024^[7]).

These challenges have been exacerbated by the forced displacement of the population, including both students and teachers. According to one study, between September 2021 and 2023, the number of teachers declined by 22.7% in the eastern region, 18.8% in the southern region, and 7.3% in the northern region of Ukraine (Bozhynskyi et al., 2024^[8]). In front-line areas, education is primarily delivered through distance or blended formats due to ongoing security concerns and the lack of adequate shelter infrastructure. The war has also led to a significant increase in the workload of Ukrainian teachers. Around 80% of teachers report that their workload has increased since the start of the full-scale invasion (GoGlobal, 2023^[9]). This rise is largely attributed to the need to conduct lessons in multiple formats (in-person and online), increased communication with students and their families, efforts to compensate for missed lessons caused by air raids and power outages, and the transition of students to family- or externally-based forms of education. This intensified workload, combined with prolonged work under martial law conditions, has contributed to widespread professional burnout among teachers (Kohut and Nazarenko, 2023^[10]).

Funding for education was reduced, which temporarily slowed the implementation of the reform at the basic secondary school level (Ukrinform, 2023^[11]). Due to a lack of financial resources, it was not possible to fully modernise the educational environment. Unlike the roll-out of the New Ukrainian School reform in

primary education, no centralised teacher ongoing learning was organised, and updated textbooks were not published (Ministry of Education and Science of Ukraine, n.d., p. 68^[12]). Research indicates that nearly half of Ukrainian teachers working at this level lack a sufficient understanding of the core principles and key features of the reform (Editorial Board of "NUS", 2024^[12]).

In addition, the challenges of maintaining educational provision under martial law, combined with the challenges of attracting people to the profession, have resulted in a shortage of subject teachers – particularly in rural areas and regions most affected by the war.

Responding to the demands of wartime conditions has required significant adjustments to school curricula. These include the integration of topics related to everyday resilience, civil protection, and Ukraine's evolving role as a bastion of democratic values. These show the evolving priorities of the teaching profession: the main tasks have become the preservation of children's health and life, mastering the technologies of creative development of students to increase the psycho-rehabilitation effect of the educational process (Sysoieva, 2024^[13]). These developments, in turn, necessitate updates to teachers' professional learning. Particularly important are skills in providing psychological support to students, ensuring self-care for teachers working under martial law (Nadyukova and Frenzel, 2025^[14]), operating effectively in digital environments, and adapting to flexible learning formats (Hrynevych, Linnik and Herczyński, 2023^[15]).

Ambition 1. Laying the foundations for an effective and meaningful teaching career

Lever 1. Process for selecting candidates for the profession

Overview of entry requirements

Ukrainian law¹ stipulates that individuals are eligible for employment as teachers if they are fluent in Ukrainian; possess high moral standards; are in satisfactory physical and mental health enabling them to fulfil professional duties; and

- either hold a degree in education or a professional teaching qualification,
- or have completed higher education in any (including non-educational) discipline (Verkhovna Rada of Ukraine, 2020^[2]).

This means Ukrainian legislation allows access to the teaching profession via three main educational pathways: completion of a pedagogical education; completion of higher education in a non-pedagogical discipline; acquisition of a professional teaching qualification.

The pedagogical education route in Ukraine consists of two pedagogical degree programmes: "primary education" and "secondary education". The primary education programme does not include subject specialisations, its graduates teach in primary schools (Grades 1–4). The secondary education programme includes subject-specific training, enabling graduates to teach one or two (occasionally three) school subjects. These graduates work in Grades 5–11. There is no distinct educational track for teachers of lower secondary (Grades 5–9) and specialised upper secondary (Grades 10–11, and starting from 2027 also Grade 12). Subject teachers may teach in any grade. Teachers of certain subjects (e.g. physical education, foreign languages) often teach across both primary and secondary levels.

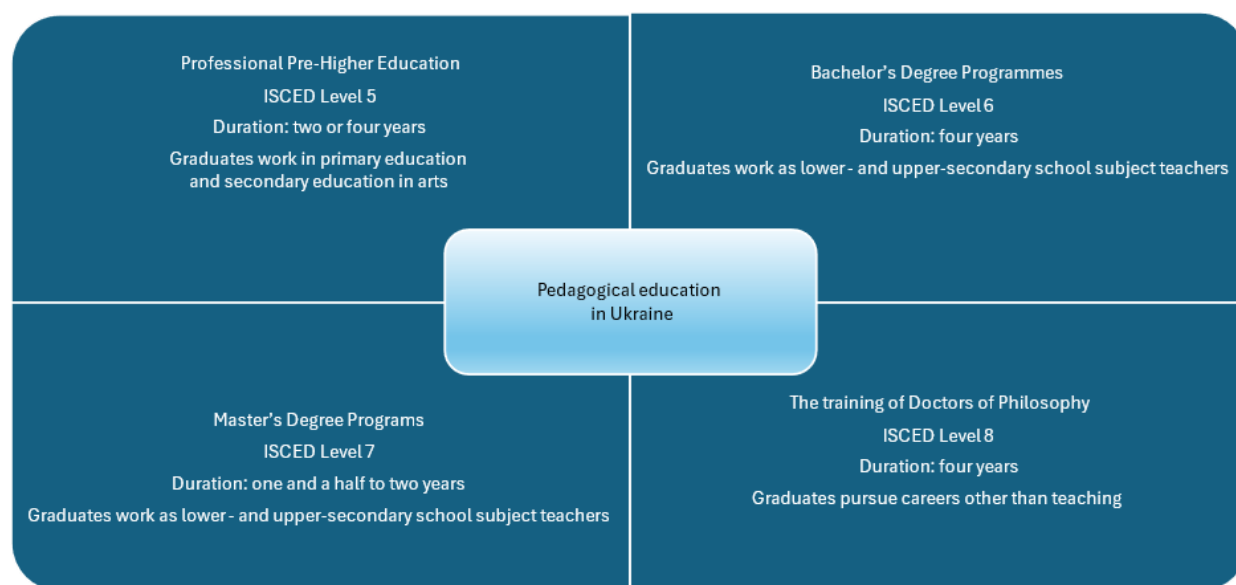
Pedagogical education in Ukraine follows a concurrent model. It can be obtained at four levels of education (Figure 6.1):

1. Professional pre-higher education. This level leads to the qualification of "Professional Junior Bachelor", equivalent to International Standard Classification of Education (ISCED) Level 5.

Graduates in primary education may work as primary school teachers. Graduates in secondary education at this level typically become teachers of the arts (e.g. fine arts, music). Training lasts four years for students entering after Grade 9, and two years for those entering after Grade 11.

2. Bachelor's degree programmes in pedagogical fields last four years (240 European Credit Transfer and Accumulation System (ECTS) credits) for students entering after Grade 11 (ISCED Level 3). This is reduced to three years (180 ECTS credits) for those who already hold a professional junior bachelor qualification. Both professional junior bachelors and bachelors may work as primary school teachers or teach art subjects in secondary school. To be employed as a subject teacher in other disciplines in Grades 5–11(12), it is necessary to hold at least a bachelor's degree in the relevant field.
3. Master's degrees in pedagogy typically last 1.5 years (90 ECTS credits), though some may run for two years (120 ECTS credits), with an emphasis on academic research. The importance of master's degree for teachers is dictated by a stable tradition. Although it is not required for entry into the teaching profession, it holds significant weight later in a teacher's career for obtaining a higher qualification category (Ministry of Education and Science of Ukraine, 2022^[16]). Master's programmes in pedagogy are also open to career switchers who have completed their bachelor's degree in a different field.
4. The training of "Doctors of Philosophy" in the field of education has become noticeably popular: between 5.8% and 7.9% of all applicants to postgraduate programmes in Ukraine from 2020 to 2024 were enrolled in relevant courses (EDEBO, n.d.^[17])². However, individuals holding a PhD in education typically pursue career paths other than teaching in secondary schools.

Figure 6.1. Four levels of pedagogical education in Ukraine



Source: Adapted from (Lantero et al., 2022^[18]), *Guidelines on Fast-Track Recognition of Ukrainian Academic Qualifications*, <https://education.ec.europa.eu/document/guidelines-on-fast-track-recognition-of-ukrainian-academic-qualifications> (accessed on 25 October 2025) and (Eurydice, 2025^[19]), *Organisation of the education system and of its structure*, <https://eurydice.eacea.ec.europa.eu/eurypedia/ukraine/organisation-education-system-and-its-structure> (accessed on 25 October 2025)

Attracting candidates

The popularity of pursuing a pedagogical education in Ukraine presents an ambiguous picture. On the one hand, statistics indicate that approximately one in ten Ukrainian students obtaining a bachelor's or a

master's degree chooses a pedagogical major. The proportion of such applicants in Ukraine is slightly above the European average. In particular, calculations based on available data suggest that in 2023, at the bachelor's level, 9.8% of students in Ukraine were enrolled in pedagogical programmes, compared to the European average of 7.4%; at the master's level, this figure was 10.6% in Ukraine, compared to the European average of 9% (Eurostat, 2023^[20]; EDEBO, 2025^[21]).

On the other hand, the teaching profession – and, by extension, pedagogical education – is widely perceived as lacking in prestige. As a result, pedagogical education programmes attract a relatively small number of highly motivated and well-prepared applicants. Most students apply to pedagogical programmes as a 'last choice', when their national entrance test results are too low to gain admission to more competitive programmes in other fields.

This is reflected in the financing model of bachelor programmes (see Box 6.2). Fields such as medicine, law, international relations, and public administration require a notably high score on national multi-subject entrance test if applicants are to compete for publicly-funded places. In education, the score is not as high. Specifically, according to our calculations based on requested data from the Unified State Electronic Database on Education (EDEBO), in 2024 the average competitive score to enter public-funded places was 147.1 for bachelor programmes in primary education, 153.1 secondary education, 176.9 for law, 165.8 for medicine, and 171.0 for public administration. The average entry score for fee-paying students was 135.7 (primary education programmes), 141.7 (secondary education), 161.8 (law), 158.1 (medicine), 160.1 (public administration) (Unified State Electronic Database on Education, 2024^[22]).

Among applicants admitted to bachelor's programmes in 2024 across all fields of study, 66.6% are paying for their education independently, while 33.4% are publicly funded. In this context, the low prestige of pedagogical training is reflected in the inverse ratio of tuition-paying to publicly-funded students in teaching-related disciplines: only 27% of future primary school teachers and 33% of students enrolled in the "Secondary Education" specialisation pay for their studies. The proportion of self-funded students in science, technology, engineering and mathematics (STEM) teaching specialisations is even lower – ranging between 16% and 26%.

A connected challenge here is also ensuring those studying on pedagogical degree programmes arrive in the actual classroom. According to a monitoring study conducted by the "Almenda' Centre for Civic Education" NGO in 2024–2025, a significant proportion of students in the final years of pedagogical degree programmes do not intend to work in schools. Specifically, among bachelor's degree students, only 32.7% plan to enter the teaching profession, 26% do not plan to do so, and 41.3% remain undecided. Among master's degree students, 35.5% intend to work as teachers, 5.9% do not, and 58.6% are still uncertain (Almenda Center, 2025^[23]). This is again a reflection of the challenges around how the teaching profession is perceived. It also suggests that efforts to raise the status of the teaching profession alongside initiatives to support initial teacher education are important; for instance, simply raising the minimum entrance requirements for pedagogical specialities – as has been done in Ukraine for legal and medical education – without efforts to also enhance the status of the profession may not be fully effective.

Box 6.2. Admission to bachelor's programmes in Ukraine

Admission to bachelor's programmes at Ukrainian universities requires candidates to sit a national multi-subject entrance test, covering several school subjects. Each test result is converted into a score from 100 to 200 points, based on a standardised scale. For the 2025 entrance test, 100 points corresponds to correct answers on approximately 15% of the questions. An applicant's competitive score is calculated as a weighted average of the test scores across several subjects.

A minimum competitive score of 100 points is required to take part in the admission competition for higher education institutions. Scores also have implications for how studies are funded. Professional pre-higher and higher education in Ukraine is financed in two ways. The first is public funding from the state or local budgets, under which students do not pay for their studies. These publicly-funded places are available to applicants with higher entrance exam scores, as well as to individuals from certain preferential categories, including orphans, persons with disabilities, and combatants. Approximately 40–45% of students studying at public expense receive a modest academic scholarship (equivalent of approximately 30–50EUR per month).

To compete for public-funded places, a score of at least 130 points is required. A score of at least 150 points is mandatory for applications in such fields as medicine, law, international relations, and public administration (Ministry of Education and Science of Ukraine, 2025^[24]).

The second financing model involves students or their parents paying tuition fees directly. This route is typically pursued by less academically competitive applicants.

Individuals who fail to pass the national entrance test (or do not sit it) are, with certain exceptions, not eligible to apply for bachelor's programmes. However, they may apply for vocational or professional pre-higher education (ISCED Level 5) or choose to retake the test in the following year.

Source: (Ministry of Education and Science of Ukraine, 2025^[24]), *On Approval of the Procedure for Admission to Higher Education in 2025*, approved by Order No. 168 of 10 February 2025, <https://zakon.rada.gov.ua/laws/show/z0312-25#Text> (accessed 24 July 2025)

A further challenge to consider here is the fragmentation of teacher training across numerous universities and professional colleges, many of which admit only small numbers of students to pedagogical programmes. For example, within the “secondary education” specialisation 35 out of 80 universities enrolled fewer than 50 applicants to their bachelor's programmes; 26 out of 60 universities enrolled fewer than 50 master's students; 23 out of 39 institutions enrolled fewer than 50 students at the professional junior bachelor's level. In “primary education” 41 out of 58 universities admitted fewer than 50 applicants to their bachelor's programmes; 34 out of 43 universities had fewer than 50 master's entrants; 20 out of 47 institutions offering professional junior bachelor's programmes also enrolled fewer than 50 applicants (EDEBO, n.d.^[17]). An insufficient number of students in an educational programme renders its delivery financially unsustainable and poses challenges to maintaining educational quality. Besides, institutions running under-enrolled programmes are often unable to expel underperforming students. This situation frequently leads to lower academic standards and a decline in the enforcement of academic integrity.

A final challenge to note relates to the presence of teachers in the system without certain minimum qualifications. In general, there is not widespread practice in Ukraine of hiring teachers without a pedagogical background, but there are some instances. Based on analysis of available data from the Institute of Educational Analytics (Institute for Educational Analytics, 2025^[25]), as of September 2024, out of 70 394 primary school teachers, only 270 (0.4%) did not have a pedagogical qualification. Among 39 544 teachers of music, fine arts, physical education, defence of Ukraine, and labour training, 2 192 (5.5%) lacked a pedagogical qualification. Of 161 953 subject teachers working in Grades 5–11, 3 213 (2.0%) had

no prior pedagogical training. In particular, the presence of a small number of teachers with only secondary education points to the fact that there can be challenges in recruiting qualified personnel (Table 6.1). This practice runs contrary to Ukrainian legislation, which does not allow hiring people with secondary education for teaching positions. It is also worth noting that, in some cases, schools are compelled to hire students enrolled in pedagogical programmes on temporary contracts.

Table 6.1. Number of teachers and school principals in Ukraine, by level of education, as of September 2024

Category	Total	Master	Bachelor	Professional Pre-Higher (ISCED 5)	Upper Secondary
Primary education teachers (Grades 1–4)	70 394	61 606	2 337	6 334	117
Teachers of Grades 5–11 (12), except those specified in the next line	161 953	155 460	4 657	1 556	280
Teachers of music, fine arts, physical education, defence of Ukraine, technology education	39 544	33 987	2 157	3 188	212
School principals	11 572	11 536	21	15	0
Deputy principals	18 418	18 268	115	35	0

Note: Calculated based on available data from the Institute for Educational Analytics

Source: Based on (Institute for Educational Analytics, 2025^[25]), *On the number and composition of pedagogical staff, 2024-2025*, https://iea.gov.ua/wp-content/uploads/2025/05/byuleten-83-rvk-2024_2025.xlsx (accessed 10 October 2025)

As mentioned, while these numbers of those in classrooms without appropriate qualifications are relatively low, further reducing this number to ensure that all students have teachers with the necessary pedagogical expertise could be an important area of future attention. This sets Ukraine apart from many European countries, where the teaching profession is regulated and access is not permitted without appropriate professional training. Given Ukraine's chosen path of European integration, it is anticipated that, over time, the approach to teacher qualifications will be aligned with European standards. Consequently, individuals without formal pedagogical education will be required to obtain a recognised pedagogical qualification.

Additional routes to the classroom

The above discussion of pedagogical qualifications invites consideration of the additional routes that exist to the classroom. According to Ukrainian law³, individuals who do not possess a pedagogical degree or professional teaching qualification – but hold higher education in another field – may be employed as teachers for a period of one year (Verkhovna Rada of Ukraine, 2017^[1]). Upon the completion of this year, they are required to undergo appraisal to continue working in the teaching profession.

This provision allows three types of institutions to assess competencies of teachers with at least one year of work experience and confer a pedagogical qualification: qualification centres, institutions of postgraduate education, and higher education institutions. However, in practice, there are challenges in each case to this implementation. Postgraduate education institutions do not actually award professional qualifications, as the procedure for doing so has not been clearly defined by law. Similarly, higher education institutions are not legally authorised to confer full professional qualifications to individuals who are not enrolled as their students. Nonetheless, Ukrainian higher education institutions do award professional qualifications – particularly pedagogical ones – to their bachelor's and master's students. They may also issue partial professional qualifications, which confirm that an individual possesses a portion of the

competencies required by the professional standard. However, the content and recognition of such partial qualifications are unclear to labour market participants. As a result, the practice of awarding them to teachers remains uncommon.

With regard to qualification centres, individuals with any professional pre-university qualification (ISCED 5) or higher education who have worked as a teacher for at least one year may apply to qualification centres, which assess their competencies and award them a professional teaching qualification. At present, confirmation of qualifications in this manner is not a mandatory requirement for entering the teaching profession in Ukraine, as there is no such provision in existing legislation. Moreover, as of June 2025, no accredited qualification centres (National Qualifications Agency of Ukraine, n.d.^[26]) exist in Ukraine that are authorised to award professional teaching qualifications or recognise teaching qualifications obtained abroad.

As such, the opportunity to obtain a professional teaching qualification through this route is expected to become available in the future, once the relevant institutions begin operating. These qualification centres may be independent bodies or established as structural units within higher or postgraduate education institutions. What matters most is not their organisational form, but the development – by the government and the educational community – of consistent, recommended approaches to the procedures and tools used for the proper assessment of applicants' competencies for a professional teaching qualification.

Overall, this alternative route into the teaching profession has the potential to enable a broader pool of specialists from other sectors of the economy to be recruited into teaching – particularly useful given the current shortage of teachers in specific subjects. However, gaps in the regulatory framework governing this route allow individuals to teach without undergoing any psychological or pedagogical training or practical teaching experience.

Table 6.2. Number of applicants to bachelor's programmes for teacher learning in Ukraine in 2018–2024

Major	2018	2019	2020	2021	2022	2023	2024
Primary education	4 341	4 026	3 489	3 109	3 635	4 410	2 230
Secondary education, all subject specialisations	14 685	13 677	12 566	12 674	12 689	13 385	9 000
Secondary education, main subject specialisations							
Ukrainian language and literature	2 172	2 013	1 826	2 006	1 610	1 440	1 227
English language and foreign literature	1 555	1 433	1 929	1 833	1 753	1 501	1 278
History and civil education	1 665	1 532	1 478	1 432	1 501	1 241	1 118
Mathematics	641	607	719	737	613	698	464
Biology and human health	1 061	937	808	1 018	715	1 269	540
Chemistry	218	219	181	145	121	171	119
Geography	539	576	567	728	522	924	389
Physics and astronomy	224	307	257	249	218	279	167
Computer science	377	394	435	478	616	1 459	506
Physical education	2 790	2 612	2 134	1 924	2 666	1 832	1 641
Sciences	65	79	109	127	113	185	108
Defence of Ukraine						109	44

Note: Orange highlights programmes with up to 500 applicants. Blue highlights programmes with between 501 and 800 applicants.

Source: Based on (EDEBO, 2025^[21]), *Number of applicants for higher education - Open Data - Register of Subjects of Educational Activity*, <https://registry.edbo.gov.ua/opendata/educators/> (accessed on 24 July 2025)

Considering the role of alternative routes to the classroom may be especially significant in light of shortages in certain subjects. The popularity of pedagogical programmes varies significantly across subject specialisations (see Table 6.2). While training programmes for primary school teachers, as well as those

focussed on languages and the humanities, tend to attract a relatively higher number of applicants, the issue of low enrolment in programmes preparing future teachers of mathematics and other STEM subjects remains unresolved⁴.

One solution to the shortage of teachers in STEM disciplines could be allowing students in non-pedagogical bachelor's programmes (e.g. in mathematics, physics, chemistry) to obtain an additional qualification in pedagogy, psychology and teaching methodology. Hence, Ukrainian higher education offers two pathways for training subject specialists for the secondary level. One, as mentioned earlier, involves the relevant specialisations within bachelor's and master's programmes in secondary education. The other is through "classical" academic programmes – in subjects such as mathematics, philology, history and archaeology, physics, chemistry, biology, and others. Unsurprisingly, the depth of subject training provided in secondary education programmes is generally inferior to that of "classical" degree programmes in philology or other subject areas. It could be that a block of pedagogical courses is made available to those in the "classical" route, after which universities could confer a formal teaching qualification.

There is not currently this type of infrastructure in place for this type of alternative route into the classroom. A review of educational programmes in "classical" disciplines at Ukraine's largest higher education institutions shows that universities rarely offer such an additional pedagogical qualification to students of STEM programmes (Rii, Kryzhanovska and Nikolaiev, 2024^[27]). Moreover, these "classical" programmes – as well as the equivalent specialisations within the "Secondary Education" field – attract a very limited number of applicants, particularly compared to certain other subjects (see Table 6.3). Consequently, the introduction of a pedagogical specialisation would be unlikely, on its own, to resolve the shortage of teachers in these subject areas, but it may be effective alongside other measures.

Table 6.3. Number of applicants to bachelor's programmes for STEM and most popular non-STEM specialities in Ukraine in 2018–2024, persons

Major	2018	2019	2020	2021	2022	2023	2024
Biology and chemistry	1 336	1 200	1 121	1 267	831	665	574
Chemistry	443	417	449	467	319	415	268
Physics and astronomy	219	275	292	313	272	376	223
Geography	414	256	255	504	309	235	201
Mathematics	366	377	446	504	375	352	248
Management	11 130	10 547	12 263	10 495	10 473	14 032	13 234
Law*	20 784	19 613	16 595	15 128	17 580	21 030	9 851
Computer Science and Software Engineering	10 932	12 073	13 060	15 530	15 848	15 610	12 349

Note: *Data on the number of applicants to legal education programmes is underestimated, as information on a portion of students in this field is not recorded in the EDEBO system.

Source: Based on (EDEBO, 2025^[21]), *Number of applicants for higher education - Open Data - Register of Subjects of Educational Activity*, <https://registry.edbo.gov.ua/opendata/educators/> (accessed on 24 July 2025)

Points of reflection for the future on the process for selecting candidates to the profession

- There may be value in broadening pathways for entry into the teaching profession to help ensure a larger and broader pool of quality candidates. Some potential avenues for the future could include:
 - Traditional universities offering students of classical disciplines (e.g. physics, mathematics) an elective pedagogical module of the educational programme to enable students to acquire an additional professional qualification of teachers. This may be especially significant where there is a shortfall of qualified subject teachers, such as that observed in "Secondary Education" in STEM specialisations.

- Introducing one-year dual teacher preparation programmes at universities for candidates with non-pedagogical degrees. Unlike many other countries, Ukraine does not have ‘fast-track’ courses for individuals with higher education who wish to start a teaching career, and this could be worth exploring as a short-term solution to addressing shortages in specific areas. The programme would need to carefully combine psychological and pedagogical training with acquiring a formal teaching qualification. Attention to longer-term retention mechanisms, alongside such an initiative, may be important too.
- In the pursuit of ensuring quality selection and preparation, there may be value in further exploring how teachers’ competencies are assessed before their full entry into the classroom. Current legislation permits individuals with any form of higher education – not necessarily pedagogical – to work as teachers. There is no system in place to ensure that such teachers receive appropriate psychological and pedagogical training. Some potential avenues for the future could include:
 - Ensuring the role and capacity of qualification centres in the current system. Although Ukrainian legislation allows individuals without a pedagogical education to demonstrate the required psychological and pedagogical competencies through an assessment for the award of a professional qualification, as of June 2025, there are no qualification centres in Ukraine authorised to carry out this function.
 - A regulated qualification assessment with uniform requirements before employment as a teacher. Such an assessment should aim to align with the national teacher professional standards and include activities that test the level of relevant professional competencies. As per international models, the exam could include both theoretical and practical components and include such forms of assessment as written test on the subject and pedagogy; a written case analysis; a review of portfolio materials according to predefined criteria; and a structured interview with evaluators. Different modalities could be envisioned, such as implementation by qualification centres, higher education institutions, by a single overseeing state body, or a mixture of these. Notably, this type of initiative would need to be carefully considered alongside the number of applicants to the profession due to its potential implications on teacher supply.
- Indeed, efforts to improve the status of the profession – as exemplified by this report and the accompanying peer-learning event – have an important role to play. The limited prestige associated with the profession impacts interest in teacher training programmes at universities. In the longer term, if measures are successful in improving the status of the profession, some potential avenues could include:
 - Adjustments could be envisioned to the entrance score for the teaching profession so that applicants with a high level of academic performance may be enrolled in pedagogical programmes. In the current situation raising the threshold score could have negative effect, as it will further reduce the number of students enrolled in pedagogical majors.
 - If realisable, a future iteration of this may also include a test on the motivation for entering the teaching profession (e.g. certain psychological traits, desire to master the profession, etc.).

Lever 2. Nature of preparation for the classroom and profession

Defining the content and structure of initial teacher education

State requirements regarding the content and structure of initial teacher education programmes in Ukraine are defined in two types of documents:

- educational standards (including standards for professional pre-higher education (Ministry of Education and Science of Ukraine, n.d.^[28]) and higher education (Ministry of Education and Science of Ukraine, n.d.^[29]),

- and professional standards (National Qualifications Agency of Ukraine, n.d.^[30]).

Educational standards outline the set of learning outcomes and competencies that students in initial teacher education programmes are expected to acquire. Institutions providing professional pre-higher and higher education are legally required to fully incorporate these standards into their educational programmes.

Professional standards, on the other hand, define the competencies required of individuals in a given profession, generally irrespective of the means or form of education through which those competencies were acquired (Box 6.3). When designing educational programmes, universities and professional colleges are expected to take professional standards into account, but these serve a secondary role in comparison to the legally binding requirements of educational standards. As such, professional standards do not carry the same authoritative weight as educational standards for universities.

Box 6.3. Teacher professional standards

Teacher professional standards specify that a qualified teacher must possess the following integral competencies:

- Linguistic and communicative competence
- Subject-specific and methodological competence
- Information and digital competence
- Psychological competence
- Emotional and ethical competence
- Competence in pedagogical partnership
- Inclusive competence
- Health-preserving competence
- Prognostic competence
- Organisational competence
- Evaluative and analytical competence
- Capacity for lifelong learning

Source: (Ministry of Education and Science of Ukraine, 2024^[31]), “Professional Standard ‘Teacher of a General Secondary Education Institution’”, approved by Order No. 1225 of 29 August 2024, https://register.nqa.gov.ua/uploads/0/646-ilovepdf_merged.pdf (accessed 24 July 2025)

The professional standards “takes into account the context of the Russian–Ukrainian war and changes in the educational environment... identifies the need for teachers to possess the competencies required to ensure student safety, maintain psychological stability, adapt educational content to new realities, and instil in students values related to civic responsibility, patriotism, and the protection of the democratic system.” It further “emphasises the ability of teachers to respond to the challenges faced by the general secondary education system during wartime, including forced displacement, loss of access to educational resources, and the necessity of working in blended or distance learning formats” (Potapova, Okhredko and Kuberska, 2025, p. 7^[32]).

Educational standards for the “Primary Education” major, which prepares future primary school teachers, were approved at various educational levels at different timepoints⁵. Notably, none of the educational standards for the “Primary Education” major fully reflect the requirements of the professional standards, as all were adopted prior to the latter’s approval. Furthermore, these educational standards do not include

competencies specifically aimed at addressing the challenges associated with wartime conditions (Potapova, Okhredko and Kuberska, 2025, p. 7_[32]).

For the “Secondary Education” major, which prepares teachers for lower and specialised upper secondary schools, an educational standard has been approved only for the professional junior bachelor’s level (in September 2021). As of June 2025, there are no approved educational standards at the bachelor’s, master’s, or doctoral levels for this major.

Alongside the professional and educational standards, additional guidance on the content and structure of initial teacher learning programmes is provided in the Concept for the Development of Pedagogical Education (Ministry of Education and Science of Ukraine, 2018_[33]), adopted in 2018. However, there are signs that this document has not been effectively implemented in practice. For example, the Concept stipulates that continuous pedagogical practice should be a mandatory component of teacher training. It recommends a total of 30 ECTS credits for in-school pedagogical practice at both the bachelor’s and master’s levels. However, this recommendation is largely not implemented in current teacher training programmes (see Table 6.4), where the volume and distribution of different blocks of educational components vary significantly, with in-school placements often falling below the 30 ECTS credits. Indeed, a review of bachelor’s and master’s degree programmes in the “Secondary Education” major published on the websites of various Ukrainian universities reveals a wide variation and a lack of unified approaches to determining the number of hours (credits) allocated to general education, subject-specific training, psychological and pedagogical training, methodological and research training, and pedagogical practice (see Table 6.4).

Table 6.4. Volume of the psychological and pedagogical component in bachelor’s programmes in the “secondary education” major at selected higher education institutions, in ECTS credits

Higher Education Institution	Psychological and Pedagogical Disciplines						Pedagogical Practice		Total Psychological and Pedagogical Training	
	Mandatory		Elective		Total		Min	Max	Min	Max
	Min	Max	Min	Max	Min	Max				
Ivan Franko National University of Lviv (8 programmes)	15	51	8	20	23	71	9	30	32	101
Volodymyr Hnatiuk Ternopil National Pedagogical University (24 programmes, excluding Physical Culture)	27	64	6	42	33	106	12	30	45	136
Uzhhorod National University (12 programmes, excluding Physical Culture)	23	53.5	0	48	23	101.5	9	25.5	32	127
Mykhailo Drahomanov Ukrainian State University (6 programmes, available on the website)	33	42	9	30	42	72	15	24	57	96
Hryhorii Skovoroda Kharkiv National Pedagogical University (17 programmes, excluding Physical Culture and Fine Arts)	29	49	51	51	80	100	21	21	101	121
V. N. Karazin Kharkiv National University (9 programmes)	28	44	0	35	28	79	6	26	34	105
Yurii Fedkovych Chernivtsi National University (5 programmes)	13	50	9	30	22	80	15	30	37	110

Source: Based on educational programmes descriptions on the respective universities’ websites: (Ivan Franko National University of Lviv, n.d._[34]), <https://www.lnu.edu.ua/> (accessed 2 December 2024); (Volodymyr Hnatiuk Ternopil National Pedagogical University, n.d._[35]), <https://tnpu.edu.ua/> (accessed 2 December 2024); (Uzhhorod National University, n.d._[36]), <https://www.uzhnu.edu.ua/> (accessed 2 December 2024); (Mykhailo Drahomanov Ukrainian State University, n.d._[37]), <https://udu.edu.ua/> (accessed 2 December 2024); (Hryhorii Skovoroda Kharkiv National Pedagogical University, n.d._[38]), <http://www.hnpu.edu.ua/> (accessed 2 December 2024); (V. N. Karazin Kharkiv National University, n.d._[39]), <https://karazin.ua/> (accessed 2 December 2024); (Yurii Fedkovych Chernivtsi National University, n.d._[40]), <https://www.chnu.edu.ua/> (accessed 2 December 2024)

Accordingly, one of the main challenges faced by universities is the lack of information from the state regarding expectations for such content. The novelty of the revised professional standard for teachers, combined with the fact that educational programme developers are not legally required to align with it, and the absence of approved bachelor's and master's level higher education standards in the "Secondary Education" major, has resulted in higher education institutions enjoying broad autonomy in determining the content of training for future teachers of lower and specialised upper secondary schools.

Moreover, descriptions of educational programmes in "Secondary Education" often do not place specific emphasis on the implementation of the New Ukrainian School reform or on addressing the challenges posed by the war. Although such topics are frequently included in the syllabi of individual academic disciplines (Potapova, Okhredko and Kuberska, 2025^[32]), there is a lack of a cross-cutting approach that would ensure the overall relevance of these programmes to these modern contexts.

Micro-credentials

A relatively new concept in the Ukrainian educational landscape is that of micro-credentials, which the Council of the EU defines as "the record of the learning outcomes that a learner has acquired following a small volume of learning" (Council of the European Union, 2022^[41]). According to EU recommendations (Council of the European Union, 2022^[41]), European countries should adopt policies on micro-credentials as a flexible instrument for addressing current labour market challenges and for promoting the principle of lifelong learning.

In 2024, the National Qualifications Agency issued "Recommendations on the Implementation of Micro-Credentials in Ukraine" (Semigina et al., 2024^[42]), which outlined general principles and approaches for their introduction. This document served as one of the sources for a recent Order (No. 966) of the Ministry of Education and Science of Ukraine⁶. Among other provisions, the Order recommends that when designing educational programmes, institutions should consider combining pedagogical subject specialisations with optional educational components, including the possibility of awarding a second full or partial professional qualification and/or micro-credentials.

However, Ukrainian educational legislation does not yet define or formally recognise the concept of micro-credentials. As a result, the practical implementation of this policy remains insufficiently regulated at the national level.

Points of reflection for the future on the nature of preparation for the classroom and profession

There could be value in fostering greater clarity around the requirements of initial teacher education, including the consistency between educational and professional standards for teachers and ensuring any gaps are addressed. One potential avenue here could be adopting bachelor's and master's level standards of higher education for "Secondary Education" and updating existing standards for "Primary Education" to reflect the current teacher professional standards would both help to ensure greater clarity and consistency.

Connected to this is the challenge of ensuring that the contemporary realities of schools are reflected in the content of initial teacher education (ITE) programmes. This may include:

- The implementation of the New Ukrainian School reform, in particular competency-based teaching practices and the development of social and emotional skills.
- Modern information technology tools, in particular, providing high-quality remote and blended learning, modern digital resources, and the use of artificial intelligence to support personalised learning pathways.

- How to respond to the context of the war including: the diagnosis and overcoming of educational losses, protection of human rights in war, approaches to trauma-informed learning, civic and cultural integration of participants in the educational process with different experiences of war, and the specificities of education in the liberated territories.

Similarly, a more balanced integration of theory with practice in ITE could be valuable, such as by expanding the grounded practice component. For instance, this could entail developing network of schools and practice bases at universities, inspired by the Finnish experience (see Chapter 2), with dedicated mentor teachers in such schools being supported to work with trainee teachers. More broadly, ways to support the participation of practicing teachers in the system of teacher education could be fruitful in terms of supporting a focus on practice as well as capturing contemporary realities in schools.

Increased clarity on content may also be coupled with mechanisms for monitoring the implementation of this content. For instance, this may occur through the form of assessment of student teachers, as outlined in Lever 1. Other forms that could be explored include evaluation processes of ITE courses, as set out in Chapter 2 and its examples from different systems.

Finally, it may be that there are also options to align initial teacher education with the wider career structure. Potential avenues here for exploration could be:

- Career ambitions pathways may be integrated into ITE programme design, enabling students in the teaching profession to envision their career progression toward leadership or specialised roles in particular areas, similar to the experience in Singapore. For instance, this approach could include training of those with bachelor's degrees in specialised master's programmes which may include special modules on textbook authorship, mentoring, innovative pedagogy of other teachers and school leadership.
- Micro-credentials based on EU Council recommendations for selected educational programme components may be introduced as a flexible tool to support competency development and to serve as a potential pathway to new professional roles. This will allow for their mutual recognition in the case of internal or external student mobility, as well as their recognition and offsetting of relevant micro-credentials obtained by students or trainees in informal education systems.

Lever 3. Attending to the transition into the classroom and school

As set out in Chapter 2, the professional adaptation of early-career teachers is a critically important stage in the development of a teaching career. In Ukraine, two key mechanisms for supporting novice teachers are legally established: pedagogical internship (Ministry of Education and Science of Ukraine, 2022^[43]) and supervision (Ministry of Education and Science of Ukraine, 2025^[44]), intended to provide systematic support and promote the professional growth of early-career teachers. These foundations have the potential to provide important support to new teachers, however the challenge ahead is to ensure it is consistently implemented effectively.

Pedagogical internships

A pedagogical internship is defined as “a system of measures aimed at supporting a newly-appointed pedagogical employee of an educational institution in the performance of their professional duties and the acquisition (or improvement) of their professional competencies” (Ministry of Education and Science of Ukraine, 2021^[45])⁷. The internship is mandatory for all newly-appointed teachers, regardless of their prior education or qualifications, and lasts for one year from the date of appointment.

The main purpose of the internship is to facilitate the professional and pedagogical adaptation of novice teachers to the realities of teaching practice. As outlined in the official Regulations on Pedagogical Internship, the key tasks of this mechanism include:

- Establishing a system of support for professional adaptation to teaching conditions, fostering a positive microclimate and a sense of socio-psychological comfort, and developing friendly relationships with educational stakeholders.
- Developing professional competencies, pedagogical skills, and readiness for independent teaching.
- Instilling respect for the profession and a conscientious attitude towards professional responsibilities.
- Supporting the formation of a mentoring institution within educational institutions and the development of professional communities among teaching staff.

At the heart of the internship is mentoring, in which an experienced teacher provides guidance and support to a novice colleague. The mentor assists the new teacher in navigating the professional environment, resolving early challenges, and developing an individual teaching style. While mentors may receive a salary supplement of up to 20%, this amount is not legally guaranteed. Schools often face challenges in appointing a suitable mentor, especially if there is no teacher available with at least five years of subject-specific teaching experience. The current teacher remuneration system does not support the recruitment of mentors from other institutions, nor does the law permit the contracting and payment of such services externally.

Supervision

Supervision is defined as a set of measures for providing voluntary professional assistance and support to teachers, aimed at improving their teaching performance and supporting continuous professional development (Ministry of Education and Science of Ukraine, 2024^[46])⁸. In contrast to the internship, supervision is a voluntary for teachers who wish to improve their professional skills.

The main tasks of supervision in Ukraine are as follows (New Ukrainian School, 2024^[47]):

- Supporting teachers in identifying their professional strengths, needs and challenges, as well as the resources available to motivate their professional development.
- Analysing the professional activities of teaching staff.
- Providing professional counselling and support based on the analysis of a teacher's professional performance.
- Offering assistance to prevent and overcome professional challenges.
- Supporting compliance with established professional standards.

The supervision mechanism only became formally recognised within the legal framework in late August 2024, making it difficult at present to evaluate its implementation in schools comprehensively. However, like mentoring, the success of supervision will depend significantly on the individual carrying out the supervision. At present, any teacher or academic staff member may act as a supervisor. The funding mechanism for supervision services from the state budget remains unclear. Moreover, for teachers to be willing to pay privately for such services, supervision must first demonstrate its effectiveness.

Points of reflection for the future on attending to the transition into the classroom and school

There may be value in creating space in new teachers' workload for their proactive engagement with pedagogical internships. Connected to this is the potential for greater attention on the implementation of the regulatory framework for pedagogical internships as outlined in Ukrainian legislation. The Law of Ukraine "On education" provides for a pedagogical internship as a formal induction period for beginning teachers (Verkhovna Rada of Ukraine, 2017^[1]), but there remain some challenges on implementation as set out above. Some potential avenues forward could include:

- Ensuring appropriate specialised training for teacher mentors.
- Attending to the incentives for mentors, such as ensuring clarity around the role's position in the horizontal career structure (as well as in relation to any micro-credentials or certification processes) and the nature of any additional payments for mentor teachers.
- Establishing mechanisms for systematic monitoring and evaluation of pedagogical internship to complement this and help to drive high standards.
- Exploring means of diagnosing new teachers' professional competencies to facilitate more tailored professional support.

Similarly, attending to the systematic training of qualified supervisors, as well as the mechanisms for monitoring supervisions' effectiveness, may be of value too in terms of supporting high standards of support to new teachers.

There may also be value in exploring the nature of collaboration between different stakeholders around induction. Some potential avenues could include:

- Exploring the role of professional communities and peer-support networks for novice teachers.
- Considering the development of feedback loops between schools and universities to allow insights from induction to be shared and fed into teacher education programmes to support alignment with the realities of current school practice.
- Exploring how to foster active involvement of Teacher Professional Development Centres alongside schools in supporting internships and supervision processes.

Ambition 2. Develop an infrastructure that incentivises and recognises excellence

Lever 4. A clear and attractive vertical career structure

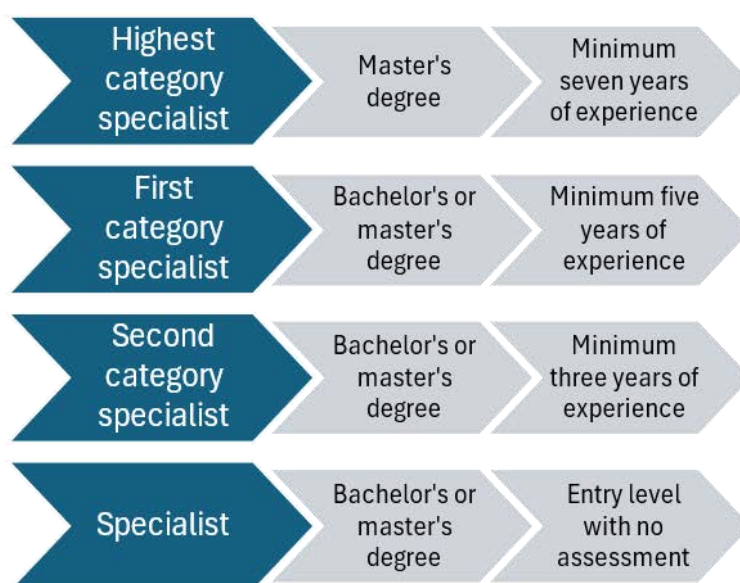
Ukraine's vertical career structure is shaped by qualification categories and pedagogical ranks. These are particularly important for remuneration, but also can entail different roles too.

Qualification categories

Ukrainian law stipulates that the system of qualification categories for teachers includes four main levels: Specialist, 2nd Category Specialist, 1st Category Specialist, and Highest Category Specialist (Figure 6.2) (Ministry of Education and Science of Ukraine, 2022^[16]).

The "Specialist" category is the entry-level classification, automatically assigned to pedagogical staff upon employment, without any assessment of their professional performance. This approach contradicts the essence of appraisal as an evaluation tool and raises questions about the very relevance of this category within the career structure. To be eligible for the specialist, second, and first categories, a teacher must hold a pedagogical qualification, while the highest category additionally requires a master's degree. The categories differ primarily in terms of required teaching experience, with minimum three years of teaching experience for the 2nd category, and minimum seven years for the highest category.

Figure 6.2. Qualification categories in Ukraine



Source: Based on (Cabinet of Ministers, Ukraine, 2022^[48]), *On approval of the Regulations on the certification of teaching staff*, <https://zakon.rada.gov.ua/laws/show/z1649-22#Text> (accessed on 7 October 2025)

Notably, regulations specify alternative pathways for individuals without a pedagogical qualification, but who have industry experience in another sector of the economy. Such individuals may be appraised without following the standard sequence of category assignment, provided they have at least two years of industry experience for the second category, five years for the first category, and seven years for the highest category. Pedagogical staff holding an educational and scientific degree, an academic degree, or an academic rank may be granted the highest qualification category directly, bypassing the sequential progression.

Qualification categories do not entail significant differences in the core professional duties of a teacher; teachers at all levels generally perform the same essential functions. The main distinction lies in the level of remuneration; teachers with higher categories receive higher salaries. However, teachers with the highest qualification category may also undertake additional roles, such as providing consultations to colleagues; supporting early-career teachers in their professional development; participating in scientific and methodological activities; developing and piloting their own instructional and methodological materials (CPDINSETT, n.d.^[49]).

Pedagogical ranks

According to the “List of Qualification Categories and Pedagogical Ranks of Teaching Staff” (Cabinet of Ministries of Ukraine, 2015^[50])⁹, there are also pedagogical ranks available to teachers: Senior Teacher and Teacher-Methodologist.

- A pedagogical rank can be awarded to teachers who:
 - Introduce and disseminate competency-based teaching methods and new educational technologies.
 - Provide professional support and assistance (e.g. mentoring, supervision) to other teachers.
 - Engage in processes aimed at ensuring educational quality and introducing pedagogical innovations and technologies.
 - Are laureates of regional, national, or international professional competitions or contests.

- Have prepared students who are winners of academic competitions or contests.
- The pedagogical rank “Senior Teacher” is awarded to educators who, following the results of a previous appraisal, have been assigned a qualification category not lower than “2nd Category Specialist” and have more than three years of teaching experience.
- The rank “Teacher-Methodologist” is granted to those with a qualification category not lower than “Highest Category Specialist,” a higher education qualification, and more than five years of work experience.

Pedagogical ranks play a practical role in career advancement: the “Senior Teacher” rank brings a 10% salary increase; the “Teacher-Methodologist” rank results in a 15% salary increase. These ranks also serve as a form of moral encouragement for experienced educators. They provide the foundation for horizontal career growth – becoming recognised experts in their subject area – and may open pathways to vertical advancement, such as movement into administrative roles, participation in educational initiatives, and greater competitiveness in the educational labour market.

One such vertical career pathway is progression to the role of Head Teacher (School Principal) in a general secondary education institution. To be eligible for this position, a candidate must be a citizen of Ukraine; possess higher education, at a minimum level of a master’s degree; have at least three years of teaching experience; be fluent in the state language; and demonstrate organisational competence (Unesco, 2024^[51]). Appointments to school principal positions are made via a competitive selection process, and the role is formalised through a six-year employment contract. While this process theoretically provides opportunities for career advancement, it represents only one part of a broader set of challenges affecting teacher career development in Ukraine.

Box 6.4. Research into teacher career paths

According to research by the civic association “Osvitoria” (Yurchenko, 2023^[52]), key challenges facing teacher career paths include:

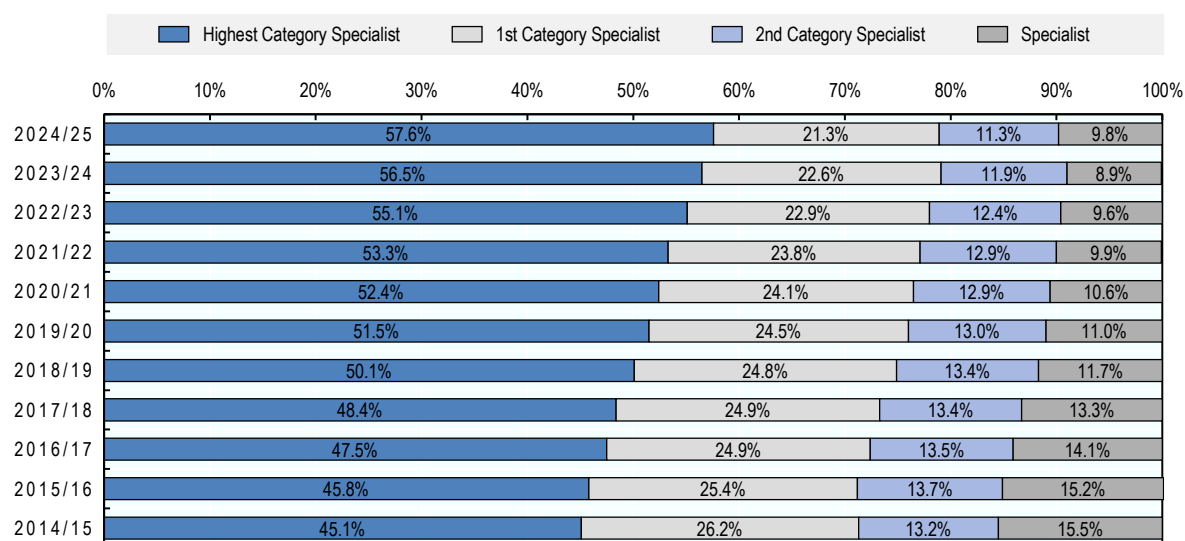
- **Overload and excessive bureaucracy:** Teachers experience a considerable burden due to numerous administrative tasks, mandatory involvement in events, competitions, and academic Olympiads. These responsibilities often divert attention from their core teaching duties.
- **Low salaries and limited financial incentives:** Even with additional allowances and bonuses, teacher salaries remain low, which diminishes the prestige of the profession and weakens motivation to remain in the school system.
- **Insufficient support for professional development:** There is a lack of access to high-quality training, mentoring, and supervision, especially for early-career teachers and in rural or under-resourced regions. Continuing professional development programmes are often delivered formally, without leading to meaningful results.
- **Psychological stress and burnout:** High levels of emotional and professional strain, exacerbated by the absence of consistent psychological support.
- **Unclear and limited career progression pathways:** Lack of transparent and structured career trajectories, with few opportunities for either horizontal development or vertical advancement. Career paths do not stimulate professional development and self-development.

Motivation and incentives

An important question surrounds the limited motivational impact of the vertical career progression system, as such systems tend to lose their incentive effect once teachers reach the “ceiling” of development. Over

57% of Ukrainian teachers, both in urban and rural areas, hold the highest qualification category (Figure 6.3). This indicates that the system is saturated with teachers who are formally classified as highly qualified. This phenomenon can be explained by a form of “compensatory mechanism”: the assignment of the highest qualification category serves as a means to increase wages within the rigidly fixed salary scale, which is legally tied to qualification categories.

Figure 6.3. Distribution of teachers with a master’s degree in Grades 1–11 (2014–2024), by qualification category



Source: Based on (Institute of Educational Analytics, n.d.^[53]), https://iea.gov.ua/wp-content/uploads/2025/05/byuleten-83-rvk-2024_2025.xlsx (accessed 10 October 2025); (Institute of Educational Analytics, n.d.^[54]), https://iea.gov.ua/wp-content/uploads/2024/02/byuleten-83-rvk-2023_2024.xlsx (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[55]), https://iea.gov.ua/wp-content/uploads/2023/05/byuleten-83-rvk-2022_2023.xlsx (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[56]), https://iea.gov.ua/wp-content/uploads/2022/03/byuleten-83-rvk-2021_2022.xlsx (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[57]), <https://iea.gov.ua/wp-content/uploads/2021/02/Inform.byuleten-pedagogichni-kadri-1.pdf> (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[58]), https://iea.gov.ua/wp-content/uploads/2020/06/Pro-chiselnist-i-sklad-pedagogichnih-pratsivnikiv-ZZSO-MON-Ukrayini-inshih-ministerstv-i-vidomstv-ta-privatnih-zakladiv_2018-2019_2019-2020-n.r.pdf (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[59]), https://iea.gov.ua/wp-content/uploads/2020/06/Pro-chiselnist-i-sklad-ped.-pratsivnikiv-ZZSO-MON-inshih-ministerstv-i-vidomstv-ta-privatnih-zakladiv_2017-2018-ta-2018-2019.pdf (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[60]), https://iea.gov.ua/wp-content/uploads/2020/06/Pro-chiselnist-i-sklad-ped.-pratsivnikiv-dennih-ZNZ-MON-inshih-ministerstv-i-vidomstv-ta-privatnih-zakladiv_2016-2017_2017-2018.pdf (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[61]), <https://iea.gov.ua/wp-content/uploads/2020/06/Pro-chiselnist-i-sklad-ped.-pratsivnikiv-dennih-ZNZ-MON-inshih-ministerstv-i-vidomstv-ta-privatnih-zakladiv-2015-2016-ta-2016-2017-n.r..pdf> (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[62]), https://iea.gov.ua/wp-content/uploads/2024/05/byuleten-83-rvk-2015_2016.xlsx (accessed on 10 October 2025); (Institute of Educational Analytics, n.d.^[63]), https://iea.gov.ua/wp-content/uploads/2024/05/byuleten-83-rvk-2014_2015.xlsx (accessed on 10 October 2025)

StatLink  <https://stat.link/yn6idx>

At the same time, the base salary for a specialist remains below the minimum wage in Ukraine, and it is only through a complex system of promotions and bonuses that teacher remuneration exceeds the minimum wage. This points to the wider challenges in the model of financial incentives in the career structure. In 2024, the average salary in the education sector was UAH 14 593 (approximately EUR 336), compared to the national average salary of UAH 21 473 (around EUR 495)¹⁰. Research by the civic association “Osvitoria” (Yurchenko, 2023^[52]) has suggested that even with additional allowances and bonuses, teacher salaries remain low, which diminishes the prestige of the profession and weakens

motivation to remain in the school system (see Box 6.4). As outlined in Chapter 3, financial incentives can be an important factor in shaping a career structure and the attractiveness of the teaching profession, alongside factors such as the ways teachers work and the opportunities that the profession provides for development.

Points of reflection for the future on clear and attractive vertical career structures

There could be value in sustaining and further coordinating any efforts to raise the social status of the teaching profession and ensure its importance is championed and promoted.

The current system of qualification categories seems to have a limited motivational potential, while the financial incentives of the system are complex and limited. As well as attending to ensuring the profession is intellectually attractive, some potential avenues going forward may include:

- A re-examination of the vertical career structure of teachers, with this linked to financial incentives. At the same time, it would be important in such a system to ensure that the transition to the next career step is based on an impartial performance evaluation of a teacher's work (see below, Lever 6), such as the certification that is currently being piloted. Such a basic certification could be mandatory for the transition from novice to certified teacher, or voluntary for advanced certification for the highest career level, which requires expertise in pedagogical skills.
- A re-examination of the system of remuneration may be one element to help shape the attractiveness of the profession. Article 61 of the Law "On Education" stipulates a base salary equivalent to three minimum wages for a teacher of a lowest qualification category (Verkhovna Rada of Ukraine, 2017^[11]). According to the law, salary increases may be implemented gradually to avoid a sharp jump and, consequently, a burden on the state and local budgets. Based on the law, it is necessary to set the wage starting with four subsistence minimums during the transition period and then increase it to the level of three minimum wages. It may also be necessary to revise the salary calculation system to account for diverse teacher responsibilities.

There could also be value in enlarging the provision of professional support for those progressing to more managerial or administrative roles in the career structure.

Lever 5. Opportunities to specialise through horizontal diversification

Ukrainian secondary school teachers do not officially hold 'parallel' positions with a different (diversified) set of job functions that would formally enable horizontal career progression through a change in professional specialisation. Instead, teachers often assume additional roles both within and outside their secondary education institutions, either paid or unpaid, and either voluntarily or due to 'operational necessity'. Let us consider some of these roles.

Mentors

Experienced teachers may serve as mentors within their schools, supporting newly-appointed teachers undergoing pedagogical internships. They may also provide supervisory support to colleagues from other educational institutions. These roles are typically remunerated. A key challenge in the implementation of these roles is the lack of systematic training to prepare teachers for such responsibilities.

Olympiads and tournaments

Ukrainian students in Grades 5 to 11 participate in Olympiads and tournaments – annual academic competitions across all major school subjects. These events are held in several stages: at the school, local, regional, and national levels (Ministry of Education and Science of Ukraine, 2024^[64]). Additionally, students aged 14 to 18 may compete in the All-Ukrainian Contest for the Defence of Scientific and Research

Projects, organised by the Small Academy of Sciences of Ukraine (Ministry of Education and Science of Ukraine, 2021^[65]). This competition also takes place in three stages: local, regional, and a final national round. According to the Ministry of Education, up to three million students (Ministry of Education and Science of Ukraine, n.d.^[66]) participate in these competitions each year, with several thousand reaching the final stage. Winners of these events often go on to represent Ukraine at international academic Olympiads and competitions.

Teachers play a vital role as mentors, coaches and academic supervisors to students participating in these intellectual contests:

- This role involves individual work with students to nurture them, the application of multiple professional competencies and participation in events for sharing experiences among teacher-mentors.
- Beyond student mentorship, teachers are also involved in organising and conducting the preliminary stages of competitions, preparing Olympiad materials and serving on judging panels.

Overall, this role represents both an opportunity and a challenge for students and teachers. These competitions aim to include all schools across Ukraine and are considered part of schools' annual reporting requirements to their governing bodies. Therefore, the state supports the organisation of these competitions, and teachers involved in their delivery gain access to specific professional development opportunities. However, some school administrations reportedly compel participation not out of motivation, but to meet reporting obligations (Matsala, 2024^[67]).

Educational experts

Successful teachers with five or more years of experience can become educational experts with the State Service for Education Quality of Ukraine:

- Upon completion of specialised training, they may participate in conducting institutional audits of secondary education institutions. An institutional audit is a procedure for external quality assurance in secondary education.
- Institutional audits are conducted regularly in each general secondary education institution, typically once every ten years. However, during martial law, regular audits are suspended; only unscheduled audits are carried out, either at the request of the educational institution or in response to complaints from participants in the educational process.
- During an audit, experts examine, evaluate, and provide recommendations for improving the educational environment, the system of assessing learners, pedagogical practices, and management processes within the institution (Ministry of Education and Science of Ukraine, 2021^[68]).
- Taking on the role of an expert within the institutional audit framework enables teachers to observe practices in schools from other regions of Ukraine and adopt useful approaches in their own work.

Resource developers

Teachers who create various pedagogical classroom materials, such as lesson plans, slide presentations, test assignments and classroom activities, can upload them to major specialist online platforms for educators. This allows their materials to be distributed and sold to anyone interested in accessing them.

The business model of these private educational platforms (the most well-known being 'Vseosvita' and 'NaUrok') is profit-orientated. Therefore, the platforms actively seek out new authors and a wide variety of content, creating a low barrier to entry for contributors. This peer-to-peer model gives educators access to a vast range of teaching resources across all school subjects and topics. At the same time, the quality of

such materials can vary significantly, and there are risks of plagiarism and other forms of academic misconduct.

Safety in a time of war

Following the full-scale invasion by the Russian Federation, many Ukrainian schools have equipped shelters on their premises to protect students, teachers, and other staff during air raids. In addition, Russian missile attacks have led to the destruction of energy infrastructure, resulting in widespread power outages across the country. During the winter months, this caused significant difficulties with heating and centralised water supply. As a response, the so-called “Points of Invincibility” have been established within the premises of individual schools (as well as other public buildings). These places are equipped with drinking water, heating, and electricity supplied by generators, providing a safe space for local residents during emergencies.

Teachers are often assigned the role of being on duty in these shelters and Points of Invincibility (Lykhovyd, 2022^[69]), including maintaining order and performing basic cleaning tasks. This work is generally unpaid and may extend beyond the teacher’s official working hours.

Private tutoring

A significant segment of the Ukrainian educational services market comprises private tutoring – supplementary individual lessons in school subjects. This is particularly prevalent among upper-secondary students preparing for the national university entrance examinations, which in 2025 included assessments in Ukrainian language, history of Ukraine, mathematics and one elective subject of the candidate’s choice (Ministry of Education and Science of Ukraine, 2023^[70]). Tutoring for primary school students is less widespread, typically aimed at re-mediating learning gaps.

For many teachers, offering private lessons in their subjects has become a customary way to earn additional income. Sessions are usually arranged on a one-to-one basis or in small groups of students. Foreign-language teachers, in addition to working with schoolchildren, often extend their clientèle to older learners. Payments for these services are predominantly off the record, though some educators register as individual entrepreneurs. In such cases, fees are paid into an official account and subject to a modest tax (6% of income as of 2025). Private tutoring has effectively become a forced stopgap for addressing students’ learning deficits, driven both by teachers’ low salaries and their need for additional income, and the absence of a formal tutoring role within state secondary schools that would integrate individual support to students with educational losses into the guaranteed educational provision.

Points of reflection for the future on opportunities for horizontal diversification

There are already some horizontal career pathways for Ukrainian teachers, with such horizontal diversification typically consisting of teachers assuming additional roles – both within and beyond their educational institutions – both for a fee and free of charge, and with varying degrees of voluntariness. Some potential avenues to further build on this going forward include:

- The range of horizontal career roles could be broadened by clearly listing the necessary additional roles that involve additional payment or a reduction of the lesson loads, such as a mentor teacher in an internship, a tutor or a head of a targeted training for other teachers, as well as those that do not involve additional payment (e.g. a head of a methodological teacher association, a supervisor).
- Introducing training programmes for additional roles of teachers may help to increase their effectiveness. This could also include the option for awarding relevant micro-credentials that recognise the additional role more formally.

Lever 6. Trusted evaluation mechanisms

The assessment of professional activity is a key component in ensuring the quality of education and professional development of teachers. In Ukraine, two primary forms of assessment are in operation: mandatory appraisal (Ministry of Education and Science of Ukraine, 2024^[71]) and voluntary certification. These mechanisms are designed to stimulate teachers' professional growth, enhance their skills, and, ultimately, improve the quality of the educational process. However, each form of assessment has its own characteristics, strengths, and limitations, which must be examined carefully within the context of contemporary educational challenges.

Mandatory appraisal

The purpose of appraisal is to promote continuous professional learning, enhance teachers' professional competence and skills, foster creative initiative, raise the status and prestige of the teaching profession, and ensure the overall effectiveness of the educational process. The key principles of appraisal include transparency and collegiality, a humane and respectful approach to teaching staff, and the thorough, objective and systematic evaluation of teaching performance.

Teacher performance in Ukraine is evaluated according to the professional teaching standard, which sets out the requirements for general and professional competencies of a teacher. The appraisal process is governed by the Regulations on the Appraisal of Teaching Staff, approved by Order No. 805 (Ministry of Education and Science of Ukraine, 2022^[16]) of the Ministry of Education and Science of Ukraine, dated 9 September 2022. These Regulations establish the procedure for appraisal as a structured system of measures aimed at conducting a comprehensive and objective assessment of teaching performance.

Appraisal is a mandatory procedure that teachers must undergo at least once every five years, although it may be conducted more frequently at the teacher's request or at the initiative of the school principal – particularly in cases where a decline in the quality of teaching is observed. Appraisal involves the awarding or confirmation of an existing qualification category or pedagogical rank. This is based on a thorough review of the teacher's performance, professional competencies, educational qualifications, and work experience. Awarding/confirmation of these categories and ranks directly influence the teacher's salary and represent the primary means of increasing it. Appraisal commissions are established at the school level for the appraisal of teachers, and at the level of the school's founding body for the appraisal of school principals.

Appraisal commissions evaluate the professional competencies of teaching staff taking into account their job responsibilities and the professional standard requirements. Teachers may voluntarily submit documents that demonstrate their pedagogical skills. If necessary, the commission appoints experts to conduct a detailed study of the teacher's practical activity, including observation of classroom activities as well as evaluation of documentation and methodological materials. The commission also assesses professional achievements, such as: implementation of competency-based teaching methods, mentorship and supervision of colleagues, participation in innovative quality assurance projects, success in professional competitions at regional, national, and international levels, preparation of winners of various student Olympiads and contests.

Voluntary certification

Certification is a voluntary procedure that enables teachers to demonstrate their professional excellence and receive corresponding benefits (State Service for Education Quality of Ukraine, 2024^[72]; Cabinet of Ministers of Ukraine, 2025^[73])¹¹. The purpose of certification is to identify and support educators with a high level of pedagogical expertise who apply competency-based learning approaches and innovative educational technologies, and promote the dissemination of these practices. It is an external assessment of the professional competencies of a pedagogical worker, carried out through independent testing, self-assessment and study of practical work experience (Osvita.UA, 2024^[74]).

The certification process consists of three consecutive stages (Ukrainian Center for Educational Quality Assessment, n.d.^[75]):

1. The first stage is independent testing conducted by the Ukrainian Centre for Educational Quality Assessment (UCEQA) and its regional branches, this stage requires participants to achieve at least 60% of the maximum possible score in order to proceed.
2. The second stage is self-assessment, when participants reflect on and evaluate their own pedagogical skills. The process is coordinated by the State Service for Education Quality of Ukraine (SSEQ) and its regional offices.
3. The third stage is review of practical teaching experience of certification participants, also conducted by the SSEQ. This stage involves an expert review of the teacher's real-life classroom practice. Typically, this includes peer evaluation (i.e. 'teachers assessing teachers').

Certification offers several benefits compared to traditional appraisal (Makarova, 2021^[76]).

- One is the impartiality of the assessment, because certification involves independent testing and evaluation by experts, which ensures a higher degree of objectivity compared to appraisal.
- A further benefit is its comprehensive approach – unlike traditional appraisal with its focus on documentation, this certification process places a greater emphasis on the practical review of a teacher's performance in the classroom.
- There are also attached financial incentives – teachers who successfully complete the certification are entitled to a 20% salary bonus.
- Besides, certification serves as formal recognition of a teacher's professional excellence. It is considered equivalent to passing the traditional appraisal and may be used as a basis for assigning or confirming qualification categories or pedagogical ranks.

Recent survey results from 2023–2024 show a generally positive perception of voluntary certification among both participants and expert assessors (New Ukrainian School, 2025^[77]):

- 90% of surveyed teachers believe that their level of professional skill was assessed objectively by experts.
- 79% of surveyed teachers would recommend the certification process to their colleagues.
- 63% of surveyed teachers expressed a desire to undergo certification again.
- 92% of expert assessors reported a positive experience in taking part in the process.
- 94% of expert assessors stated they would like to continue serving in this role in the future.

Implementation of certification and appraisal

Despite the recognised advantages of voluntary certification, appraisal remains the primary mechanism for assessing pedagogical performance in Ukraine. However, critics argue that the appraisal process is often overly formal and does not always reflect a teacher's actual level of professional competence. This leads to a decline in trust in this evaluation mechanism (FACT News, 2024^[78]).

One drawback of certification is the limited number of participants, owing to the high cost of the procedure and the shortage of trained expert assessors. In 2025, only 5 000 out of 13 000 Ukrainian teachers will be able to take part in certification: 1 500 primary school teachers; 1 000 mathematics teachers; 1 000 Ukrainian language and literature teachers; 1 000 history and civics teachers; 500 English teachers. Participation quotas have been set for each region of Ukraine. For instance, the total quota for the Rivne region is 224 teachers, including: 63 primary school teachers; 45 mathematics teachers; 47 Ukrainian language and literature teachers; 45 history and civics teachers; 24 English teachers (Rivne Institute of Postgraduate Pedagogical Education, 2024^[79]).

The presence of quotas points to capacity challenges for large-scale implementation. The structure of the current certification stages may be revised to help resolve this situation. For example, the self-assessment stage, where teachers evaluate their own pedagogical skills, could be optimised through the introduction of digital tools such as the Information and Analytical System “Evalued” (New Ukrainian School, 2024^[80]). This would necessitate an expansion of the system’s capabilities.

Meanwhile, it is also necessary to address the shortage of expert teachers. This may intersect with the career structure, outlined above, as well as ensuring sufficient financial support for their activities (e.g. remuneration, travel expenses school staff replacements during their absence). The presence of expert teachers is particularly important for effectively implementing a peer-to-peer approach at the stage of evaluating a teacher’s practical experience.

Points of reflection for the future on trusted evaluation mechanisms

The current attestation process is hindered by several shortcomings, including a lack of clear criteria and requirements, a lack of depth and a reliance on potentially subjective measures, and a bureaucratic burden on teachers in collecting and processing documentation. Since 2018, a pilot independent evaluation of teachers through certification has been introduced, incorporating external independent testing, self-assessment of the certification participant’s own pedagogical skills, and review of their practical teaching experience. Despite its promise, certification remains costly and complex for organising institutions and, accordingly, is available for a limited number of possible participants.

In light of this, there may be value in exploring alternative approaches to how attestation and certification are undertaking, which would also encompass the transitions in the vertical career structure. One option to consider may be a two-stage approach:

- First could be improving the certification mechanism. In the present-day situation of teacher professional development in Ukraine, improving the current certification system and expanding opportunities for teachers of different subjects are needed to support voluntary independent certification.
- A second stage could be certification replacing attestation. In a context of increased prestige of the profession, two types of certifications could be considered: basic certification, which would be mandatory for the transition from novice to certified teacher and for periodic confirmation of this career status; and advanced certification, which would be voluntary and would provide for the transition to a higher career level. Here, we see certification as a non-mandatory growth opportunity allowing a person to make a voluntary choice about his or her future career.

This type of certification could be orientated around the NUS and its demands. It could endeavour to cover the teacher’s impact and involvement at the national level contributions, as reflected in the portfolio (e.g. the introduction of evidence-informed practices, work on the development of a regulatory framework, or creation and field testing of textbooks) to gradually replace certification with voluntary certification, as well as to expand the list of categories of teachers who are certified.

It may be possible to explore a more comprehensive system of evaluation criteria of teacher’s practice. For instance, covering teaching quality and other professional activities. It may take the form of:

- Diversifying the tools for evaluating the quality of teaching.
- Introducing reflective practices (similar to the experience of Singapore), including ongoing and end-of-year interviews with school leadership.
- Promoting self-reflection and self-assessment based on the national teacher professional standards to support teacher’s own individual professional development planning.

At the same time, it is worth noting the importance of considering administrative burden throughout an evaluation process, with this being one of the challenges in the current appraisal system.

Ambition 3. Facilitate quality, meaningful ongoing learning that enables the profession to grow and face future challenges

Lever 7. Provision of relevant learning opportunities

A network of dedicated institutes is the primary institutional arm responsible for ongoing professional learning for in-service teachers in Ukraine. Institutes of Postgraduate Pedagogical Education (IPPEs) operate in every regional centre of Ukraine and serve their respective regions. This stable network also functions as a tool for implementing educational policy, as it can reach a wide educational audience. As part of the implementation of the New Ukrainian School reform, mass professional learning for primary schools was carried out using this network, with specially trained staff prepared for the purpose.

Until 2017, the Ukrainian system operated according to principles common in post-Soviet countries: centralised planning, limited choice of service providers – since the IPPEs held a monopoly in this field – and episodic training. Licensing conditions served as artificial barriers that prevented private providers from entering the education services market. This contradicted global trends towards diversification and competition in teacher professional development.

Following an update to education legislation in 2017, the market for professional development was de-monopolised – teachers were granted the right to choose their providers, and the range of organisations eligible to offer training was expanded. Since then, training may be offered not only by licensed IPPEs, but also by unlicensed “other providers of educational activity, including individual entrepreneurs and legal entities” (Resolution of the Cabinet of Ministers of Ukraine, 2019^[81]). As a result, the professional development market has expanded significantly. Current providers include public organisations, private companies, and individuals – often teachers themselves – who design and deliver training based on their own methodological approaches. Today, Ukrainian teachers may enhance their qualifications either through the IPPE network or other providers. In the case of the latter, due to the absence of a licence, training outcomes must be validated by a separate decision of the school’s pedagogical council.

Alongside this de-monopolisation, an annual obligation for professional development was introduced and its required volume increased. According to the Law “On Education,” the amount of training guaranteed and funded by the state has doubled and now amounts to a minimum of 150 hours over five years.

The opening of the professional development market has broadened both the content and the formats of training programmes. The public sector has introduced international programmes and practices to support Ukrainian teachers, including in the area of social and emotional learning (EdCamp DoSEN, n.d.^[82]). Online formats, developed by both the state and public sector, have expanded, playing a crucial role in supporting the education system during the COVID-19 pandemic and the full-scale war. Moreover, professional development platforms now exist in Ukraine to centralise access to professional learning opportunities (Box 6.5).

The financing system for professional development has also been updated and diversified. Funding sources now include state and local budgets, contributions from educational institutions, teachers’ personal funds, and support from individuals and legal entities. As for state funding, a targeted subvention for teacher professional development was introduced as part of the implementation of the New Ukrainian School reform. The law also established the principle of ‘the money follows the teacher’ (New Ukrainian School, 2025^[83]), meaning that budget funds should be allocated directly to teachers, who then decide how to spend them on professional development.

However, this principle has not yet been implemented in practice. At present, an annual subvention from the state budget is allocated for professional development, but it is not fully utilised. In practice, funds allocated for advanced training within the framework of the targeted subvention for the New Ukrainian School are distributed exclusively among regional IPPEs, in proportion to the number of teachers

responsible for the completion of Grades 5–11(12) of general secondary education (Ministry of Education and Science of Ukraine, 2025^[84]). The institutes lack sufficient capacity to fully utilise these funds. For example, in 2024, UAH 100 million was allocated for the professional development of teachers under the subvention, but only 52% of the amount was actually used (Ukrinform, 2025^[85]). At the same time, public funds are not allocated to private providers of professional development services for teachers, which may have the potential to help provide more up-to-date professional learning content for more participants or support alignment between teachers' needs and content.

Box 6.5. National Platform for Professional Development Opportunities for Teaching Staff

One effort to centralise access to training opportunities is the National Platform for Professional Development Opportunities for Teaching Staff, known as EdWay, developed by the public organisation EdCamp Ukraine. Since 2021, the free-to-use platform has brought together over 67 000 users and 315 training providers, offering 1 286 professional development opportunities. Platform users can filter options by specific competencies in the teacher professional standard, as well as select preferred formats and training volumes.

To expand access, the EdWay platform is in the process of being transferred to state ownership and integrated into the digital ecosystem “Mriia” (EdCamp Ukraine, n.d.^[86]), a unified platform and database of opportunities for all education system participants being developed by the Ministry of Digital Transformation and the Ministry of Education and Science of Ukraine.

The EdWay platform has established terms of use for all users (EdWay, 2021^[87]), including providers submitting advanced training programmes. When submitting a programme, an individual or organisation must complete a form aligned with the requirements of Resolution No. 800 (Cabinet of Ministers of Ukraine, 2019^[3]). Platform moderators then review the submission and either approve or reject it based on compliance with these standards. Additionally, the terms of use stipulate that programme content must be accurate, reliable, non-discriminatory, non-deceptive, ethical and must not infringe upon intellectual property rights. Nevertheless, the platform disclaims responsibility for the accuracy or completeness of the information provided by training providers.

Points of reflection for the future on provision relevant professional learning opportunities

The Law of Ukraine “On Education” has laid the groundwork for broadening opportunities for professional development of teachers (Verkhovna Rada of Ukraine, 2017^[1]). Since 2017, a targeted subvention for in-service professional development has been introduced, and teachers have more choice in choosing providers and determining their learning pathways. There are also digital platforms now in place to help structure professional development offerings. Going forward, further consideration may be given to:

- The type of structures that allocation the educational subvention for teacher professional development. At present, the current IPPE network, which is the sole recipient of this targeted subvention, lacks the capacity to deliver the full volume of required in-service training opportunities, and alternative structures may support greater reach and flexibility.
 - One option for distributing the subvention for professional leaning could be to split it into two parts. One of them could be allocated to the network of IPPE to support targeted courses related to the implementation of current national educational policy. The second part, which is implemented on the principle of ‘funding follows the teacher’, is for teachers to take courses with other providers. Prior to the doubling of the minimum number of hours to 150 over five years, in-service training was provided only by IPPEs, with a minimum of 75 hours per five years for one teacher. Thus, the distribution of the education subvention in the ratio of 50%, in

particular, for mandatory targeted courses, usually in the IPPE, and 50% for teacher's choice is logically justified.

- An alternative option would be to allocate a portion of the education subvention through a tender-based process among providers, which is in line with the approach in England (United Kingdom). In this case, it is important to open access to these tenders to all providers who are able to prove the quality of their offering.
- A third option is that the entire education subvention is distributed on a 'funding follows the teacher' model. This mechanism can be ensured through the use of a unified digital platform of quality professional learning offerings, if, within the platform's functionality, teachers have the opportunity to receive funds for professional learning in their personal accounts and use them for the quality offerings they need. The same approach is used for the first option, when part of the educational subvention is distributed on the principle of 'funding follows the teacher'.

Underlying approaches to the models of the educational subvention above is the question of balancing alignment between individual needs, school needs, and system needs. It means that monitoring of alignment across these may be important, as set out in Chapter 4. Similarly, the question of the quality of delivery is ever-present, and is discussed more below.

An upgrade of the technological infrastructure, in particular in rural areas, and enhancement of teachers' digital competencies may be significant for enabling access to professional development opportunities in remote and blended learning during wartime.

Lever 8. Delivery of quality learning opportunities

In Ukraine, the requirements of advanced training programmes for teachers have been clearly stated in 'Resolution No. 800' (Cabinet of Ministers of Ukraine, 2019^[3])¹². Key areas of professional development, including the enhancement of professional competencies, the development of students' cross-curricular skills, the creation of inclusive environments, the improvement of digital and informational literacy, and the advancement of managerial competencies for educational leaders and administrators. With regard to the content of these programmes, governmental requirements are largely technical, focussing on elements such as volume, format, types of activities, and duration. Resolution No. 800 also stipulates that programmes must specify the competencies teachers will acquire, as well as the "academic and professional opportunities resulting from completion of the programme". However, the resolution does not define any criteria for assessing the quality of programme content.

According to Resolution No. 800, the only procedure for assessing the quality of advanced training programmes of non-licensed providers is through recognition of their results by the school's teachers' council. However, such recognition is often granted formally, without rigorous analysis of programme effectiveness. Moreover, programmes offered by licensed institutions, such as IPPEs, do not require this recognition. In practice, the responsibility for quality assurance rests with the providers of advanced training, or with individual organisations that independently establish certain norms and standards (see Box 6.5).

The de-monopolisation of the advanced training market in Ukraine has partially contributed to improving the quality of training programmes. A significant reform was the transformation of district methodological centres into centres for the professional development of pedagogical staff (CPDPSs), which now operate alongside private entities, fostering competition. However, this reform does not eliminate the need to build the professional reputation of service providers, introduce micro-credentials as a credible quality assurance tool, and establish a unified system of quality criteria applicable to all training providers. In sum, in the context of deregulation, the removal of licensing barriers, combined with the annual requirement for teachers to undertake professional development has created a complex environment. The mere provision of services does not guarantee their quality.

More broadly, there is also a significant disconnect between research-backed characteristics of effective professional development and the actual design and delivery of most programmes in Ukraine (see Ambition 4). This issue is exacerbated by a lack of data on the effectiveness of various training formats, limited integration of international research findings into local practice, and an absence of systematic evaluation of how professional development impacts students' academic outcomes.

Points of reflection for the future on the delivery of quality professional learning opportunities

A challenge within the professional learning landscape is the variable quality of in-service trainings as there is currently no comprehensive system for assuring the quality of in-service training offerings. This also means that professional learning may not always meet the current priorities of the New Ukrainian School reform and the evolving demands of wartime education. Some of the potential avenues that could be further explored include:

- Ensuring uniform quality standards are in place to guide delivery, and to support any efforts to monitor and evaluate the delivery. Integrating research-backed characteristics of effective professional development into the provision will be important here.
- Considering mechanisms to assess providers' reputation and the quality of delivery.
 - The voice of the profession may be important here, such as the evaluation of professional learning by participants, or the monitoring by a dedicated expert group of stakeholders (e.g. school leaders). This may also systematically consider not only the content of the programme when assessing the quality of in-service training courses, but also the methods of implementing the educational process.
 - Introducing micro-credentials may be fruitful as a tool for quality assurance and recognition of skills acquired through professional development offerings. Only those educational in-service training programmes that offer and implement systematic assessment of newly acquired professional skills could be eligible to award micro-credentials. To this end, it is also necessary to develop an appropriate regulatory framework and operational mechanisms for integrating micro-credentials into the system of professional development of teachers.
 - A digital platform could play a role too. To provide quality assurance, such a platform should not only display a list of professional development offers from various providers, but also include the possibility of expert verification of the quality of the educational programme implemented by the provider. The latter is a next step in updating the functionality of EdWay platform (see Box 6.5).

Strengthening the role of the network of Centres for Professional Development of Educators (CPDEs) in communities could be helpful in supporting teachers and educational managers in identifying quality learning offerings, facilitating the creation of individual professional development plans aligned with the national teacher professional standards, and collecting feedback from teachers on the quality of providers' courses. The school principals plan the professional development and reflect on it, and thus could provide input and feedback when the teachers choose their professional development courses.

Lever 9. Incentives to engage in professional learning

Several legislative mechanisms underpin teachers' participation in professional development activities:

- Article 59 of the Law of Ukraine "On Education" stipulates the continuous participation of teaching staff in advanced training programmes (Verkhovna Rada of Ukraine, 2017^[1]). As previously mentioned, according to Resolution No. 800, the total volume of professional development for teaching and academic-pedagogical staff in general secondary and vocational (vocational-

technical) education institutions must amount to at least 150 hours over a five-year period (Cabinet of Ministers of Ukraine, 2019^[3]).

- The law also guarantees state funding for teachers' advanced training (Cabinet of Ministers of Ukraine, 2018^[88])¹³.
- The law also allows teachers to independently choose the form, type, direction, and provider of educational services for their advanced training (Verkhovna Rada of Ukraine, 2017^[1]).

Both material and non-material incentives also play a role in motivating teachers to enhance their knowledge, skills, and competencies.

- Material incentives include salary increases linked to career progression. Completion of professional development programmes can lead to the attainment of a higher qualification category, which in turn increases remuneration.
- Non-material incentives include opportunities for networking and engagement in professional communities (Verkhovna Rada of Ukraine, 2017^[1]). These networks facilitate experience-sharing and provide psycho-emotional support.
- Further motivational factors include responding to changes in the general secondary education system following the implementation of the New Ukrainian School reform, as well as the challenges presented first by the COVID-19 pandemic and later by the full-scale invasion of Ukraine by the Russian Federation. Such events have necessitated the acquisition of new knowledge, skills, and adaptive capacities, prompting many teachers to pursue further training (see Box 6.7).

A study on the implementation of the NUS reform in pilot educational institutions included an analysis of teacher motivation for professional development (State Research Institution "Institute of Educational Analytics", 2021^[89]). Notably, 63.8% of teachers cited self-realisation as the main motivating factor, pointing to a hunger to learn among many in the profession. The importance of financial motivation also grew significantly: in 2021, 41.0% of teachers linked advanced training to the opportunity for higher earnings. There has also been a marked increase in interest in career progression and professional growth based on autonomy and individual educational needs. Most teachers evaluated changes to the professional development system positively, particularly in relation to the ability to choose among service providers, the flexible format of distance learning, and the practical relevance of courses. These innovations not only expanded access to educational opportunities but also became essential tools for self-education and self-realisation, reinforcing teachers' sense of professional autonomy. However, not all teachers were equally engaged in the process, highlighting the need to strengthen intrinsic motivation, particularly through improvements in quality, accessibility and formal recognition of professional growth (State Research Institution "Institute of Educational Analytics", 2021^[89]).

Box 6.6. Pilot projects under the NUS framework

Participation in pilot projects under the NUS framework has also served as a powerful stimulus for teachers' professional development. For example, in 2025, as part of the preparation for 'Academic Lyceums' at the upper secondary level – the final stage of the NUS reform implementation – principals of 30 pilot lyceums took part in a training programme for educational managers. The number of these pilot lyceums will be widened to some 150 in 2026.

This programme was developed by the savED charitable foundation in partnership with the Ministry of Education and Science of Ukraine, with support from the international NGO Finn Church Aid Ukraine (FCA). The OECD has also facilitated international partnerships between Ukrainian pilot lyceums and international schools as a means of support, knowledge exchange and knowledge creation.

The programme included modules on educational human resource (HR) management, fundraising, strategic project management, psycho-emotional support, career guidance for adolescents, analysis of international education reform cases and marketing communications (Ministry of Education and Science of Ukraine, 2025^[90]).

Points of reflection for the future on incentives to engage in professional learning

Education reform and the challenges of war have created an urgent need for Ukrainian teachers to acquire new skills and competencies, and in some cases there are signs of teachers seeking out learning opportunities to grow in line with these new demands. Ukrainian teachers have a number of external and internal incentives to participate in professional learning. A key challenge remains ensuring that teachers see the 150 hours of in-service training guaranteed by Ukrainian law as an opportunity, rather than a burdensome obligation. Some of the future avenues that could be explored in relation to this include:

- Ensuring alignment between professional learning and meeting the demands of the New Ukrainian School reform and overcoming the challenges of war gain could be impactful, helping to ensure professional learning is attractive and viewed by teachers as relevant to them.
- A voluntary teacher certification may be a tool to support career progression. If professional learning can be means of attaining the necessary competencies for this, it could motivate engagement. Hence, it is not participation in professional learning that is assessed, but the skills professional learning builds which is important for real but not formal professional development.
- In particular, building on this point, micro-credentials may be beneficial as a structured assessment tool that validates the person's acquisition of particular new skills during the process of obtaining a qualification. Courses that end with such assessment and lead to achievement of micro-credentials could serve as a basis for assuming additional roles within a teacher's horizontal career progression.
- Digital services and platforms could play significant role as a motivational tool for further professional growth. For example, a teacher's profile on the platform, with their consent, could display micro-credentials they have earned, which would nurture their career progression and be considered for new roles.

Ambition 4. Create an ecosystem for collective learning and sharing to ensure opportunities for excellence for all

Lever 10. A culture of collaborative inquiry within and across schools

Fostering a culture of professional inquiry within teacher communities requires the establishment of systemic conditions that integrate collaboration and reflection systematically into the daily work of schools. Such a culture does not emerge spontaneously. It is cultivated through clearly defined procedures, supported structures, and strategic management decisions that transform inquiry from a sporadic activity into a consistent educational norm in schools. It is also supported by access to high-quality sources of knowledge, as set out in the next two levers.

Research has suggested that one key barrier to achieving cultures of collaborative inquiry within schools is the limited implementation of pedagogical autonomy. According to the Center for Innovation in Education, although autonomy exists in formal terms, teachers are often unable to exercise it effectively due to time constraints, insufficient professional competencies, and a lack of effective motivational mechanisms (Pro.Svit, 2020^[91]). This underscores the need for comprehensive support from educational management bodies to create conditions that enable both the development of autonomy and the establishment of a research-orientated culture within every educational institution.

This is reflected in the fact that few teachers create their own educational resources or have experience in publishing in professional publications, as reflected in the 2024 institutional audit of schools conducted by the State Service for Education Quality of Ukraine (SSEQ). Only about half of teachers have such practices, and even fewer are engaged in methodological development and dissemination of their own experience.

Significantly, there are examples of impactful cultures of collaborative inquiry being fostered in some schools. As part of the piloting of the New Ukrainian School reform in upper secondary education, launched on 1 September 2025, 30 lyceums were selected to jointly research and try updated approaches to the organisation of the educational process. These approaches include the development of individual learning trajectories and subject specialisation for students in Grades 10–12 (Ministry of Education and Science of Ukraine, n.d.^[92]). The pilot initiative is accompanied by ongoing monitoring and scientific support, as one of its objectives is to scientifically justify and experimentally verify the educational and methodological materials aligned with the new educational standards (Lviv Regional Institute of Postgraduate Pedagogical Education, 2025^[93]). A pre-pilot phase has supported the preparation for this work, including with the OECD facilitating partnerships between participating pre-pilot Ukrainian schools and international schools, which will be further expanded. These types of partnerships with international schools as well as the connections between schools aim to foster more inquiry and exchange within and between schools.

Nevertheless, a widespread culture of collaborative inquiry remains largely underdeveloped. This is due to insufficient training among school leaders and teachers, and the lack of necessary resources to facilitate structured collaboration.

Points of reflection for the future on nurturing cultures of collaborative inquiry within and across schools

As a result of different pilots of the NUS reform, active professional communities of teachers began to form, sharing experiences, establishing horizontal connections and sharing common values. At the same time, there is a need to foster a culture of collaborative inquiry and horizontal professional communication more widely. Going forward, some potential avenues to explore could include:

- Ensuring that schools have the resources (e.g. time, financial resources) to foster this culture, including access to high-quality digital sources, as well as the know-how and skills for building inquiry will be relevant here.
- Nurturing in-service training opportunities for collaborative inquiry and mutual pedagogical reflection for school leaders who organise horizontal professional development. School leaders are the main agents of change and innovation at the local level. If teaching staff are ready for change, but the school leader does not support them, it demotivates the teachers.
- Promoting the development of potential leaders of professional learning communities to implement innovations in teaching practice and for scaling up, development of subject associations that are designed to improve educational processes, similar to the experience of Estonia. Similarly, fostering school-level cultures through specific in-service training opportunities not only at the individual level, but also at the level of teams of schools to build capacity in joint growth of development strategies and plans for implementing changes at the teaching staff level.

More broadly, the presence of collaborative inquiry may be an expectation in any school development plans or become one of the criteria for the quality of an educational institution, which is monitored by the State Service of Education Quality of Ukraine. In particular, inquiry could be orientated around the implementation of the NUS (e.g. competency-based teaching) or responding to particular challenges from Russia's war of aggression (e.g. educational losses). This may also enable the disseminating of practices that demonstrate successful outcomes and provide feedback to education authorities on content components and methodologies that need to be further attended to.

Lever 11. Developing engagement with research

The development of culture of using scientific research within Ukrainian schools is a key prerequisite for the meaningful renewal of educational content. Evidence from scientific research can provide valuable knowledge for adapting practices as well as supporting processes of iterative innovation and refinement.

Promisingly, in pilot institutions testing new state standards, a research-informed culture and an empirical approach to planning educational changes is gradually emerging. Innovative educational activities are typically carried out through pedagogical experiments or innovative educational projects conducted at national or regional levels in line with the regulations approved by the Ministry of Education and Science of Ukraine (Ministry of Education and Science of Ukraine, 2000^[94]; Ministry of Education and Science Ukraine, 2023^[95]). They involve collaboration with research institutions, and data are gathered through pedagogical research based on the outcomes of these initiatives.

A notable example is the piloting of the New Ukrainian School reform at primary level, which began in 2017 and involved 143 schools across the country. The piloting has since expanded to include lower secondary schools (Grades 5–9) and specialised upper secondary schools. The Institute of Educational Analytics regularly conducts monitoring studies covering various aspects of the NUS pilot, including the quality of educational and methodological support, the usage of teaching aids and classroom equipment, the effectiveness of teacher training, the assessment practices for student learning outcomes, and the overall perceptions of the pilot process among stakeholders (Denysiuk et al., 2024^[96]).

There are also growing numbers of teachers who are actively involved in data collection, analysis and interpretation for both national and international research studies such as PISA, TALIS, SSES, as well as initiatives coordinated by the SSEQ and the Institute of Educational Analytics. Schools increasingly serve as testing grounds for innovative methods and tools, including: methodologies, digital solutions, blended learning models, and national platforms such as the All-Ukrainian Online School and “Mriia”. Participation in these types of initiatives can help foster more research literacy and engagement among teachers and schools.

However, participants in the 2024 institutional audit of schools conducted by the SSEQ noted that the vast majority do not cooperate with research institutions and do not implement projects or experiments of the all-Ukrainian or regional level (e.g. Democratic School, Innovative Schools of the Cherkasy Region, or piloting of new educational standards) (Verhun, 2025^[97]). Only 8% of institutions participate in all-Ukrainian or regional educational experiments. More broadly, fostering a culture of research engagement where research is a key part of the conversation remains a widespread challenge still. The audit found that the issue of participation in research and innovation activities is discussed at meetings of pedagogical councils in only 23% of schools (Verhun, 2025^[97]).

The challenge of building a widespread culture of research engagement in Ukrainian schools is reflected in the ongoing work to bring what happens in classrooms in line with some of the pedagogical approaches that have been advocated for in research studies. In most institutions, teachers do not employ pedagogical tools aimed at developing key competencies, such as level differentiation, STEM education, robotics, and project-based or inquiry-based learning. Instead, traditional teaching approaches continue to dominate, limiting students' ability to apply their knowledge in real-life contexts (Verhun, 2025^[97]).

This trend is further reflected in teachers' selection of educational materials, particularly those with an integrated component. "A textbook may be considered integrated if it facilitates the co-ordination and unification of separate elements within the curriculum into a coherent whole within a single subject" (Zasiekina, 2020^[98]). Such textbooks combine the content of several traditional subjects around a common theme or competence, which contributes to a holistic perception of knowledge, and their innovation is manifested in the combination of printed materials with interactive online tools, multimedia content, educational games and adaptive exercises. Despite the availability of such resources, the majority of teachers still prefer traditional textbooks, which often fall short of competency-based education standards. Notably, integrated textbooks, designed to promote interdisciplinary connections and critical thinking, failed to surpass 10 000 copies during competitive selection¹⁴ (Ministry of Education and Science Ukraine, 2025^[99]), indicating a low level of readiness among the teaching staff to adopt innovative methods.

Thus, although Ukrainian schools are increasingly actively involved in scientific and analytical initiatives, there remains a need to go from pockets of excellence to a position where the entire system is nurtured by rich, meaningful engagement with research. Support from management structures and research institutions, as well as the further development of a professional community of teachers capable of critically reflecting on research findings, are all important going forward.

Points of reflection for the future on developing engagement with research

Developing infrastructure between stakeholders around research may help foster more engagement among teachers and schools. For instance, some avenues that could be considered include:

- There may be some advantages to networks of research schools that conduct research in partnership with universities and other research institutions, in a similar vein to England (United Kingdom). This could also see collaboration between clusters of schools, research institutions, in particular Institutes of the National Academy of Educational Sciences of Ukraine, Institute for Educational Analytics, Ukrainian Institute of Education Development and departments in the Ministry of Education and Science to work on finding solutions to particular issues and can disseminate their findings and experience.
- Leveraging the 'early adopters' of change, in particular, pilot schools in the implementation of the New Ukrainian School reform, which implement field testing of innovations and new approaches, share their experiences and motivate others. Indeed, ensuring that there are feedback loops for the research work of expert teachers to contribute to wider professional learning may be relevant, as per Finland's experience.

- It could be that certain digital tools, such as platforms, can engage various actors in educational projects. For instance, this may be helpful in securing funding for particular projects or facilitating collaboration between stakeholders.

Effective research infrastructure is dependent on quality research to work with. An area of further exploration may be how to enhance the supply of quality evidence for teachers to draw upon, such as through synthesis work. For instance, it could be a case of expanding the capabilities of Ukrainian institutions that already provide educational monitoring and research, such as the Ukrainian Centre for Educational Quality Assessment, the State Service of education quality of Ukraine, the Institute of Educational Analytics. In particular, it may be useful to expand the range of their research and include synthesis work on teaching methods and technologies, similarly to England (United Kingdom)'s experience.

Teachers are rarely involved in expert or research activities, except for those who are employees of schools piloting NUS or new standards. This raises the question of building research literacy more widely, and it could be that the following are further explored:

- The development of research skills may be integrated into the initial teacher education programmes and ensured that these skills are further developed through ongoing professional learning opportunities. For instance, developing skills in interpreting the research evidence and using it in their own practice.
- Similarly, professional learning for school leaders may help their ability to cultivate teachers' research skills, and a culture of using research to improve practice at the school.

Lever 12. Systematising and disseminating knowledge

An essential avenue for knowledge dissemination is intra-community interaction within the educational environment, in particular between the work of methodological associations, professional communities and various educational initiatives. There are some strong foundations in Ukraine already in this regard.

For instance, the institutional foundation for the systematisation and dissemination of knowledge among educators is the network of Centres for the Professional Development of Pedagogical Workers. Notably, these have been established in most communities across Ukraine.

- According to the Regulation on the Centres for the Professional Development of Pedagogical Workers, such communities are associations that facilitate continuous professional growth (Cabinet of Ministers of Ukraine, 2020^[100]).
- The functions of these centres include providing consultations, delivering training sessions and webinars, developing professional development programmes, supporting teacher appraisal, and facilitating the creation of professional communities (UIED, 2025^[101]).
- These teacher communities are based on the principles of humanism, voluntariness, personal growth trajectories, and shared values.
- As of 1 October 2024, there were 364 such centres in Ukraine, with 351 operational. Positive developments have been noted in several regions, including Ternopil, Kharkiv, Chernivtsi, and Chernihiv. However, a decline in active centres was observed in six front-line regions, particularly in Zaporizhzhia and Kherson.

Civil society organisations also play a key role in supporting connections and the dissemination of emerging knowledge from new initiatives. For example, the public organisation EdCamp Ukraine, which works in the field of social and emotional learning (SEL), has become the national centre for conducting an international comparative study on social-emotional skills (SES) in Ukraine. At the same time, EdCamp Ukraine has formed a robust educational community comprising over 50 000 teachers, with representation in at least every second school across the country. As a result, research findings are widely disseminated within

schools, and educational institutions collaborate through events organised by EdCamp Ukraine focussed on SEL topics. Monitoring studies highlight the high effectiveness of EdCamp events: nearly 80% of participants report the immediate implementation of acquired knowledge and skills into their teaching practice, particularly in the area of supporting students' social-emotional development. In response to challenges identified through research, the community also provides the necessary support resources (EdCamp, n.d.^[102]).

This type of ecosystem-based approach to knowledge dissemination can be seen also in the training of teachers during the implementation of the New Ukrainian School reform at the primary school level. From 1 September 2018, Ukraine introduced a new more competency-based model based on the State Professional Standard "Primary School Teacher of a General Secondary Education Institution" (Ministry of Social Policy of Ukraine, 2018^[103]). To ensure that teachers were adequately prepared for these changes, a centralised in-service training programme for primary school teachers was launched. Approximately 22 000 teachers participated annually. The training was delivered using a blended learning model, comprising a 60-hour online course supported by the EdEra NGO, combined with three in-person sessions of equal duration (a total of 120 hours), held at regional Institutes of Postgraduate Pedagogical Education (IPPEs) with the involvement of trainers trained by the Osvitoria NGO. A defining feature of this approach was not only its large-scale reach, but also the establishment of active professional teacher communities. These communities enabled peer-to-peer learning, exchange of best practices, and the formation of shared values among educators. The success of this initiative was made possible through close co-operation between state institutions and civil society, showcasing the effectiveness of an ecosystem model for teacher professional development on a national scale.

As mentioned, it is also promising to see the development of digital platforms that can support teachers' access to quality knowledge at greater scale (see Box 6.7).

Box 6.7. Digital resources that foster knowledge dissemination

Another key tool in the dissemination of knowledge is access to external digital resources, which provide teachers with flexible learning opportunities, participation in professional events, and access to curricula and communities. Examples include:

- All-Ukrainian Online School – A state-run platform offering free video lessons, quizzes, and summaries for Grades 5–11 (Ministry of Education and Science of Ukraine and Smart Education, n.d.^[104]).
- Diia.Osvita (Diia.Education) – Diia, a Ukrainian government app and web portal that provides a wide range of government services, including Diia.Education which is a national 'edutainment' platform that offers free courses, digital-literacy training, reskilling and upskilling opportunities to people of all ages (Diia.Osvita, n.d.^[105]).
- EdCamp Ukraine – A national public organisation that hosts teacher conferences, professional development events, conducts research, contributes to educational policy reform, and cultivates professional communities (EdCamp Ukraine, n.d.^[106]).
- Prometheus – Ukraine's largest platform for massive open online courses, including specialised training programmes for educators (Prometheus, n.d.^[107]).
- EdEra – Ukrainian edtech company that develops educational content and tools for government, businesses, and civil society. It creates online courses, e-textbooks, and collaborates with NGOs and government bodies (EdEra, n.d.^[108]).
- New Ukrainian School Portal – One of Ukraine's leading educational media platforms. Originally created to promote the NUS reform, it now covers a wide range of topics related to preschool

and general secondary education, and shares best international education practices (New Ukrainian School, n.d.^[109]).

- Osvitoria.Media – An educational media outlet dedicated to modern teaching practices and methodologies, with a strong focus on the development of both teachers and parents (Osvitoria Media, n.d.^[110]).
- Teach for Ukraine – A public organisation affiliated with the international Teach For All network, which includes over 60 member countries. The organisation works to engage young leaders in schools to promote equal access to quality education (Teach For Ukraine, n.d.^[111]).

Points of reflection for the future on systematising and disseminating knowledge

In the area of systematising and dissemination of knowledge, Ukrainian education has developed a range of external resources, mostly digital, both government-led and created by the public sector. Such resources give Ukrainian teachers access to professional development programmes, educational events, professional communities and their activities. Institutional support for the systematisation and dissemination of knowledge is delivered through the Teacher Professional Development Centres (TPDCs), which are designed to offer guidance for teachers, facilitate them in their professional development, and disseminate successful educational practices.

At the same time, one challenge is the access to this type of knowledge in a coherent and coordinated way. Some avenues that could be further explored here in response to this challenge include:

- Digital platforms may develop as a unified gateway for sharing experience and dissemination of educational research evidence for teachers to learn about, in particular, in relation to the implementation of the NUS reform and overcoming the challenges of war. For instance, the online platform All-Ukrainian Online School of the Ukrainian Institute for Educational Development (UIED) could be expanded and become such a place for this purpose.
- Promoting the use of peer-to-peer learning approach may expand through the exchange of experience on digital educational platforms, which encourages teachers to voluntarily share their own methodological materials with some preliminary selection and curation to ensure a quality offering. More broadly, it may be that such materials could be recognised in the certification evaluations of teachers' practice or some incentive payments can be provided for them.
- Ensuring the digital infrastructure is in place for all to access such platforms may need further exploration, with some concerns of the suitability of the infrastructure in some areas. It may be fruitful to allocate specific funding to this to ensure access to high-quality digital resources and up-to-date research findings.

There may also be value in looking at how more coherence can be provided to horizontal sharing among key stakeholders. For instance, some areas that could be explored include:

- Promote professional associations and co-operation among initial teacher education universities, education authorities, and schools, including through joint educational forums to share experiences and establish close co-operation, such as the systemic National Teacher Education Forum in Finland.
- It may be valuable to cooperate with other stakeholders within and beyond the community — e.g. employers, interested businesses — to offer a wider range of educational activities for students, including job/career exploration, involvement in various activities, participation in projects for the benefit of the local community, and volunteer engagement. Initiatives such as networks could help schools better consider the needs of communities, including aspects such as the expectations of the local labour market from graduates.

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Notes

¹ Article 22, Law of Ukraine “On Complete General Secondary Education”

² Throughout this text, the reference “Source: EDEBO” should be understood to mean that the relevant statistical data have been calculated on the basis of information from the Unified State Electronic Database on Education (EDEBO). This database contains information on educational programmes, institutions, students, postgraduate students in Ukraine, and more. A portion of anonymised data from EDEBO is publicly available via the internet at <https://registry.edbo.gov.ua/opendata/>. Other data, used for analytical purposes in this document, was obtained by the authors from the State Enterprise “Inforesurs” (the administrator of EDEBO), in response to formal data requests.

³ Part Five of Article 58 of the Law of Ukraine “On Education”

⁴ Further in this text, statistical information on teacher learning is provided primarily at the bachelor’s level, as it represents the main stage within the system of initial teacher learning.

⁵ Professional pre-higher education — July 2021; bachelor’s level — June 2022; master’s level — April 2024; PhD — June 2022.

⁶ Dated 3 July 2025, entitled “On the Approval of Methodological Recommendations on the Specifics of Implementing Educational Programmes in the Field of Knowledge A ‘Education’”.

⁷ The pedagogical internship in Ukraine is regulated by the Law of Ukraine “On Complete General Secondary Education” and the Regulations on Pedagogical Internship, approved by Order No. 1128 of the Ministry of Education and Science of Ukraine, dated 25 October 2021.

⁸ Supervision is regulated by the Procedure for Providing Professional Support and Assistance to Pedagogical Employees (Implementation of Supervision), approved by Order No. 855 of the Ministry of Education and Science of Ukraine, dated 14 June 2024.

⁹ Approved by the Resolution of the Cabinet of Ministers of Ukraine No. 1109, dated 23 December 2015.

¹⁰ These conversions of UAH to EUR are based on the average annual exchange rate for 2024, which was UAH 43.4 per euro.

¹¹ Certification of teaching staff is regulated by the relevant regulations approved by the Government Resolution No. 1190 dated 27 December 2018.

¹² The requirements for advanced training programmes are established by the Resolution of the Cabinet of Ministers of Ukraine, No. 800, dated 21 August 2019, “Some Issues of Advanced Training of Pedagogical and Scientific-Pedagogical Workers”.

¹³ Funding is allocated in accordance with the Procedure and Conditions for Providing a Subvention from the State Budget to Local Budgets to Ensure High-Quality, Modern and Accessible General Secondary Education of “New Ukrainian School”.

¹⁴ The textbook selection procedure involves a step-by-step process that begins with teachers familiarising themselves with electronic versions of textbooks, cover designs, and appendices. After that, teachers collectively select the most appropriate textbooks, approving the choice at a meeting of the pedagogical council, and the results are transmitted to education management bodies. The final stage is a competitive evaluation, the results of which determine the textbooks recommended for printing and use in schools. The Ukrainian Institute for the Development of Education coordinates this process. What every teacher should know about choosing textbooks. URL: <https://mon.gov.ua/news/shcho-maie-znaty-kozhen-uchytel-pro-vybir-pidruchnykiv>.

Annex A. Reflection questions for a process of change

In line with this paper offering a ‘vision’ of professional learning, the below reflection questions are framed with a view to the future and what could be possible, but they could also be transformed to focus on the present and gauge the current state-of-play.

Ambition 1: Build the initial foundations of an effective and meaningful career

Process for selecting candidates for the profession

- How can entry requirements to initial teacher education (ITE) and, in connection, certification, best drive quality standards forward?
- How do the pathways to the classroom foster quality as well as a diversity of candidates?
- What could be the role of incentives to further attract quality candidates?
- How can the overall selection process send positive signals about the teaching profession?

Nature of the preparation for the classroom and profession

- What blend of mechanisms and criteria could best promote quality provision of initial teacher education?
- How is information on the quality of provision used to support further learning and development among ITE providers?
- What could be the balance of quality control with supporting the need for innovation and flexibility?
- How can teachers be supported to develop a coherent, foundational body of theory and knowledge?
- How might teachers be supported to go beyond theory to develop their actual classroom practice through quality experiences?
- How can coherence across the content taught by teacher educators, as well as other actors like schools, be ensured?
- How can the knowledge-base and content of ITE programmes be updated in tune with evolving research and evidence?
- How can teachers be supported early on to develop a critical, reflective stance towards their own practice to help its ongoing development?
- How might foundational research and inquiry skills be developed to support a teacher’s lifelong learning?
- How might a teacher’s self-concept and sense of purpose as part of the profession also be attended to in ITE?

Attending to the transition into the classroom and school

- How can resources and capacity-building be used to promote quality coaching and mentoring experiences?
- What can support all teachers to have access to quality induction supports?
- What structures could enable meaningful dialogue between ITE providers and schools to ensure that different needs are being appropriately met?
- How might induction contribute to wider learning in the school by introducing new ideas or questioning received wisdom for greater clarity?

Ambition 2: Develop an infrastructure that incentivises and recognises excellence

A clear and attractive vertical career structure

- How can each stage of the career, as well as the connections between them, be clearly defined and communicated?
- How could clearly defined teachers' standards with their expected knowledge, skills, values and attitudes underpin career stages?
- How can ownership and shared understanding of teaching standards and career structure be developed?
- What may be the role of financial incentives in career progression?

Opportunities to specialise through horizontal diversification

- How can specialist roles be clearly defined, from tasks to expectations for success?
- What features of the career structure may incentivise teachers' engagement with specialist roles?
- How can effective fulfilment in specialist roles be ensured through sufficient resources, from time to financial resources to the learning opportunities that may be included?
- How are specialist roles accredited to teachers to ensure they are suitable for this role?

Trusted evaluation mechanisms

- How can evaluation positively shape career progression?
- What is the potential to draw upon multiple measures, including the perspectives of different stakeholders, to address the complexity of teaching in evaluating?
- What mechanisms can support the reliability and validity of measures? (e.g. capacity of evaluators, auditing, calibrating)
- What can be the role of more regular and ongoing internal formative evaluation in supporting professional learning?
- How can learning needs from summative evaluations be identified and harnessed for positive formative experiences?
- How can both individual needs be identified for teachers, but also wider school and system needs by school or system leaders?

Ambition 3: Facilitate quality, meaningful ongoing learning that enables the profession to grow and face future challenges

Provision of relevant learning opportunities

- How can the responsibility for the provision of professional learning opportunities be best addressed for high-quality learning?
- What could be the criteria and mechanisms for promoting high-quality offers of professional learning?
- What might be the right balance of agency and choice with more mandatory learning opportunities?
- How can the provision be designed with evolving needs of teachers, schools and the wider system?

Delivery of quality learning opportunities

- What resources could support structured communities of learning? (e.g. time, physical resources, capacity-building)
- How could data and research provided to support structured communities of learning?
- How can a culture of learning be embodied in the system and across schools?
- What opportunities can exist for quality coaching and mentoring in the system?
- What enablers in terms of resources (e.g. time) and capacity-building could support effective coaching and mentoring?

Incentives to engage in professional learning

- How can potential barriers to teachers' participation in professional learning be addressed? (e.g. cost, time)
- What could be the approach to recognising and tracking professional learning participation?
- How could recognising and tracking support sustained reflection on professional learning?
- What measures can be used to gather teachers' and school leaders' views on professional learning opportunities and drive further improvement?
- How can the overall quality across the provision and delivery of professional learning be monitored and used?

Ambition 4: Create an ecosystem for collective learning and sharing to ensure opportunities for excellence for all

A culture of collaborative inquiry within and across schools

- How can structures (e.g. networks) and roles (e.g. facilitators) be used to support collaborative inquiry within and across schools?
- What resources can be provided to support collaborative inquiry within and across schools? (e.g. time, physical resources [e.g. videos, digital tools], capacity-building)
- How can practitioners enact agency over defining tangible focusses for collaboration?
- What role can data, research and artefacts play in grounding collaboration?
- How can additional actors be drawn upon to enhance collaborative processes?
- How could best practice around collaboration be more widely shared?

Developing engagement with research

- What mechanisms can support access to high-quality, relevant research?
- How could teachers and school leaders be supported to develop their research literacy?
- What system-level infrastructure would enable a culture of effective, sustainable research engagement?
- What role could teachers and school leaders play in shaping the research agenda?

Systematising and disseminating knowledge

- How could new knowledge from school-level actors be coherently and systematically captured?
- What could be the role of different actors in generating and curating new knowledge?
- What tools could enable an evolving knowledge base to be accessible to a wide range of actors?

Key implementation considerations

- How do the career structure and professional learning opportunities coherently build upon the foundations of the system of ITE?
- How can the different roles of stakeholders across ITE and professional learning be defined and coordinated?
- How might the voices of different stakeholders – across teacher professional learning – be heard and considered to foster buy-in and ownership?
- How might the capacity of teacher educators and professional learning providers be supported and developed to increase their impact in ITE and in-service learning?
- How are the distinct needs of leaders met and considered in the career structure and system of professional learning, whilst retaining coherence with those of teachers?
- How can leaders be supported to become enablers of strong collaborative cultures in their school and more widely?
- How may the potential effects of market forces on the provision of professional learning be best understood and responded to?
- What could be the role of pilot programmes in supporting potential change?
- What could be the potential of identifying and supporting early adopters when implementing change?
- What might be the role of patience and incremental steps in the face of a time-consuming process of changing cultures?

Annex B. Summary of Education Systems

Box B.1. Overview of Chile

Entry to the profession via alternative pathways

Chile has sought to support individuals with diverse backgrounds to enter the profession. Continuing education programmes target individuals with prior training – be it university or technical. This may mean professional experience too but is not necessary. Thus, there is a need to meet one of two requirements:

- Possess an academic degree or professional title.
- Possess a higher-level technical degree.

Some 2 000 individuals were trained in one of these programmes in 2024. There are some 50 different programmes for different education levels and modalities, such as the Mathematics and Computer Science Pedagogy programme, Basic Education programme, and Pedagogy in Technical and Vocational Education programme.

These programmes also offer a means of individuals working as teachers, or other roles, in the school system to pursue a pedagogical degree if they do not have one yet.

Monitoring initial teacher education(ITE)

Universities are required to administer two diagnostic assessments to ITE students. This aims to furnish providers with information to improve their offer to students.

- One assessment is at the beginning of the programme and administered by the universities themselves. This aims to support diagnosis of support and levelling mechanisms for students, if necessary.
- A second assessment is administered centrally by the Ministry of Education one year before graduation. The “National Diagnostic Assessment” is based on the specific defined standards for ITE that define the essential core of disciplinary and pedagogical knowledge that education professionals are expected to possess once they have completed their initial training.
 - The assessment aims to understand the knowledge that has to that point been acquired by students.
 - It is a mandatory requirement to complete this assessment, with results submitted to the National Accreditation Commission.
 - Improvement plans may be established based on this, meaning that there is a degree of central oversight of providers.

A key actor is the Center for Pedagogical Improvement, Experimentation, and Research (CPEIP). They promote collaboration between various public entities and generate knowledge to support ITE providers.

Moreover, recently in March 2025, the Initial Teacher Training Strengthening Agenda has been established to implement strategies and actions to strengthen ITE through an inclusive, equitable, and territorial approach.

Career structure

Chile introduced wholesale changes to its system of professional learning and teacher career structure in 2016. Progression across its five-stage career model hinges upon a certification process that is competency-based and regulated by an external component, as well as teachers' years of experience. Interestingly, this was recently further reformed to try to create more efficiency and reduce the burden on teachers.

Competencies are specified in the national teaching standards. Evaluation consists of two external components: the teacher completes a standardised written assessment on their pedagogical knowledge and prepares a professional portfolio. The latter seeks to evaluate different aspects of the pedagogical practice of classroom teachers, based on multiple direct evidence of the work inside and outside it, including samples of work (e.g. lesson planning), a recording of their teaching, and proof of collaborative work with colleagues.

Teachers are expected to move from the first stage ('Initial') to the second stage ('Early') and, under the new reform, onto the third stage ('Advanced'). Teachers have the opportunity to engage in the evaluation process between these stages twice. This means that evaluation serves as a mechanism to remove underperforming teachers from the profession if they fail the examination more than twice (Santiago et al., 2017^[21] pp. 240, 253). In some cases, teachers can re-enter the profession after a two-year break and try the evaluations again.

Progression to the two highest stages ('Expert I' and 'Expert II') are voluntary. Thus, those at the 'Advanced' or higher levels do not need to engage in evaluation processes but, instead, they must participate in the 'Deepening Cycle'. This cycle lasts for four years and seeks to actively promote professional learning. During this, teachers must complete one of four actions:

- Participate and pass at least a program, course or support peer action, developed or certified by CPEIP.
- Take and pass at least one of the evaluation instruments (portfolio or assessment).
- Participate through a program of active participation on an activity of the Teachers' Network of Teachers.
- Become a mentor for one or more new teachers under the induction or mentoring processes recognised by CPEIP.

Under their new salary and career re-structuring in 2016, all beginning teachers were assigned to the lowest stage while current teachers were designated a step based on their years of experience and their previous results in multiple teacher evaluation processes (Santiago et al., 2017^[21] p. 240). In the more recent reforms, an additional provisional category was created for teachers who did not have evaluations from recent years, who have several years to complete an evaluation with which they would essentially 're-join' the new progression ladder.

The reforms centred on creating a high-stakes evaluation system that, as mentioned, could see the removal of underperforming teachers. The evaluation was also intensive in terms of measures. Central to its successful implementation was it being a multi-stakeholder affair, with the government creating a tripartite technical committee that provided a platform for central and local authorities alongside the country's main teacher union to develop consensus positions.

Use of standards

The Standards for the Teaching Profession (STP) serve as Chile's primary benchmark for defining the teaching that children and young people need to develop their full potential. It serves as a common reference point across the entire education systems, supporting pre-service and in-service teacher professional learning and enabling dialogue across stakeholders.

Induction

Considerable attention has been focussed on recognising the importance of the first years as a teacher. The Induction and Mentoring Program provides dedicated support to new teachers and their integration into the education community.

Induction is a right of all beginning teachers who are appointed or contracted to a state-funded institution. It last for 10 months during the school year, and beginning teachers receive an allowance for the time spent on induction.

Beginning teachers have an assigned mentor who is an experienced teacher (at the upper stages of the career structure: Advanced, Expert I or Expert II) and who has received specific training to be able to lead and support effective induction.

- Mentors may be from the same institution or another.
- Mentors may oversee up to three beginning teachers.
- Mentors are also entitled to receive allowances for each beginning teacher they mentor and support.
- Mentors must design, implement and evaluate a mentoring plan based on the beginning teachers' needs. Guidance is published by the Ministry to support this, including a rubric for self-assessment with beginning teachers.

The induction process requires a commitment of between four and six hours a week for the beginning teacher and mentor to meet a reflect on practice and challenges, as well preparing classroom work, analysing learning outcomes, and considering integration into the education community. Notably, the emphasis is on mentoring and formative development, rather than supervision and evaluation.

High-performing schools have the right to implement and manage their own induction processes.

A commissioned evaluation of the system of professional learning noted that induction was highly valued, but yet to fully permeate the school culture. It also did not have wide coverage yet. This has been a valuable evaluation for further refinement of the system of professional learning, including induction. For instance, now there are now ongoing efforts to enhance the relevance of induction to specific, more localised contextual needs. This includes trying to better match beginning teachers with mentor teachers from that particular local context.

Provision of professional learning

In 2014, teachers at all levels had 2 006 annual working hours, of which 75% were designated to classroom activities, which constituted both a heavy workload and a high proportion of contact hours by international comparison.

Motivated in part by a desire to ameliorate these working conditions, Chile introduced a new System for Teacher Professional Development in 2016.

- The new law initiated a reduction in the proportion of teaching time to 70% of the workload by 2017 and to 65% of the workload by 2019.

- This provides teachers with more time to engage in non-teaching activities, fosters teachers' engagement in the school and provides greater opportunities for collaboration among peers since teachers are required to stay at the school during some of their non-instruction hours.

The online portal “Desarrollo Docente en Línea” provides access to free, regularly updated trainings. Facilitated by CPEIP, it aims to provide ongoing professional learning support that is differentiated to teachers depending on their levels in the career structure and needs.

- Teachers can search for learning opportunities that correspond to their roles in schools and apply online.
- Providers of the courses then inform participants of their selection or not.
- Teachers receive a certificate of completion once they have met the approved criteria.

Monitoring professional learning opportunities

The Ministry undertakes the certifying of professional learning opportunities. This relates to activities, programmes or courses that are aimed at teachers and to support their learning. Certification can be given to professional learning providers be they public, private or non-profit institutions, as well as national or foreign.

Clear procedures are set forth by the Ministry, with a required set of documents needed as well as follow-up reports for tracking. The process unfolds in three steps: an administrative evaluation stage, a technical evaluation stage (of the quality and relevance of the proposed activity) and a legal evaluation stage.

The Center for Pedagogical Improvement, Experimentation and Research periodically reports and updates the training actions certified since 2016, with these publicly available online. In this respect, teachers and school leaders are provided with valuable information on providers and where to find quality professional learning.

Digital tools

A virtual assistant chatbot is being developed, that focusses on answering questions of new teachers using generative artificial intelligence. It is due for release in 2025.

Notes: Chile shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (Santiago et al., 2017^[1]), *OECD Reviews of School Resources: Chile 2017*, <https://doi.org/10.1787/9789264285637-en>; (CPEIP, 2025^[2]), *Certificación Acciones Formativas [Certification of Professional Learning Activities]*, <https://www.cpeip.cl/certificacion-acciones-formativas/> (accessed on 24 April 2025); (CPEIP, 2025^[3]), *Estandares Formación Docente [Teacher Training Standards]*, <https://www.cpeip.cl/estandares-formacion-docente/> (accessed on 24 April 2025); (CPEIP, 2025^[4]), *Estudiantes Pedagogía [Pedagogy Students]*, <https://www.cpeip.cl/estudiantes-pedagogia/> (accessed on 24 April 2025); (CPEIP, 2025^[5]), *Formación Continua Docente [Continuous Professional Learning Standards]*, <https://cpeip.cl/formacion-continua-docente/> (accessed on 24 April 2025); (CPEIP, 2025^[6]), *Formación Inicial Docente [Initial Teacher Education]*, <https://www.cpeip.cl/instituciones-formadoras/> (accessed on 24 April 2025)

Box B.2. Overview of England (United Kingdom)

Entry requirements

'Qualified Teacher Status' (QTS) is a minimum requirement needed to teach in publicly funded primary, secondary and special schools in England (United Kingdom) and is awarded as part of Initial Teacher Training (ITT) courses. Academies schools (which are publicly funded) can employ teachers without QTS, though the majority of teachers employed by academies do hold QTS. Private schools are able to employ teachers without QTS, though many chose to employ teachers holding QTS.

There are entry requirements to be achieved prior to the Initial Teacher Training course commencing and separate requirements to be awarded QTS that will be met over the course of the Initial Teacher Training Course.

For entry into primary and secondary teaching courses, candidates must have passed Maths and English at the upper secondary level at the equivalent minimum grade (GCSE grade 4, or recognised equivalent). In addition, primary school teachers must also hold a minimum qualification (GCSE grade 4) in a science subject.

To be awarded QTS, candidates must have a bachelor's degree – or equivalent qualification – in any subject. For postgraduate routes to QTS, candidates must have achieved a bachelor's degree as an entry requirement. For undergraduate routes, trainees may complete a bachelor's degree in primary or secondary education with QTS as part of their ITT course.

Routes to Qualified Teacher Status

There are three routes to QTS:

1. The postgraduate fee funded route, which normally runs for one year full-time:
 - Postgraduate Initial Teacher Training courses typically last nine months full-time and combine theoretical learning with at least two classroom placements in schools.
 - Courses have to be paid for (approx. 9 500 GBP per year [~11 000 EUR]), though there are some bursaries or scholarships to incentivise participation in certain subjects. A system of loans also exists.
 - Some of these courses lead to QTS, and others to QTS with a combined postgraduate certificate in education (PGCE).
 - It is also worth noting that only providers who have been accredited by the Department for Education (DfE), or providers who work in partnership with an accredited provider, may offer courses that lead to QTS. Not all teacher training courses in England lead to QTS, and candidates should check this before committing.
2. The postgraduate salaried route, which normally runs for one year full-time and includes the School Direct salaried programme, the High Potential ITT programme (HPITT) and, since 2018/19, the Postgraduate Teaching Apprenticeship (PGTA):
 - These allow individuals to earn a salary while they work towards QTS.
 - These vary in length from nine to twelve months typically, with most of the time spent in school placements over theoretical learning. They follow a more apprenticeship-based approach.
 - These are highly competitive and in high demand.

An example of a salaried route into teaching is via Teach First. Teach First is a charitable organisation that also delivers so-called 'high potential initial teacher training'.

- This is an employment-based route into teaching for high-performing graduates and those who are looking to change careers.
- Similar to the above, participants on the programme also earn a salary while they work towards qualified teacher status (QTS) with a postgraduate certificate in education (PGCE).
- Teach First is part of the Teach for All network, which offers similar alternative pathways into teaching in a range of countries.
- The undergraduate route, which normally runs over three or four years and includes teacher training via an undergraduate degree route and the new Teacher Degree Apprenticeship pilot which is a salaried route (and is accepting applications from 2024/25 onwards).

In addition to the three routes outlined above, a more direct assessment-only programme can also be pursued for those that have worked as an unqualified teacher for a significant amount of time and can demonstrate that they already meet the standards required to be awarded QTS.

Following the award of QTS, individuals can apply to teaching positions to complete a mandatory two-year induction period (see below) as a so-called 'Early Career Teacher (ECT)'.

Incentives

A system of 'teaching internships' are available. They are designed to offer a means of understanding what it is like in the classroom for those interested in the profession.

The internships last three weeks in June or July (the academic year ends in July in England (United Kingdom)). Interns are also paid, receiving 425 GBP (~500 EUR) per week.

Paid internships are available to those studying for an undergraduate or master's degree in the following subjects: chemistry, computing, languages, maths, physics. They are thus targeted towards specific subjects where there are notable shortages. Those interested in teaching from other subjects can visit schools to get unpaid experience in the classroom.

Defining core content of preparation and induction

From 2025, the new Initial Teacher Training and Early Career Framework (ITTECF) sets out the entitlement of every trainee and early career teacher at the start of their career. In particular it outlines:

- The core body of knowledge, skills and behaviours that define great teaching.
- The types of support that should be provided both when preparing teachers and in delivering induction support, so they receive a three or more-year introduction to the core body and mastering it.

A key goal of the new framework has been to combine the previously separate frameworks on teacher preparation, core content, and beginning teachers. The development process has also sought to:

- Integrate the latest evidence based on a recent public call for evidence, including relating to early cognitive development, oracy, supporting pupils with special educational needs and disabilities (SEND), and social and emotional learning. Evidence has been assured by the Education Endowment Foundation (EEF).
- Build on lessons learned over recent years delivering implementation of the previous frameworks.

The framework is designed to work across subjects and age groups to promote a shared body of knowledge and skills in the profession. The framework promotes coherence, such that early career

teachers will purposely revisit elements from their preparation programmes to go deeper into these aspects. The framework is structured around two types of content:

- **‘Learn that...’ statements:** These are primarily knowledge-based statements that are informed by the best available educational research (e.g. practice guides, rigorous individual studies, high-quality reviews and syntheses).
- **‘Learn how to...’ statements:** These are drawn from the wider evidence base, including both academic research and guidance from expert practitioners.

Notably the framework is a minimum entitlement to training, and not a full curriculum. Accredited providers of ITT or training to beginning teachers make decisions about the form this takes in terms of coherent content, as well as additional elements. Hence, the framework informs curriculum design and is not an assessment framework – with the published teachers’ standards serving as the primary reference for any performance-based assessments in the system at the end of ITT and induction respectively.

To support the translation of the framework into actual learning opportunities in ITT or supporting beginning teachers, guidance is included in the framework. This includes ensuring that teachers engage in:

- Observation of practice, including discussing and analysing these with expert colleagues.
- Receiving clear, consistent and effective mentoring, grounded in evidence.
- Opportunities to practise using specific approaches through rehearsal and refinement.
- Receiving feedback on their application of particular approaches, with a focus on reflection and critique for improvement.

Accreditation and Quality Control of Initial Teacher Training Providers

Initial Teacher Training providers are accredited by DfE.

- Where a provider of ITT has been accredited by DfE, this means the quality of their provision has been assessed against a specific set of assessment criteria, which cover areas including course design, coherence across the provision, evidence base and evaluation.
- This process also involves assessing how the provision meets the requirements set out in the ITT criteria and other supporting literature, such as the ITTECF.
- The accreditation process is led by DfE, in collaboration with ITT sector experts.

A second key body in quality control is that of an inspectorate. Ofsted (Office for Standards in Education, Children’s Services and Skills) inspect services providing education and skills for learners of all ages in England (United Kingdom). Ofsted have a statutory duty to inspect ITT provision that leads to Qualified Teacher Status (QTS).

Ofsted inspects ITT providers under its Initial Teacher Education inspection framework. Inspection outcomes from Ofsted are valuable in several ways:

- They provide trainees and prospective trainees with an expert, independent assessment of how well an ITT provider is performing and the quality of ITT and training offered.
- They also provide information to ministers and system leaders about the work of ITT providers and the extent to which an acceptable standard of teacher training is being provided.
- More specifically, the English Department for Education can withdraw a provider’s accreditation if the quality of their provision (as shown through Ofsted inspection reports) falls below the expected level.

- Inspections also provide information for and encourage the improvement of individual ITT providers and the education system as a whole.

Induction overview

As mentioned above, once QTS is gained, teachers can apply to complete a mandatory two-year induction in a school as a 'Early Career Teacher (ECT)'. This is essentially qualified teachers' first teaching job.

Induction should be equivalent to two full-time school years (six semesters) and can be split across different schools. Performance during the induction and in the final assessment for successfully completing it is measured against the teachers' standards. There are appropriate bodies for quality assurance of this (e.g. teaching school hubs), with each ECT having a named contact at their appropriate body for raising any concerns.

It is encouraged to start induction as soon as possible after gaining qualified teacher status, but there is no set time limit for starting or completing an induction period. There is only one chance to complete statutory induction. Hence, without satisfactorily completing induction a qualified teacher cannot be employed as a teacher in a relevant school in England.

Working in some schools (e.g. private sector, an academy or free school) means that teachers are not legally required to satisfactorily complete an induction period although many opt to. Induction must only be undertaken in appropriate settings where the school can provide the right types of supports. For instance, some settings, such as schools that are in 'special measures' based on inspection, cannot conduct induction.

Features of induction

During the two-year induction, ECTs are entitled to a reduced timetable; they can teach a maximum of 90% of the typical timetable in their first year, and a maximum of 95% in the second year. This is in addition to the timetable reduction that all teachers receive for planning, preparation and assessment time.

An induction tutor and a mentor should be appointed, held by different individuals. Alongside this, a suitable monitoring and support programme must be put in place as part of the induction. This is expected to include: a programme of training aligned to expected frameworks and the teachers' standards; regular one-to-one mentoring, support and guidance from a designated induction tutor, being observed at regular intervals and receiving written feedback, and observing colleagues (see more below). There should also be periodic professional reviews of progress with the induction tutor, which not a formal appraisal but monitoring of the ECT's progress against the teachers' standards. It is designed to prepare them for their formal assessments.

Formal assessments should occur twice during induction, namely at the end of their first and second year of induction. These should be evidence-based assessments with a report developed summarising it, conducted by school principal or induction tutor. The final assessment will see the school principal make a recommendation to the appropriate body as to whether the ECT has satisfactorily completed their induction period.

Notably, there is funding to cover mentor sessions during timetabled hours and school leaders are expected to ensure mentors have the time and means of fulfilling their roles.

Background to National Professional Qualifications

England (United Kingdom) has a series of 'National Professional Qualifications' (NPQ) for teachers and school leaders. These were reformed in 2021 to ensure they were underpinned by the best available evidence of effective teaching and leadership practice. The qualifications aim to:

- Improve career progression
- Improve quality and consistency of leadership
- Improve retention of teachers and leaders
- Improve the quality of teaching

The NPQs provide training and support for teachers and school leaders at all levels, from those who want to develop expertise in high-quality teaching practice, such as behaviour management, to those leading multiple schools across trusts. The NPQ frameworks build on the evidence base and expert guidance already established in the Department's Initial Teacher Training and Early Career offer. Formally taught through a blend of face-to-face, online and independent learning against defined frameworks, the qualifications are both a learning opportunity as well as recognition of the teacher or school leader's knowledge and competence in a specific area.

Overview of NPQs

The suite includes a series of what are referred to as 'Specialist National Professional Qualifications' (SNPQs), which enable teachers and school leaders to broaden and deepen their expertise in specialist areas.

- NPQ in Leading Teaching (NPQLT): Participants learn how to lead the teaching and learning of a subject, year, group or phase. Qualification launched in Autumn 2021.
- NPQ in Leading Behaviour and Culture (NPQLBC): Participants learn how to create a culture of good behaviour and high expectations in which staff and pupils can thrive. Qualification launched in Autumn 2021.
- NPQ in Leading Teacher Development (NPQLTD): Participants learn how to become a teacher educator and successfully support teachers in their school to expand their skills. Qualification launched in Autumn 2021.
- NPQ in Leading Literacy (NPQLL): Participants learn how to develop literacy across their school and recognise the importance of literacy for pupils' academic achievement, well-being and success in life. Qualification launched in Autumn 2022.
- NPQs for Leading Primary Mathematics: Participants learn how to lead primary mathematics teaching across a school, key stage or phase. Qualification launched in Spring 2024.

The suite also includes a series of 'Leadership National Professional Qualifications' (LNPQs):

- NPQ in Senior Leadership (NPQSL): Participants develop their leadership knowledge and expertise to improve outcomes for teachers and pupils in their school. Qualification launched in Autumn 2021.
- NPQ in Headship (NPQH): Participants develop the knowledge that underpins expert school leadership and apply it in order to become an outstanding headteacher. Qualification launched in Autumn 2021.
- NPQ in Executive Leadership (NPQEL): Participants develop the expertise needed to become an outstanding executive leader, leading change and establishing supportive networks to drive improvement across the group of schools or multi-academy trusts. Qualification launched in Autumn 2021.
- NPQ in Early Years Leadership (NPQEYL): Participants develop their leadership knowledge and skills to improve delivery of high-quality early education and care as well as staff development, while also implementing improvements in their setting. Qualification launched in Autumn 2022.

- NPQs for Special Education Needs Co-ordinators (SENCO): A new leadership level qualification to support the development of SENCOs, school leaders or aspiring SENCOs. Qualification launched in Autumn 2024.

LNPQs typically last for 18 months and demand a minimum of 75 hours of engagement, while the SNPQs last for 12 months with a minimum of 50 hours of engagement. The NPQ for SENCOs is a minimum of 85 hours.

Provision of NPQs

The actual provision and delivery of NPQs is outsourced by the English Department for Education to an approved set of providers. Selection of providers is rigorous, with distinct criteria needing to be met. Indeed, in general there is strong oversight by the Department for Education:

- For each NPQ, the Department for Education has developed and published a framework to support organisations developing and delivering NPQs. This can also support teachers applying for, or participating in, an NPQ; school leaders who are encouraging staff to undertake an NPQ.
- The framework sets out what participants should know, and know how to do, after completing an NPQ.
- A strong emphasis in the frameworks is the evidence base that underpins the qualifications and, more broadly, alignment with the accompanying frameworks that have been developed for Initial Teacher Training and Early Career Framework. The framework sets clear 'key evidence statements' of expected learning, as well as so-called 'practice statements' on the more practical skills to develop.
- The content of the guidance for NPQs and the supporting evidence of the content framework are independently assessed and endorsed by the Education Endowment Foundation (EEF). The latter review each guidance and content framework to ensure they draw on the best available evidence and that it is interpreted with fidelity.

Teachers can search for the providers of different NPQs online through the central government website. To apply, teachers need to:

- First register with the Department for Education. This allows for oversight over funding allocations (see below) and is a means of keeping one's professional record up-to-date and, eventually, issuing NPQ certificates.
- Apply directly to the provider who then confirm registration for the NPQ.

As mentioned, to support engagement, teachers can also apply for funding to cover the costs of NPQ course fees. In particular, current eligibility criteria for funding is orientated towards teachers and school leaders in the top 50% of state-funded settings in England that have the highest proportion of disadvantaged students. Funding is also currently available for the NPQH, NPQ for SENCOs and NPQLPM.

Delivery of NPQs

In terms of delivery, as mentioned this consists of three strands: face-to-face learning, synchronous online learning, and asynchronous independent learning too that is led by the individual teacher or school leader. There is though variation among providers in terms of delivery. For instance, features such as mentoring or attending conferences may also be elements of how certain providers deliver NPQs.

To successfully complete the NPQ there is an assessment. Assessment of the qualification was reformed in 2021. Previously, a school-based project was undertaken as a means of documenting learning, but this proved relatively burdensome for teachers. The new approach to assessment sees

participants in NPQs complete a fictional case study. This case study aims to recreate an authentic school-based situation aligned with the content of the NPQ. They are carefully developed to incorporate hypothetical data and resources and are open-book to allow candidates to use previous readings and work from the NPQ course.

Notably, some providers also undertake diagnostic assessments at the beginning of courses to gauge participants' existing levels of knowledge. These assessments rely on digital tools.

Transition lessons

While the NPQs are notably substantial in terms of their commitment as well as the breadth of content many of them cover, there is a resemblance to certain features of micro-credentials. Particularly those of the 'Specialist' qualifications – which seek to support and recognise the diversification of skills – have some overlap with the broad understanding of micro-credentials in other spheres (e.g. higher education – see Chapter 3).

The use of assessment and, in particular, the reform of this to be less demanding and more manageable for teachers, reflects the potential need for ongoing refinement and monitoring in such an initiative. The ongoing efforts to evaluate the NPQs provision is also indicative of this iterative approach to its implementation in general.

The use of diagnostic assessment points by some providers signals the potential of digital tools to gauge teachers' levels of competence. Notably these are used in a low-stakes sense; for instance, at the beginning of a course as a formative tool to understand prior knowledge, rather than a more high-stakes summative assessment at the end as a measure of completion. They provide a potential indication of how digital tools could be drawn upon.

The Department for Education (DfE) in England (United Kingdom) knows that evidence is not static, and insights evolve. Therefore, DfE is now undertaking a formal review of the existing NPQ suite. The review includes, but is not limited to, consideration of the following key areas:

- further best-practice for teaching pupils with Special Education Needs and Disabilities (SEND)
- ensuring that NPQs genuinely support progression through different levels of leadership
- addressing how leadership practices can support workload reduction
- further best-practice, particularly at executive leadership level, in operational aspects of leadership, such as how to manage a budget and deploy a workforce effectively

The department will work in partnership with the Education Endowment Foundation (EEF) to update the evidence base underpinning the qualifications. It is expected the review of the suite of 10 NPQs will take approximately 18 months. The existing suite of NPQs will remain in place until the review is complete.

Note: England (United Kingdom) shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (CFE Research, 2023^[7]), *Emerging findings from the NPQ evaluation: Interim report 1*, <https://assets.publishing.service.gov.uk/media/63cfdb418fa8f53fe33dadb3/>

Emerging findings from the evaluation of National Professional Qualifications Interim report 1.pdf (accessed on 19 March 2025);

(CFE Research, 2024^[8]), *Emerging findings from the NPQ evaluation: Interim Report 2*, https://assets.publishing.service.gov.uk/media/66e17cc5caa02d92e72c8d54/NPQ_evaluation_interim_report_2.pdf (accessed on 19 March 2025);

(Department for Education, 2020^[9]), *National Professional Qualification (NPQ): Leading Teacher Development Framework*, https://assets.publishing.service.gov.uk/media/66c4908b057d859c0e8fa7c4/NPQ_Leading_Teacher_Development_.pdf (accessed on 19 March 2025);

(Department for Education, 2024^[10]), *Outcomes of the review of the Initial Teacher Training Core Content Framework and Early Career Framework*, [https://assets.publishing.service.gov.uk/media/661d24ba08c3be25cfbd3e62/](https://assets.publishing.service.gov.uk/media/661d24ba08c3be25cfbd3e62/Outcomes_of_the_review_of_the_Initial_Teacher_Training_Core_Content_Framework_and_Early_Career_Framework.pdf)

Outcomes_of_the_review_of_the_Initial_Teacher_Training_Core_Content_Framework_and_Early_Career_Framework.pdf (accessed on 19 March 2025); (Department for Education, 2025^[11]), *Apply for a national professional qualification (NPQ)*, <https://www.gov.uk/guidance/apply-for-a-national-professional-qualification-npq> (accessed on 19 March 2025)

Box B.3. Overview of Estonia

Background of a new career structure

In 2013 Estonia transitioned from an attestation system of professional learning and career progression to one that was more competency-based.

Previously, teachers were expected to complete a set number of hours of professional learning (160 hours every five years) with which attestation at different career levels would be determined. This was a complex and resource-intensive system. Teacher attestation processes focussed on the acquisition of qualifications with no professional standards as a reference, serving more as an academic exercise than concentrating on the core work of teachers.

Table A B.1. Overview of the Estonian career structure

Position	Notable features
Teacher (level 6)	- Applies only to pre-primary teachers upon entrance in the teaching profession, following the completion of an initial teacher education programme (at bachelor's degree level) or following the recognition of professional qualifications for this level by the teacher professional body. - This career stage is awarded indefinitely.
Teacher (level 7.1)	- Awarded upon entrance in the teaching profession, following the completion of an initial teacher education programme (at master's degree level) or following the recognition of professional qualifications for this level by the teacher professional body. - This career stage is awarded indefinitely.
Senior Teacher (level 7.2)	- Awarded to a teacher who, in addition to conducting teaching activities, supports the development of the school and of other teachers and is involved in methodological work at the school level. - Teachers choose to apply to this stage. - Awarded for five years, after which a new application must be submitted for this career stage.
Master Teacher (level 8)	- Awarded to a teacher who, in addition to conducting teaching activities, participates in development and creative activities in and outside his or her school and closely co-operates with a higher education institution. - Teachers choose to apply to this stage. - Awarded for five years, after which a new application must be submitted for this career stage.

Source: (Santiago et al., 2016^[12]), *OECD Reviews of School Resources: Estonia 2016*, <https://doi.org/10.1787/9789264251731-en>

Some horizontal diversification also exists alongside the vertical structure. A teacher can become the head of a subject section, co-ordinate a subject in the school, or be invited by the school director to management roles such as deputy director. Other possible roles include mentor of a beginning teacher, curriculum development or co-ordination of professional development at the school.

There were also considerable efforts to raise teacher salaries. Salaries increased, in real terms, by 30% between 2005 and 2012 (for lower secondary teachers with 15 years of experience), with further increases in 2013 and 2014. This saw the average teacher salaries move to around the level of the Estonian average salary. The increase in teachers' salaries in the last few years in Estonia has been the most significant among the OECD countries for which data are available. This was despite a context of no shortages in the workforce, but the need to consider the motivation levels of teachers and the way the workforce attracts candidates.

That said, career progression decisions are made independently of financial decisions. Promotions are accompanied by a guaranteed salary increase, but the exact amount depends on the number of teachers at each level; local authorities share out salaries from a lump salary sum. This means that there is not an upper limit on promotions but at the same time salary expenditure remains constant and predictable for the authorities. This also means that there are not strong financial incentives.

Transition process of a new career structure

Two distinct bodies now oversee the definition of expectations and the appraisal process:

- The development of the teacher professional standards is the responsibility of the Estonian Qualifications Authority.
- The certification processes to reach the different career stages are the responsibility of a teacher professional organisation (the Estonian Association of Teachers). Notably, this has given ownership of the certification process to teachers themselves.

A strength of the new career structure is associated with a set of teacher professional standards, which define the competencies associated with each career stage. In this new model, professional learning is now incentivised as a means of gaining the competencies needed to access higher stages and perform new roles. Developed by the aforementioned Estonian Qualifications Authority, the standards cover the following aspects with key competencies defined for each of these for each career stage:

- I. planning and preparation
- II. the learning environment
- III. instruction and supporting learning
- IV. reflection and professional development
- V. student counselling and communication with parents
- VI. methodological work and mentoring at school level (for Senior and Master Teachers)

However, it has proven challenging to build engagement and use with these standards in consistent ways (see final section below). The adoption of the standards has been characterised by limited understanding, particularly of certain roles.

Transition process of a new professional learning model

Co-ordination of the provision of professional learning has also been to some extent recentralised. Municipalities (or private providers of schools) previously had 3% of the 'salary budget' earmarked for professional learning. This has moved to 1% now, with the remaining 2% centrally managed. Institutions such as the Education and Youth Board of Estonia have accredited providers to deliver specific professional learning activities aligned to wider system goals.

Since 2015, earmarked funds for professional development activities of teachers and school leaders are no longer linked to the 'salary budget'. Instead, these are determined on the basis of a per student model (12 EUR per student per year).

This means that there are two funding mechanisms and modes of choice:

- Teachers and school leaders have the choice of how to use the direct funding provision for professional learning that they have. Thus, teachers can choose particular learning activities aligned with their professional development plan, under the oversight of school's director who guides choices and validates them.
- Learning activities from central providers (e.g. Education and Youth Board of Estonia) as part of the 2% are free of charge for teachers to engage with.

There is an online database of professional learning activities (“Juhan”) though it is not exhaustive. This provides a detailed overview of courses (e.g. expected outcomes, facilitators etc.), but not necessarily information to evaluate its effectiveness.

An ongoing challenge is the relevancy of professional learning. A lack of relevant programmes was cited as one of the main barriers to participating in professional development for teachers in Estonia (Cordingley et al., 2022^[13]).

Overall, there is no central co-ordination of professional learning. There are a range of different institutions that teachers are free to choose from with the allocated funding. For instance, higher education institutions, teacher education institutions, individual schools, teachers’ professional organisations, municipalities and private companies.

Initial teacher education

In recent years Estonia has concentrated considerable efforts on enhancing the quality of its teacher workforce. New teacher standards were introduced in 2013, which were developed through an interactive process representing a considerable range of stakeholders. New standards have been introduced since these with the latest entering force as of 1 September 2025.

The standards are an important tool; institutions need to demonstrate that their curriculum prepares teacher candidates for all the requirements stated in the professional standards, with an explicit matching occurring between the learning outcomes of the curriculum, modules and courses on the one hand, and the teacher standards on the other (Révai, 2018^[14]).

There are two major initial teacher education providers in Estonia: Tartu and Tallinn University. There are also two other institutions that only focus on preparing teachers of certain areas (music, arts and theatre). For those entering at the primary level, programmes of ITE consist of mostly concurrent studies. At the lower or upper secondary level, it generally consists of a consecutive programme.

On the back of concerns that a teaching career was not a very popular choice among upper secondary graduates, Estonia has developed a series of programmes to further incentivise talented candidates to join the teaching profession. This includes:

- A scholarship programme: This seeks to motivate students with talent for teaching to apply to teacher education specialisations and to acquire a teaching qualification. Students who enter the teaching profession within 18 months of completing their education receive an additional allowance. The programme was also extended to professional support specialists.
- The “Youth to School” programme (“Noored Kooli”), part of the “Teach for All” programme: This seeks to raise interest in teaching and education by awarding scholarships to a select group of university students. Recipients teach at a school for two years, while at the same time participating in teaching and leadership training. After the programme, students can choose to keep working at school, return to university or work elsewhere.

There is a strong focus on linking theory and practice in ITE in Estonia. In particular, ITE seeks to prepare teacher candidates for the realities of the classroom through a range of school-based experiences. These occur in the two years of the master’s programme, and include:

- Observations to help teacher candidates relate their theoretical knowledge to practice.
- Teaching practice of 6-10 weeks in a school, organised through a range of partnerships with schools.
- A university-based supervisor and a school-based mentor to support reflection on practice, and to provide general oversight or guidance.

- In some cases, simulated pedagogical practice also occurs in a university setting (e.g. the University of Tartu).

There is also a notable focus on research skills too, grounded in bridging theory and practice. This is evidenced by the writing of a master's thesis, for instance. Or, in some settings, building teacher candidates' research capacity through involvement in research projects (e.g. the University of Tartu).

There is no standardised curriculum for ITE, but institutions use the aforementioned teaching standards to inform the content of ITE programmes. The standards define a broad range of knowledge areas teachers need (e.g. development psychology, basics of research, basics of reflection, age-related peculiarities) and competencies, as well as particular tasks. Analysis has suggested that the breadth of tasks teachers undertake demands a wide knowledge base that is both theoretical – stretching across general pedagogical knowledge and more specific content-based pedagogical knowledge – and including knowledge of research methods (Révai, 2018^[14]).

Teachers beginning their careers in school have the opportunity to voluntarily participate in a 12-month induction programme. This is funded by the Ministry of Education and Research and the induction programme is provided by both of the public comprehensive universities that also provide ITE.

The induction programme hinges on the allocation of a mentor to supervise the work of the beginning teacher. The mentor is a teacher at the receiving school and is appointed by the school principal. There are certain criteria that the mentor must fulfil too, namely having at least three years of experience in pedagogical work and having passed a specific training in supervision.

In terms of tasks, the mentor provides a support for beginning teachers as well as supervising their work. The beginning teacher will, in particular, prepare an individual development chart which contains self-reflections of their experience. The mentor is also required to provide feedback to the ITE provider from which the new teacher received their qualification. In this respect, there is a feedback loop between schools and institutions.

The beginning teacher also has access to tailored workshops that address the frequent challenges of the first years in a school. These seek to address experiences and explore solutions, be that individually or jointly with other teachers.

Two contexts of appraisal

Estonia is characterised by a blend of internal formative appraisal/assessment and an external summative appraisal which is used for certification in the formal career structure.

1. Regular appraisal for performance management:
 - School directors appraise on a regular basis. Others may participate (or lead) such as deputy directors or chairs of subject sections.
 - Primary aim is formative and to inform competency development. That said, may inform salary levels as these are defined at the school level. Though seems rare in practice.
 - High degree of autonomy with criteria defined by directors (e.g. aims, criteria, methods). Typically involves observation and may have a written evaluation report at the end.
2. External appraisal for certification
 - This is a voluntary teacher appraisal process that teachers can undertake to be certified at higher levels on the career ladder.
 - It is organised by the Estonian Association of Teachers, and is thus external to the school. Teachers can apply for certification at any of the levels twice a year (April and November).
 - It involves an evaluation based on two instruments:

- A portfolio prepared by the teacher which includes a self-evaluation, examples of own work, lesson plans and might also include appraisals by the school director.
- A 30-minute interview with the three-member appraisal committee, who also analyses the portfolio. Classroom observation is not required but either the candidate or the appraisal committee can request it.
- Appraisal committees are formed from a pool of about 30 experienced teachers organised by the Estonian Association of Teachers.
- The certification process may also involve some feedback to the candidates which they can use for their own professional development.

The second certification process is mostly disconnected from the first, more school-based teacher appraisal. These processes do not inform each other, except when the teacher applying for certification includes the results of school-based appraisal in his or her portfolio.

Potential transition lessons

The transition process highlights some potential challenges, particularly with regard to communication and buy-in:

- There were some initial challenges with the degree schools embraced the more competency-based career structure. For instance, the professional standards developed by the Estonian Qualifications Authority were not typically used for formative teacher appraisal. Accordingly, a secondary knock-on effect of this was that internal teacher appraisal practices were likely to vary across schools in terms of the criteria applied and the way the results are used for professional development and teacher rewards. This may also have allowed formative and summative appraisal processes to be better connected.
- The challenges of communication and buy-in appear pertinent when considering that a recent review of the system suggested that the roles of senior and master teacher are not universally understood (Cordingley et al., 2022^[13]). Moreover, the review suggested that there was a need to better leverage the two more senior roles of senior and master teacher for fostering more collaboration.
- These challenges around the engagement of teachers and schools may also be connected to the nature of incentives. Notably, the absence of financial incentives has meant that most teachers show little interest in attaining the certifications and engaging in the professional learning that allow access to higher levels.
- With regard to professional learning, as for any system monitoring its effectiveness is challenging. Participation and satisfaction rates are the primary means of assessing professional learning activities.
- One challenge is more systematic dissemination of learning from professional learning activities. For instance, if several teachers take part in an externally provided programme, how can the learnings from this programme be shared more widely among their colleagues and embedded to have greater impact in a school?

Source: (Santiago et al., 2016^[12]), *OECD Reviews of School Resources: Estonia 2016*, <https://doi.org/10.1787/9789264251731-en>; (Cordingley et al., 2022^[13]), *Recommendations for the Professional Development of Teachers and School Leaders in Estonia*, https://www.britishcouncil.ee/sites/default/files/policy_recommendations_for_the_professional_development_of_teachers_and_school_leaders_in_estonia_eng.pdf (accessed 15 February 2025); (Crehan, 2016^[15]), *Exploring the impact of career models on teacher motivation*, <http://unesdoc.unesco.org/images/0024/002462/246252e.pdf> (accessed 21 February 2025); (Révai, 2018^[14]), *What difference do standards make to educating teachers?: A review with case studies on Australia, Estonia and Singapore*, <https://doi.org/10.1787/f1cb24d5-en>

Box B.4. Overview of Finland

Entry to the profession

In Finland, teacher qualification requirements are defined in legislation, specifying the qualification requirements for various teaching positions, as well as language skills. For instance, subject teachers (lower secondary and general upper secondary) are required to have:

- A master's degree (which is necessary at all levels bar early childhood).
- Credits in pedagogical studies (60≤ credits), which can consist of aspects such as practice, educational sciences and subject didactics and include a strong focus on research skills.
- And, at the upper secondary level, in one subject to be taught at least the basic, intermediate and advanced credits of that subject (120≤ credits), as well as credits in other subjects to be taught (60≤ credits). At the lower secondary level, the subject requirement is slightly lower at (60≤ credits).

There are two potential routes to qualification:

- **Concurrent route:** Individuals may undertake teacher education programmes for early childhood education and care teachers, class teachers, and special needs teachers, and study and careers advisers. Those pursuing becoming subject teachers undertake a blend of courses as outlined above.
- **Consecutive route:** Individuals may first complete a master's-level university degree or university of applied science degree, and then undertake pedagogical studies as non-degree studies. Similarly, they also undertake additional studies depending on their planned role as a teacher, for instance in additional subjects to be taught, special needs teacher studies, or study and careers adviser studies.

There is also a degree of selection into these programmes. Indeed, teacher education programmes are in high demand in Finland.

- To take the subject teacher route as an example, students typically apply to a subject programme and then apply during their studies for the right to study pedagogical studies.
- Their selection can be based on their matriculation exam scores as well as a written exam, and during their studies an aptitude test organised in an interview format.
- For those applying to be, say, an early childhood education and care teacher or class teacher, the selection is similar, including a nationally planned and coordinated written exam ("VAKAVA").

Initial teacher education

There are eight universities and five universities of applied sciences with responsibility for teacher education in Finland. Teacher education is organised by different groups depending on one's future career, such as class teacher (primary education), subject teachers (lower secondary education or general upper secondary education), or special needs teachers, among others.

Finland's preparation of teachers has a strong focus on the development research skills.

- Teachers are typically required to complete a master's degree, including a master's thesis.
- More broadly, ITE is characterised by a strong focus on scientific knowledge from research evidence, as well as a focus on the thinking processes and cognitive skills used in conducting research.

- On the one hand, this seeks to develop the type of reflective thinking that is necessary in the profession and ongoing, self-driven learning.
- This also seeks to foster a strong level of research literacy that prepares teachers with the capacity to undertake their own research and to critically evaluate their work.

ITE in Finland is also characterised by concerted efforts to foster robust pedagogical content knowledge. Teacher education is a shared responsibility between the teacher education faculty and the academic subject faculty. This combination means that there is substantial attention to the type of more subject-specific pedagogy necessary for robust pedagogical content knowledge. It avoids treating pedagogy as generic and devoid of particular subject contexts (e.g. questioning approaches in different subjects). This notably applies to the preparation of upper-grade teachers, but there are also some opportunities for classroom teachers as the primary level to complete more studies in a certain subject.

A focus on more theoretical pedagogical studies is coupled with guided teaching practice. Candidates to the profession practice in specific teacher training schools during their studies. These schools provide a space not only for candidates to the profession to practice but also support experiments, research and continuous professional learning too. Such practice is not an additional 'extra', but an essential part of achieving the credits to complete specific programmes.

Underlying this is a blend of both autonomy but also evaluation, transparency and co-operation. There is no single 'curriculum' for ITE programmes and universities have autonomy in determining this and delivering it. Similarly, autonomy covers admissions and assessment of candidates. At the same time, there are also mechanisms to promote quality, particularly through self-driven means:

- The Finnish legislation for universities and universities of applied sciences contain provisions which oblige HEIs to participate in external evaluation of their activities and quality systems.
- According to the Universities act, universities must evaluate their education, research and artistic activities as well as the effectiveness thereof.
- The universities must also regularly participate in external evaluations of their activities and quality assurance systems. Notably, the universities must publish the results of the evaluations they have organised.
- The Ministry interacts continuously with higher education institutions and holds agreement negotiations with them.

There is also the Finnish Education Evaluation Centre (FINEEC). This is an independent authority responsible for the national evaluation of education, from early childhood to higher education. It undertakes three key evaluation types, which are all guided by a principal of 'enhancement-led evaluation': quality audits of higher education institutions (HEIs), thematic evaluations and engineering degree programme reviews.

Overall, when it comes to policy-planning for teacher education, there are high levels of trust and a strong belief in broad and regular collaboration that sources diverse expertise.

Delivery of professional learning

There is a strong emphasis on research skills in Finland with its orientation towards teacher and school autonomy and, thus, problem-solving. This is established early on, with initial teacher education including strong research-based aspects such as group problem-solving activities and a research-based dissertation. This aims to equip beginning teachers with the skills for continuous learning going forward, both in terms of developing themselves but also in terms of supporting their colleagues.

A further arm is the National Teacher Education Forum. This also seeks to foster teachers' active engagement in research, with a wide range of stakeholders participating.

In Finland, programmes of university and school collaboration have been trialled. Jyväskylä University in collaboration with its local municipality saw the development of a project between trainee teachers, their supervisors and practicing teachers in a local primary school. This three-day initiative focussed on innovate teaching methods including first-hand, inter-subject experiences like project-based learning.

Online learning is widely available to Finnish teachers for them to engage with. The alignment with teachers' needs is led by teachers, who have agency in their engagement with this.

The high potential of tutoring is also drawn upon. Between 2016 and 2020, earmarked government funding supported the development of regional networks of 'Tutor Teachers'. The initiative aimed to address contemporary and future challenges in education by supporting teachers through peer mentoring and training. For instance, benefits were particularly noted in relation to teachers' use of digital tools in their practice – one of the goals of the initiative – while tutors themselves also benefited from the opportunity and helped mobilise good practice more widely (Pennanen et al., 2021^[16])

The availability of funding and the allocation of time in teachers' schedules for their engagement was particularly key. Indeed, by 2020, many providers had taken steps to ensure that the tutor teachers could continue following the end of allocated funding (Pennanen et al., 2021^[16]). Finland is currently undertaking research into how widely the model, and other digital supports, are in use across the system.

In Finland, there are also tentative means of tracking the contributions of professional learning to teaching and student learning. The data system "Koski" allows the real-time tracking of student progress, which can be analysed alongside that of the professional learning engaged in by teachers. This allows for an evaluation of the contributions of learning opportunities (Mikkonen, n.d.^[17]). Such information may be relevant for quality insurance.

Collaborative inquiry - Networks

A range of networks support knowledge-building and sharing in Finland. For instance, the National Agency for Education oversees the "Lighthouse network of leadership" to support school-to-school collaboration. It includes typically monthly webinars, including a presentation and subsequent discussion, as well as occasional in-person events or seminars. Numerous similar examples could be included, such as the National Network for Bilingual Education ("Kaksari"), coordinated by the City of Helsinki and funded by the National Agency for Education, or the Early Childhood Education Development Network ("Loisto"), coordinated by the Finnish National Agency for Education.

Indeed, an orientation towards collaborative inquiry permeates the entire system. An example of this can be seen in the National Teacher Education Forum, set out below. A further example is the "LUMA Centre Network". This has operated since 2013 and is a network of 11 universities and which focusses on the subjects of science, maths, technology and environmental studies ("LUMA" subjects), and in particular approaching them in a multi-disciplinary way. It aims to foster a high level of knowledge in these subjects, as well as a high number of professionals across Finland in fields associated with these subjects. To this end, it works to:

- Develop research-based solutions for teaching LUMA subjects
- Support the work of teachers and their continuous professional learning
- Inspire children and youth towards studying these fields and having them as hobbies.

The LUMA Centre Network has built a collaborative 'ecosystem' around these goals, including ranging from museums and libraries to the business sector and science centres. During 2021 and 2024, some 185 000 people participated in its activities each year and the network provided in-service training to some 4 600 professionals.

National Teacher Education Forum

The National Teacher Education Forum seeks to meet the new needs of teacher education in the context of the 2020s through broad co-operation based on research and knowledge.

- It consists of representatives of the universities, providers of education and municipalities, teachers and leaders, students, and other relevant stakeholders. All higher education institutions responsible of providing teacher education and all relevant stakeholders are involved. There is also a small international advisory board.
- The Forum was first coordinated by the Ministry for Education and Culture (2016-2023), with the University of Helsinki since responsible with the Ministry participating and financing the Forum.
- There are four main sub-sections:
 - Student admissions in teacher education
 - Teachers' expertise development
 - Developing teacher education through co-operation and networks
 - Pedagogical leadership in educational institutions
- It undertakes activities such as National and local seminars, workshops, national brainstorming, thematic working groups, literature reviews, reports, benchmarking, and expert consultations. Indeed, it provides extensive funding for collaborative teacher education development projects.
- Notable outputs include research-based development projects and reports and the collaboratively developed national guidance ("The Teacher Education Development Programme").

Its programme growing forward includes the implementation of the aforementioned guidance, a national science, technology, engineering and mathematics (STEM) strategy and action plan, creating a future vision and scenarios for teacher education, research-based reports on digitalisation and AI, and designing and implementing a research-based quality assurance system.

In particular, in March 2025 the Teacher Education 2050 vision was launched. It is the first time such a vision has been created in Finland, being developed from extensive collaboration with all teacher education units and key stakeholders.

In the face of different challenges, including those yet known, the Vision seeks to focus on the development of teacher education, including:

- Enhancing attractiveness and retention through predictions and student admissions
- Continuous development of teacher expertise
- Strengthening teacher education through collaboration and networks
- Skillful leadership enhancing educational institutions to become learning and evolving communities

Underpinning this is the sense that in a more complex and unpredictable world, teachers' strong expertise and professional agency – their will, capabilities and skills – to learn independently and together with others throughout their teacher career are essential. In this changing context, research-based teacher education and its' continuous research-based development are crucial.

Notes: Finland shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (Mikkonen, n.d.^[17]), *KOSKI - a National Registry and Data Transfer Service for Study Rights and Completed Studies*, https://www.oph.fi/sites/default/files/documents/Mikkonen_koski_en_25012021.pdf (accessed on 26 February 2025) ; (OECD, 2019^[18]), *Working and Learning Together: Rethinking Human Resource Policies for Schools*, <https://doi.org/10.1787/b7aaf050-en>; (OECD, 2019^[19]), *A Flying Start: Improving Initial Teacher Preparation Systems*, <https://doi.org/10.1787/cf74e549-en>; (OECD, 2024^[20]), *Education Policy Outlook 2024: Reshaping Teaching into a Thriving Profession from ABCs to AI*, <https://doi.org/10.1787/dd5140e4-en>; (Pennanen et al., 2021^[16]), *Tutkimus Perusopetuksen Tutoropettajatoiminnasta ja sen Vaikutuksista [Research in Basic Education: Tutor Teacher Activities and its Effects]*, https://www.oph.fi/sites/default/files/documents/Tutkimus_perusopetuksen_tutoropettajatoimi (accessed on 11 July 2024)

Box B.5. Overview of Ireland

A continuum of standards

The Teaching Council is the professional standards body for teachers in Ireland. It was established on 28 March 2006 on a statutory, and it is governed by the Teaching Council Acts. The Council is funded by the teaching profession and has an independent regulatory function. It has a dual mandate to both regulate and promote the teaching profession. Concretely its work in teacher education is defined in the Teaching Council Acts to encompass:

- Reviewing the standards of entry to the teaching profession in terms of the knowledge, skills and competences required for teaching.
- Reviewing and accrediting programmes of initial teacher education.
- Setting standards to support the professional learning of newly qualified teachers during the induction phase.
- Advising the Minister regarding entry requirements to programmes of initial teacher education.
- Assessing the qualifications of teachers who qualify overseas to ensure they are of a comparable standard with Council-accredited programmes.

A key feature is the 'Continuum of Teacher Education' that it published in 2011. This is a recognition of the fact that learning never ends, and that teachers are lifelong learners. It encompasses initial teacher education ("Céim" – Irish for 'step'), induction ("Droichead" – Irish for 'bridge') and continuing professional development ("Cosán" – Irish for 'pathway').

Standards of Initial Teacher Education

In Ireland, the Teaching Council is very much a partnership-based organisation and as part of the review of the policy on initial teacher education undertook four rounds of consultations, constituting some 12 meetings, with different stakeholders. This included a considerable of stakeholders, ranging from teacher unions to student unions, parent bodies to principals' bodies, and examination bodies to government bodies, to name but some of the participants.

This led to a range of adjustments from its initial criteria and guidelines for initial teacher education of 2011. There was a notable shift from 'criteria' to 'standards', as well as a range of additional features:

- Core elements that are interwoven throughout all elements of ITE to ensure that these skills and competencies really have time to become embedded into the student teacher's thinking. These core elements are inclusive education, global citizenship education, digital skills, professional relationships and working with parents, professional identity and agency, creativity and reflective practice, and literacy and numeracy.
- The conceptual framework that seeks to support the individual programme design by ITE providers. Hence, accredited providers (see below) draw upon this framework as well as the wider Continuum and Code of Conduct of the Council to develop programmes of ITE that fit their particular context, mission and ethos. This framework promotes a research-informed approach in ITE programmes that supports students in engaging with and in research.
- The school placement, which is a central part of ITE and is based on a partnership approach between Higher Education Institutions (HEI) and schools. This includes direct teaching of 200 hours as well as experiencing a variety of placement settings, so they have a wider experience to draw on. Notably, there is also an expected scheduling of an 'extended period' of school placement that constitutes a minimum of 10 weeks in one school in the second half of the programme. This allows for greater immersion in a particular context.

Accreditation of Initial Teacher Education providers

The Teaching Council has overseen the procedures for accrediting programmes of ITE. Notably, this procedure, published in 2021, has been developed through an iterative, multi-dimensional process, including:

- Surveying higher education institutions among other stakeholders
- Learning from international case studies (including Scotland and Ontario (Canada))
- A literature review
- A pilot programme with ongoing focus group discussions and consultations

The accreditation process sees the Teaching Council appoint an independent panel to review and assess the programmes of ITE. The panels consist of:

- An independent Review Panel Chairperson
- A registered teacher with expertise in the support and/or evaluation of teaching and learning at the school level in the sector in question
- An external expert in the area of teacher education or accreditation processes in another system/jurisdiction
- A supporting rapporteur

During the review process, the Teaching Council staff may also have some engagement in terms of seeking clarification or raising queries with the Review Panel.

Following their review, the Panel make a recommendation to the Teaching Council's Education Committee, either to accredit the programme, accredit with requirements, or refuse accreditation. The Committee then take a decision. Requirements need to be addressed within six months.

There are currently 14 providers of ITE, with 52 accredited programmes of Primary and Post-primary ITE. These programmes cover concurrent, consecutive and hybrid models. The providers were reviewed and accredited in the last cycle of accreditation, which started in November 2021 and lasted

some two years. The average review time from submission date by the HEI to the outcome of the Education Committee was approximately eight months, reflecting the thoroughness of the process.

Induction

There is one sole route of induction to full registration as a teacher in Ireland. It is a state-funded induction process, and needs to be completed within three years of when someone obtains 'Newly Qualified Teacher' status following their completion of ITE. It consists of a minimum of 60 days at the primary level, and 200 hours at the post-primary level, within which 'Newly Qualified Teachers' must demonstrate to:

- have engaged professionally with school-based induction and additional learning activities
- have shown their professional commitment to quality teaching and learning for their pupils/students
- have engaged in reflective practice that supports their professional learning and practice, both individually and collaboratively

The induction process has recently undergone review and revision. In 2024, a survey of some 3 500 teachers, supported by follow-up interviews and focus groups with stakeholders, found that induction, overall, is fulfilling its role in supporting newly qualified teachers albeit with some areas of challenge. Following the review, an updated Policy was redesigned and approved – including with an Irish language translation – and is due to come into effect in September 2025.

Background to professional learning approaches

In 2023, *Oide* (an Irish word meaning “teacher” or “tutor”) was introduced to consolidate professional support services for teachers and school leaders. It is funded by the Department of Education (DoE) in Ireland, and brings together four previously separate professional learning services: the Centre for School Leadership (CSL), Junior Cycle for Teachers (JCT), the National Induction Programme for Teachers (NIPT) and the Professional Development Service for Teachers (PDST). Its overarching purpose is to support the professional learning of teachers and school leaders.

Much of this professional learning occurs through 21 full-time and 9 part-time 'Education Support Centres'. Each Centre is managed by an annually elected voluntary Management Committee and serves the local education community. In total, the DoE provides annual funding exceeding EUR 60 million for professional development courses provided by *Oide* and the Education Support Centres. This funding both supports delivery and participation:

- funding includes the cost of over 360 seconded teachers providing continuing professional learning (CPL),
- and funding for substitute teacher cover in schools that can enable participation in professional learning (Department of Education, 2024^[21])

Monitoring professional learning

There are several mechanisms in place to support monitoring of the quality of professional learning:

- *Oide* make regular use of teacher evaluation. All summer courses include a teacher evaluation. *Oide* also regularly reviews the quality of in-person training courses at the primary and post-primary levels, as well as a sample of online courses.
- The DoE Inspectorate evaluates the design and facilitation of a sample of online and in-person professional learning courses each year.
- The Educational Research Centre published in 2023 a framework intended to support the evaluation of different forms of formal professional learning, either by providers or external

evaluators. In parallel, the DoE has also developed an internal framework (the Design and Quality Assurance Process) to inform the design and quality assurance process of learning opportunities provided by its support services (Gilleece, Surdey and Rawdon, 2023^[22])

Finally, there are also some further additional mechanisms that seek to support *Oide* to ensure that the centrally-provided professional learning offer is responsive to schools' needs.

- For example, link inspectors are nominated by the Inspectorate to support *Oide* during the scoping, design and review of new professional learning programmes. Crucially, these link inspectors have the ability to inform *Oide* of training needs that the Inspectorate have observed in the field.
- Similarly, the Teaching Council plays an important role in shaping professional learning opportunities. In addition to its role in licensing teachers and accrediting ITE programmes, the Council is responsible for advising the Minister for Education on teachers' CPL, promoting engagement in CPL, conducting research and raising awareness of the benefits of teachers' professional learning (OECD, 2022^[23]).
- The Teaching Council has also been responsible for important outputs such as:
 - A Code of Professional Conduct, which sets out standards and expectations for teachers' professional knowledge, skill, competence and conduct (The Teaching Council, 2016^[24]).
 - Cosán (the Irish word for "pathway") which was published in 2016 and seeks to provide a national framework guiding teachers' continuing professional learning (The Teaching Council, 2016^[25]).

These are examples of the type of feedback loops between stakeholders that can help to ensure the provision of learning opportunities meets different needs.

Delivery of professional learning

A range of professional learning opportunities are available:

- There is a concerted effort from *Oide* to promote the formation of communities of practice. These are building built into their professional learning frameworks.
- There are also efforts around building capacity in data and research-inform school planning. *Oide* offers school leadership teams a programme in relation to this. The Inspectorate also offers advisory visits to support the robust use of school-level data.

Sharing of best-practice

There are also notable efforts to facilitate the sharing of best practices widely across schools.

- The DoE has organised learning days focussed on sharing good practices. For instance, one topic has been Adapting and Managing Successful Transitions as part of Ireland's Delivering Equality of Opportunity in Schools programme.
- The Inspectorate collects and describes promising practices and programmes in its published reports of inspected schools as well as more thematic reports, including on main inspection findings from Delivering Equality of Opportunity In Schools (DEIS) programme schools. There has also been the provision of webinars by the Inspectorate through the Education Support Centre networks (see OECD, 2024^[103]).
- The Inspectorate also published a good practice guide titled "Effective literacy and numeracy practices in DEIS schools". This was intended to support the sharing of good practice among schools and teachers (Inspectorate, 2009^[26]).

- Similarly, a depository of tools on social inclusion and equality of opportunity is being developed for the *Oide* website.

Notes: Ireland shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (OECD, 2024^[27]), *OECD Review of Resourcing Schools to Address Educational Disadvantage in Ireland*, <https://doi.org/10.1787/3433784c-en>; (OECD, 2022^[23]), *Education at a Glance 2022: OECD Indicators*, <https://doi.org/10.1787/3197152b-en>

Box B.6. Overview of Japan

The career structure of teachers in Japan

Teaching is a highly attractive job in Japan. Teachers are well-paid with a competitive starting salary relative to other careers, but teaching also carries a long and well-established prestige (OECD, 2011^[28]).

The Boards of Education (BOEs) of prefectures and ordinance-designated cities are responsible for hiring teachers to work in schools in their jurisdiction. After completing initial teacher education (see below), teacher candidates apply to their local BOE who issues teaching certificates. Candidates must pass a series of multiple-stage competitive employment examinations to be eligible for a permanent teaching position in a public. These are set by the BOE but tend to be similar, with aspects such as general knowledge, professional knowledge and subject-based knowledge all being typically tested, while it may be that there are also interviews, essay writing and practical and aptitude tests.

Teachers are employed by the BOE and assigned to teach in schools in its jurisdiction. They are typically transferred to different schools every few years. Indeed, this is part of the centralised efforts to co-ordinate resources. Thus, BOEs also usually allocate senior teachers and senior vice-principals to schools with difficulties or a large number of students, and advanced skilled teachers to school with young teaching staff. In this respect, teachers are assigned based on a needs-basis, with the strongest going to those who need them the most.

Table A B.2. Overview of the career structure in Japan

Position	Notable features
Homeroom teachers / subject teachers.	• Typical starting point for beginning teachers. • One-year formal induction programme upon entry.
Chief teacher of a grade.	• After more experience, teachers can take on the role of chief teacher of a grade and assume responsibilities for managing a group of teachers.
Senior teachers / Advanced skilled teacher	• Both roles are optional in a school. • Advanced skilled teachers and senior teachers are selected by the prefectural Board of Education after recommendations from the Board of Education of the city, town, or village. • Both roles see direct teaching of students. • Senior teachers also support the principal in the effective running of the school. • Advanced skilled teacher advises other teachers and staff in order to improve education guidance for students.
Vice-principal / Senior vice-principal	• Teachers may serve as a vice-principal or, with more experience, a senior vice-principal • The senior vice-principal is optional in a school. It is part of management and accessing this position requires passing an examination at the prefecture level.
Principal	• Specific requirements for principal can be set by the BOE.

Source: (OECD, 2018^[29]), *Education Policy in Japan: Building Bridges towards 2030*, <https://doi.org/10.1787/9789264302402-en>

There are also some roles that are associated with BOEs. Some teachers are also transferred to BOEs to become teacher supervisors, who advise schools and co-ordinate training for teachers and school leaders.

Schools are required to assign a mentor teacher for every new teacher with a regular-term contract. Mentor teachers are most commonly selected by principals from among vice principals or senior teachers at the school. Time is freed up in their schedules to allow them to engage in mentoring as part of a strong commitment to supporting quality induction for beginning teachers to make sure they have the necessary skills to succeed.

In Japan, a teacher's career is characterised by working in different schools; teachers do not necessarily progress through their career in only one school. There are mandatory rotation mechanisms requiring teachers to relocate to different schools periodically. This is centred on the idea that transferring good teachers to schools that are struggling will ensure all students benefit from equally qualified teaching staff, and also enhance capacity in schools by spreading good practice. Rules for such mechanisms are decided by local education authorities

Initial teacher education

In Japan, ITE is provided by approved universities or departments of universities. There are hundreds of universities and departments that have met the necessary requirements to provide ITE programmes. They must undertake self-evaluations and undergo accreditation processes by evaluation and accreditation organisations certified by the Ministry of Education, Culture, Sports, Science and Technology (MEXT) at least every seven years.

In order to enter ITE programmes, individuals need to pass entrance examinations and be enrolled as students at universities with ITE programmes. In order to be able to become a teacher, candidates need to complete all required teacher-preparation courses and a practicum, in addition to an associate or a bachelor's degree. A key actor here is the local Boards of Education (BOE). Following completion of an ITE programme, teacher candidates must apply to the local BOE. The BOEs of prefectures and ordinance-designated cities are responsible for certificating and hiring teachers to work in schools in their jurisdiction. They administer their own competitive employment examinations for assigning teachers as permanent teachers at public schools.

More broadly, there have been efforts in recent decades to build more collaboration between universities and BOEs to help raise the quality of initial teacher education programmes, as well as that of ongoing professional learning.

BOEs in Japan are required by law to provide induction training for one year for all beginning teachers with a regular-term contract. Candidates who have been successful with the examinations of their BOE to be hired, are on a probationary status in their first year of employment. The induction training process typically includes assigning a mentor teacher for every new teacher, who are commonly selected by the principal of the school from among their pool of vice principals or senior teachers in the school. It often includes in-school professional learning opportunities too.

More broadly, Japan has seen more concerted efforts in recent years to foster collaboration and closer connections between different stakeholders. For instance, the system of affiliated schools in Japan are an example of strong university-school partnerships that seek to support practical insights into the classroom. More than 250 affiliated schools provide opportunities for on-site innovative teaching experiences for teacher candidates, as well as serving as a space for collaborative research between the school and university. Such experiences can mean that induction is not beginning teachers' first exposure to the realities of the classroom.

Professional learning and career progression

Professional learning focusses on both extending and renewing teachers' practice, skills and beliefs. An example of this is the "Lesson Study" approach set out below.

Until recently, the Teaching Certificate Renewal License – introduced by Japan in 2009 – meant that teachers needed to renew their teaching certificates by participating in at least 30 hours of professional learning programmes every 10 years. The extra burden this placed in terms of time (with programmes being undertaken during holidays or weekends) and costs (with costs of programmes being covered by teachers) were seemingly barriers in the reform's implementation (Chapple, 2023^[30]).

Similarly, it is important to note that the more senior levels of the career structure are dependent on certain forms of professional learning. Teachers wanting to become a school leader have to enter a Professional Graduate School in Education to build on their applied knowledge and develop a theoretical background. This graduate school delivers a master's degree in teaching to become a "School Leader (mid-career core teaching staff)". School leaders are then expected to master leadership theories and exhibit practical and applied skills needed to fulfil their leadership role in the school as well as bringing local communities closer. There are also specific requirements for principal as set by the BOE.

Collaborative professional learning

A notable feature of the Japanese approach to professional learning is that of 'Lesson Study'. It parallels the way that teams often work in private industry and also reflects the Japanese focus on relying on groups to get work done.

The approach essentially sees groups of teachers collectively observe and review their lessons and how to improve them (in part by analysing student errors). Typically, teachers work together to prepare a specific lesson on a topic where students have struggled. This preparation process draws upon academic research to support their work and reflection. The designed lesson is then taught by one teacher to students, while other teachers (sometimes even from other schools) observe and learn the new pedagogical approach.

The approach has a profound impact on the practice of teaching by bringing the work of teaching out from behind the closed door of the classroom. Instead, it allows teaching to be opened up to observation, analysis and even critique by colleagues. Indeed, there is a very strong degree of teacher accountability in Japan in terms of accountability to one's colleagues and ensuring that one contributes positively to the group and sustains high standards. Teachers do not want to let the group down but strive to develop superior lesson plans, teach them well, and to provide sound and useful critiques when it is their colleague's turn to demonstrate their lesson plans to them. This is all grounded in the goal of nurturing the continual, sustained improvement of teaching practice. It thus strengthens co-operation between teachers.

One further consequence of this is the nurturing of consistent quality teaching focussed on rigorous concept development. Across lessons, there is a similar pattern to questioning and how student thinking progresses.

Another further consequence of this more open format of observation and analysis is a wider cultural of sharing and critiquing with a goal of improvement for students' benefit. Hence, when new content arrives on the national curriculum, there may be groups of teachers and researchers who review research and curriculum materials and refine their ideas in pilot classrooms. This refinement may occur over a whole year before a public research lesson is hosted. Such public research lessons can be viewed online by hundreds of teachers, researchers and policy makers. The Lesson Study approach also contributes to fostering inter-school networks of teachers too.

A further dimension of collaborative learning is that of the ‘teachers’ room’ in schools (*shokuin shitsu*). All teachers in Japan are based during the school day in a common teacher’s room. Teachers have an assigned desk and space to meet together daily to prepare and collaborate. This means that teachers do not have an assigned classroom but travel from classroom to classroom while students stay in one place. Researchers have found this provide a valuable physical space for informal exchange and learning for all teachers whatever their stage of career (Ahn et al., 2018^[31]). While less structured as a form of professional learning, it seems a valuable space for fostering collaborative learning.

Use of digital tools

Japan has recently developed an extensive new digital platform to support teachers in delivering a new era of education (“*Reiwa*”). The ‘Platform for Teachers and Staff Development’ aims to support efficiency and effectiveness:

- Teachers are required to record all professional learning activities in the platform.
- There are specific tests to check knowledge and skill acquisition in professional learning activities, as well as reflective tools to support contextual translation.
- Teachers and school leaders can select from a catalogue of development opportunities those that best fit their own and their school’s needs.
- A library of video on-demand teacher development programmes developed by universities, boards of education and public-interest corporations, among others.
- Tools for school-based collaboration (e.g. sharing content) and lesson study.

Teachers can evaluate the quality of the professional learning that they engage with, including its perceived impact on their teaching. For system leaders, this has the potential to offer an aggregate view across the system.

Source: (National Center on Education and the Economy (NCEE), 2025^[32]), *Top-Performing Countries: Japan*, <https://ncee.org/japan/#teachers-and-principals> (accessed on 16 March 2025); (Schleicher, 2020^[33]), *Insights and Interpretations TALIS 2018 Teaching and Learning International Survey*, http://www.oecd.org/education/talis/TALIS2018_insights_and_interpretations.pdf (accessed 14 March 2025); (Schleicher, 2018^[34]), *Valuing our Teachers and Raising their Status: How Communities Can Help*, <https://doi.org/10.1787/9789264292697-en>; (OECD, 2011^[28]), *Lessons from PISA for the United States*, <https://doi.org/10.1787/9789264118539-en>; (OECD, 2018^[29]), *Education Policy in Japan: Building Bridges towards 2030*, <https://doi.org/10.1787/9789264302402-en>; (Chapple, 2023^[30]), *No need to renew: The end of Japan’s teacher license renewal system and the future*, <https://doi.org/10.20448/edu.v9i2.4685>; (Ahn et al., 2018^[31]), *Japan’s innovative approach to professional learning*, <https://doi.org/10.1177/0031721718815674>

Box B.7. Overview of Singapore

Career structure

The Singapore career structure encompasses the entire education system and the varied roles within this. Thus, it includes both teaching and management roles within a school as well as a range of positions in the system more broadly (see Table A B.3).

Table A B.3. Overview of Singapore's career structure

Classroom teacher		
↓	↓	↓
Teaching Track	Leadership Track	Specialist Track
Senior teacher	Subject head / level head	Senior specialist 1
Lead teacher	Head of department	Senior specialist 2
Master teacher	Vice Principal	Lead specialist
Principal master teacher	Principal	Principal specialist
	Cluster superintendent	Chief specialist
	Deputy director	
	Director	

Source: (National Institute for School Leadership, 2019^[35]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025)

There is not a specific salary grid based exactly on the tracks, though there are connections. Thus, moving up to the next level typically incurs an increase though this may vary.

While tracks are well-defined with a clear progression, there is also a degree of flexibility. There are some broad expectations (see below) on the roles that each level encompasses, but these roles are also shaped by principals and leaders who define what is expected based also on what is needed in a particular school.

There is also some flexibility between the tracks. Teachers can both permanently change trajectory and also move to another track more temporarily as a means of gaining new experiences. In both cases, these require authorisation from their supervisor. That said, it may also be that there is a degree of reassignment of individuals by the Ministry if their skillset is deemed particularly well suited to one track.

Connected to this idea of some flexibility is that there may be the option to take on certain roles from a higher level on the career ladder without being formally assigned to that level. This is a way of essentially testing and proving one's capability to undertake new skills and roles. This is highly informal and subjective, and is referred to as 'upmanning' in the system.

Classroom Teachers

The first rung on the career ladder in Singapore is that of classroom teacher. Some 60% of teachers in Singapore fall into this category.

- Teachers are part of this category for approximately five years, when they may after consultation with their supervisors move into one of the three tracks (see summaries of each below). Hence, they can take on these additional roles when there is agreement that a teacher is ready for extra responsibilities.

- Accordingly, this category may be composed of younger staff not yet ready for additional roles or those with less interest in mentoring or cluster work.
- Classroom teachers teach full-time though this includes programmed hours for meetings (e.g. team peer meetings), parent engagement and professional learning.
- There are five salary levels within this category. Moving up is contingent on evaluation and normally yields around a 10% salary increase. One-off performance-based bonuses are also available each year.

Teaching Track

The objective of the Teaching Track is to support those aspiring to remain in the classroom so they become experts in their practice and can support their peers in the classroom.

The Teaching Track consists of some 15% of teachers in the Singaporean system. This track has a particular salary scale (the “Senior Education Officer Scale”), with three levels of 15-20% increase.

Table A B.4. Overview of the Teaching Track

Position	Notable features
Senior teacher	• Mentors for all first and second years • Supervise teacher training practicums • Sometimes workshops or CPL sessions • Lose ~20% of teaching timetable for the roles
Lead teacher	• Build capacity of teachers and senior teachers (content, pedagogy, assessment focus) • Develop professional learning communities in schools • Share their expertise within clusters • Lose ~33% of teaching timetable
Master teacher	• Mentoring • Responsible for driving standards and new pedagogies across schools • ~80% of their work attached to the Ministry, serving as primary staff for the teacher professional learning of the Academy of Singapore Teachers • Teach master classes in schools for observation • Approximately 70 in this role in the system
Principal master teacher	• Essentially chief pedagogical experts for their subjects • Responsible for pedagogical innovation and research • Deliver master classes and lead TPL • Required to teach one class a week in a school • Same pay as school principals • Approximately 20 in this role in the system

Source: (National Institute for School Leadership, 2019^[35]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025)

Leadership Track

The Leadership Track aims to support those seeking school leadership position or leadership roles in the wider Ministry.

Some 25% of teachers are on this track.

Table A B.5. Overview of the Leadership Track

Position	Notable features
Subject head / level head	• Observe and coach beginning teachers • Participate in formal end-of-year evaluations • Subject heads organise the teachers of a subject • Level heads co-ordinate a given age group • Lose ~20% of teaching timetable lost
Head of department	• Departmental management, including coaching and developing teachers in subject areas • Administration, supporting the principal and vice-principals • Part of school's management team • ~33% of teaching timetable lost
Vice principal (V-P)	• Two in every school, one for education and one for administration • V-P for education: oversee all curriculum-related areas, lead evaluations for teaching staff, make decisions about teaching and learning • V-P for administration: oversee all non-curriculum areas and lead a team of executive and administrative staff. Provide strategic planning and resource management support
Principal	• Systematically appointed and rotated between schools to bring fresh perspectives and share best-practice • Lead and inspire teachers • Work closely with parents and the community
Cluster superintendent	• Mentor and support a group of principals (10-15 principals) • Observe all school-based evaluation processes to monitor them for fairness
Deputy director	• Collectively lead the ~40 departments and sub-departments in the Ministry.
Director	
Director-general of Education	• Highest position in education • Leads the entire system

Source: (National Institute for School Leadership, 2019^[35]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025)

Specialist Track

The Specialist Track's objective is to support those seeking to develop deep knowledge and skills in specific disciplines, with potential Ministry-level oversight (e.g. curriculum development, evaluation).

Less than 1% of teachers in the Singapore system are on this track.

Table A B.6. Overview of the Specialist Track

Position	Notable features
Senior specialist 1	• Do not teach but service in the Ministry and at the National Institute of Education • Less differentiation in roles and responsibilities than other functions with progress depending on quality of research • Produce scholarship similar to that of a professor, aiming to produce new subject matter knowledge • Expected to have or be undertaking a doctorate
Senior specialist 2	
Lead specialist	
Principal specialist	
Chief specialist	

Source: (National Institute for School Leadership, 2019^[35]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025)

Evaluation Performance Management System

The Evaluation Performance Management System (EPMS) relies both on clear procedures but also professional judgement. There is no specification of exactly how supervisors must assess, with an intentionally broad set of outcomes and competencies (see Table A B.7). There are three outcomes ('Key Results Areas') that teachers are judged against, though the indicators or features that are measured within this remain at the discretion of the principal and evaluators. The 'Key Results Areas' are:

- Student outcomes, which may include aspects such as quality of student learning, students' character development and co-curricular engagement.
- Professional outcomes, which may include aspects such as their own development and engagement in professional learning, and their coaching and development of others.
- Organisational outcomes, which may include aspects such as the teacher's contributions to school, to wider committees, or collaboration with parents.

The final evaluation seeks to establish if candidates have met the agreed goals and are ready for the next career step and additional roles, as well as areas where further support may be needed. The final evaluation each year sees the Principal as well as other more senior staff such as Heads of Department/Levels participate. Likewise, the Cluster Supervisor attends all evaluations for the schools under the remit to offer oversight. Notably, the Cluster Superintendent has the ability to challenge ratings. This is though seemingly rare in practice.

Table A B.7. Overview of competencies and key result areas in the EPMS process

Individual attributes	Professional mastery	Organisational excellence	Effective collaboration
Professional values and ethics	Student-centric, values-driven practice	Visioning and planning	Interpersonal skills and relationships
Self-management and development	Curriculum and content	Action management and implementation	Teamwork and team building
Analytical thinking and intellectual flexibility	Pedagogy and instruction	Culture building and people development	Internal and external partnerships
	Assessment and evaluation		

Source: (National Institute for School Leadership, 2019^[35]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025)

It is worth noting that the evaluation system rests upon professional judgement and the strong institutional expertise that exists within this system having been built up over many years. Rigorous training and the collective participation of many of the same actors together (e.g. Principals, Cluster Superintendents) helps to build this consistency. It raises the question of how to foster this type of culture, and whether more prescriptive guidance and reflective feedback loops may be needed initially to support a transition to this style of evaluation system.

Initial teacher education

In Singapore, there is one single provider of initial teacher education to prepare teachers for the classroom and profession: the National Institute of Education. Preparation seeks to develop the knowledge and skills related to teaching, as well as imbuing candidates with the purpose and values of teaching and the education system.

In 2023, the Enhanced TE²¹: Empowering Teachers for the Future Model was launched to help develop teachers that are ready for the future, insofar as they themselves can develop students' 21st-century competencies so they are also future-ready. The new model builds on the original model but with a particular focus on preparing teachers and the school systems for the fluctuating changes of the future. As outlined in PISA 2022 Volume V (OECD, 2024^[36]), the new model is rooted in:

- Three core values:
 - Commitment to the learner: believing that all children can learn, nurturing each learner holistically, and valuing diversity.
 - Commitment to the teaching profession: engaging in lifelong learning not only for one's personal and professional growth but as a role model to students.
 - Commitment to the community: being cognisant of their role in the ecosystem and contribute to society by impacting the next generation of learners.
- Knowledge of the self as a teacher, the learners, the subject content and pedagogy. In particular, there has been a focus in the new model of expanding teachers' knowledge of topics such as sustainability, global and environmental issues, and health and mental health.
- Skills for teachers of 21st-century learners such as promoting reflection and metacognition, digital and data literacy and cross-cultural literacy.
- Three core competency dimensions: professional practice, personal growth and development, and leadership and agency. These aims to help teachers to enable teachers to perform five roles, namely, shapers of character, creators of knowledge, facilitators of learning, architects of learning environments and agents of educational change.

In terms of delivery, there has been a focus in the past decade or so on promoting a learning environment in ITE that is more centred on and driven by teacher candidates rather than teacher educators. This was grounded in seeking to develop more self-directed, inquisitive learners. Simultaneously, there was also a focus on strengthening the link between theory and practice. A manifestation of these shifts is that candidates were, for instance, increasingly asked to engage in inquiry and reflection in the classroom, such as through using real-life school scenarios as a basis for this. More problem-based, blended and flipped learning have become more important.

Professional learning approach

Teachers are entitled to 100 hours of professional learning opportunities per year – an indication of the value that the system places on continuous growth and development. This may be undertaken in several ways:

- On the one hand, it may be through more formal, central channels. The National Institute of Education provides courses that focus on aspects such as subject matter pedagogical knowledge. Engagement in these may contribute towards higher degrees or advanced diplomas.
- School-based professional learning that is led by staff can account for a large amount of professional learning. This typically focusses on identifying specific problems in a school or introducing new approaches.
- Networks across schools are also an important means of encouraging and facilitating peer-to-peer learning. For instance, the Academy of Singapore Teachers was opened in September 2010 to further encourage teachers to continuously share best practices.

In terms of funding, each school has an allocated fund through which it can support teacher growth. This can include funding trips and visits abroad to learn about new approaches from other countries and to gain fresh perspectives.

Centre for Character and Citizenship Education

The Singapore Centre for Character and Citizenship Education (SCCCE) was established in 2023. It was established to advance Singapore's vision that teachers play a crucial role in guiding students to become good people and citizens. There is a deep belief in the system – as reflected across ITE and professional learning opportunities above – that every teacher is a teacher of character and citizenship education (CCE), with it a collective responsibility and part of a teacher's identity.

The Centre serves as a bridge between research and practice. It synthesises local and international scholarship, developing in 2025 a meta-framework for character and citizenship education. Its supports for professional learning at all levels manifest in different ways:

- Pre-service teachers in the *Meranti Programme* reflect on their identity and purposes as educators.
- In-service teachers in the “*HeART of CCE*” course deepen their moral conviction and relational practice to create caring and enabling school environments that cater to diverse needs.
- Through the *Facilitating Challenging Discussions* course, specialised CCE teachers develop their efficacy in leading values-based discussions on complex social issues.

Notes: Singapore shared insights in the organised peer-learning event of 13-14 May 2025 in Kyiv, Ukraine for the Ukrainian Ministry of Education and Science that have informed this box.

Source: (National Institute for School Leadership, 2019^[35]), *Singapore's Career Ladder System*, <https://documents.ncsl.org/wwwncsl/Education/Study-Group/NISLSingapore%27sCareerLadderSystem.pdf> (accessed 21 February 2025); (Schleicher, 2011^[37]), *Building a High-Quality Teaching Profession: Lessons from around the World*, <https://doi.org/10.1787/9789264113046-en>; (Schleicher, 2018^[34]), *Valuing our Teachers and Raising their Status: How Communities Can Help*, <https://doi.org/10.1787/9789264292697-en>; (OECD, 2024^[36]), *PISA 2022 Results (Volume V): Learning Strategies and Attitudes for Life*, <https://doi.org/10.1787/c2e44201-en>

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This publication provides a comprehensive overview of teacher professional learning from the initial steps of a teacher's career to supporting their sustained development and excellence. The publication considers four high-level ambitions for policymakers: how teachers are selected to and prepared for the profession, how a career structure can incentivise and reward excellence, how to effectively provision and deliver ongoing teacher professional learning, and how to develop an infrastructure that can create an ecosystem of learning across schools. For each of these, the publication synthesises OECD and wider research and highlights powerful policy examples from different systems. The publication concludes by considering the current context of teacher professional learning in Ukraine. The publication can provide relevant insights to those, like Ukraine, interested in reimagining their approach to professional learning and how to strengthen the policy environment that allows teachers to thrive.



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